Tensor networks as surrogate models of training dynamics

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Abstract

This note motivates tree tensor networks (TTNs) as surrogate models of the training dynamics of deep neural networks, and makes the underlying mathematics precise. We define tensor networks, TTNs, their edge matricizations, hierarchical ranks and hierarchical singular value decomposition (HSVD), and compare the relevant notions of tensor rank. A deep network with polynomial activations is exactly a weight-tied TTN evaluated on copies of the homogenized input. This suggests reading the HSVD of that TTN as a gauge-invariant description of the hierarchical features learned by the network, and its hierarchical ranks as a measure of learning capacity; we make both notions precise and discuss their limits. Taylor-expanding the activations of a smooth network at a training checkpoint gives a local TTN surrogate, whose fidelity to the original training dynamics is controlled by prediction and Jacobian errors. We state a global and a local surrogate hypothesis, the open questions they raise, and the experiments that would test them.

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1 Introduction

In this research note I motivate tree tensor networks (TTNs) as a surrogate model of the training dynamics of deep neural networks (DNNs). I also suggest experiments to check whether TTNs are a promising surrogate for the training dynamics of more realistic neural networks. Section 2 develops the mathematical background, and the experiments are collected in Section 7.

Why training dynamics? To be brief, my current position is that AI safety is bottlenecked by understanding. Interpretability as a field remains shallow, in the sense that current methods explain little about the behaviour of current models, or are unlikely to scale. One of the main reasons for this state of affairs is that we are still confused about fundamental questions on how neural networks work. One such question is *what kind of structure neural networks learn*. Extracting this structure from a network at the end of training seems extremely challenging. The bet behind studying training dynamics is that internal structure decomposes more naturally during training, as it is progressively learned.

Why tensor networks? We have a good understanding of deep linear networks (DLNs). For whitened inputs and a suitably aligned initialization, Saxe et al. (2014) solved the gradient-flow dynamics of DLNs exactly: the singular modes of the input–output map decouple and are learned one after the other, each after a plateau, stronger modes first (see also Saxe et al., 2019). From a small initialization, DLNs and diagonal linear networks follow saddle-to-saddle dynamics in which the rank (or the support) of the learned map increases step by step (Jacot et al., 2021; Pesme and Flammarion, 2023), and similar dynamics explain a simplicity bias across architectures (Zhang et al., 2026). We are, however, still far from understanding more complex architectures such as transformers. Even the learning dynamics of deep networks with quadratic activations remain largely open, although there are interesting results for shallow ones (e.g. Du and Lee, 2018; Sarao Mannelli et al., 2020).

TTNs provide an interesting model of more complex deep networks (Furman et al., 2026). A TTN is a tree of multilinear maps (Definition 2.7); it generalizes DLNs, which are chains, and Tucker decompositions, which are stars. Unlike a DLN, a TTN computes nonlinear functions of its inputs: it can implement any read-once Boolean formula, and it can be evaluated in time polynomial in the size of the tree and in the bond dimensions. Consequently, TTNs contain targets that are easy to evaluate but cannot be learned by gradient descent in polynomial time, such as parities, for the same reason as neural networks. Nevertheless their loss landscape is conditionally benign, and the difficulty can instead come from high-order saddle points, which are located at rank-deficient parameters (Furman et al., 2026, Theorems 4.3 and 5.2, Proposition 5.3). Hierarchical tensor formats can moreover be exponentially more compact than shallow (CP) ones (Cohen et al., 2016). At the same time, TTNs look more interpretable than generic networks because they come with a canonical, gauge-invariant hierarchical singular value decomposition (HSVD, Section 2.4). Empirically, TTNs trained from a small initialization undergo saddle-to-saddle dynamics, i.e. they learn features sequentially, consistently with the incremental hierarchical-rank bias of hierarchical tensor factorizations (Razin et al., 2022) and with the rank-deficient saddles of Furman et al. (2026). TTNs are also more expressive than DLNs, and they can display *soft hierarchical learning*: learning some features accelerates the learning of subsequent features. The following definition makes this precise.

Definition 1.1 (Hard and soft acceleration; after Diagonal TTN note, 2026, Definition 3.1)**.** Let $q_B(\vartheta)$ be an observable measuring the acquisition of a feature B, with target value $q_B^\star > 0$, and fix a threshold $\delta \in (0,1)$. For an intervention $\mathcal{I}$ on training (for example, clamping the parameters that compute another feature A to a learned or to a blocked value, all else being equal), let

$$\tau_B(\delta \mid \mathcal{I}) = \inf\{t \geq 0 : \; q_B(\vartheta_{\mathcal{I}}(t)) \geq \delta\, q_B^\star\} \in [0, \infty].$$

Learning A *accelerates* learning B if $\tau_B(\delta \mid A \text{ learned}) < \tau_B(\delta \mid A \text{ blocked})$. The acceleration is *hard* if $\tau_B(\delta \mid A \text{ blocked}) = \infty$, and *soft* if both times are finite.

In a DLN with decoupled modes, the only such effect is the symmetric facilitation between the factors of a single mode. In TTNs, acceleration between distinct features can be computed exactly in simple cases (Diagonal TTN note, 2026, Sections 5–6): readouts of nested sub-products ("staircases") produce hard prerequisite gating, and residual paths produce soft acceleration. For instance, in a three-leaf channelwise residual TTN with residual strength ρ and composition strength γ, learning the lower core (teacher value α) divides the acquisition time of the upper core by $1 + \gamma^2\alpha^2/(2\rho^2)$ (Diagonal TTN note, 2026, Proposition 6.1). Expressivity alone is not the mechanism: with fixed orthogonal leaf features, a homogeneous diagonal TTN has exactly the amplitude dynamics of a diagonal DLN, and acceleration comes from parameters shared by several output features (Diagonal TTN note, 2026, Proposition 4.1 and Section 7). For a broader review of hierarchical feature learning and of the observables proposed to track it, see Literature review (2026).

Two hypotheses. While TTNs have properties that make them attractive as a model of deep networks, we would like to first establish how far their training dynamics is from that of more realistic architectures. If TTNs are a good model of the training dynamics of neural networks, then according to a range of observables the training dynamics of TTNs should track the training dynamics of more realistic architectures.

To state this precisely, fix a distribution ν of input–label pairs $(x, y) \in \mathbb{R}^{d_0} \times \mathbb{R}$, with input marginal ν_x, the square loss $\mathcal{L}(f) = \frac{1}{2}\mathbb{E}_\nu[(f(x) - y)^2]$, and an optimizer with its hyperparameters, for instance full-batch gradient descent $\vartheta_{n+1} = \vartheta_n - \eta_n \nabla_\vartheta \mathcal{L}(f_{\vartheta_n})$, or its gradient-flow limit in the time variable $t = \sum_n \eta_n$. Training a model class $\vartheta \mapsto f_\vartheta$ from an initialization produces a trajectory $t \mapsto f_{\vartheta(t)}$. By an *observable* we mean a quantity that can be evaluated along the trajectory of any architecture. The most robust observables are computed from the predictors $f_{\vartheta(t)}$ alone: the loss $\mathcal{L}(f_{\vartheta(t)})$; the function-space distance $\|f_{\theta(t)} - g_{\vartheta(t)}\|_{L^2(\nu_x)}$ between two models; and, in a teacher–student setting where the target $f^\star = \sum_{m=1}^{M} \sqrt{\lambda_m}\, \chi_m$ is built from features χ_m that are orthonormal in $L^2(\nu_x)$, the feature coefficients, the acquisition times and the order of acquisition,

$$a_m(t) = \frac{\mathbb{E}_{\nu_x}[f_{\vartheta(t)}(x)\, \chi_m(x)]}{\sqrt{\lambda_m}}, \qquad \tau_m(\delta) = \inf\{t \geq 0 :\ a_m(t) \geq \delta\} \quad (0 < \delta < 1), \tag{1.1}$$

the order being the permutation of $\{1, \dots, M\}$ that sorts $\tau_1(\delta), \dots, \tau_M(\delta)$. Other observables depend on the parameterization but still live in a space common to all architectures, such as the empirical tangent kernel on a fixed set of inputs (Remark 1.4). Quantities attached to a particular representation, such as the HSVD of the TTN of Section 2.6, are diagnostics rather than observables: they can be compared across models only through a common representation (Remark 2.39). The hypothesis can be formulated at two levels. The first one is global.

Hypothesis 1.2 (Global surrogate)**.** Fix the data, the loss function and the hyperparameters, a family of observables, a tolerance $\varepsilon > 0$ and a training horizon $t_{\max}$. There exist a TTN architecture and an initialization scheme such that, for typical initializations of both models, the observables of the TTN trajectory $g_{\vartheta(t)}$ and of the DNN trajectory $f_{\theta(t)}$ differ by at most ε for all $t \in [0, t_{\max}]$, in the same time variable and without post hoc time warping, and discrete observables, such as the order in which features are learned, coincide.

The second one is local. Even if the global version fails, we could still use TTNs as *local* surrogates of the training dynamics. Consider a DNN with smooth activation functions and Taylor expand its activations to some order around a training checkpoint. The resulting network is polynomial and, as shown in Section 2.6, it can be represented exactly as a TTN.

Hypothesis 1.3 (Local surrogate)**.** Let $\theta_j = \theta(t_j)$ be a checkpoint of DNN training and $\hat{f}_j$ the TTN surrogate built at θ_j (Definition 2.44). Fix a tolerance $\varepsilon > 0$ and a horizon $H > 0$ that is not small compared with the time scale on which features are acquired. When the surrogate is started from θ_j, with the same optimizer state, data and hyperparameters as the DNN, its trajectory $\tilde{\theta}$ satisfies

$$\sup_{0 \le s \le H} \|f_{\theta(t_j+s)} - \hat{f}_j(\tilde{\theta}(t_j+s), \cdot)\|_{L^2(\nu_x)} \le \varepsilon,$$

and the chosen observables agree within ε on $[t_j, t_j + H]$.

If ε exceeds the initial prediction error, some positive horizon always exists by continuity; the content of the hypothesis is that the horizon is long enough to cover non-trivial learning, such as the acquisition of a feature. A sequence of such local surrogates, rebuilt at successive checkpoints, could provide an interpretable "atlas" of training even when no single TTN is globally faithful (Surrogate note, 2026).

Note that even if these hypotheses fail, solving the training dynamics of TTNs could still be highly instructive in the same way that studying the training dynamics of DLNs is instructive although their training dynamics differs from more realistic architectures. For example the saddle to saddle dynamics regime is a regime that we can find in more realistic architecture and they teach us about the theoretical gap between simple toy models and more realistic architectures

Remark 1.4 (Comparing different parameterizations)**.** A DNN and a TTN have different parameters, but their dynamics can be compared in function space. Let $x_1, \dots, x_n$ be the training inputs, $F_\theta = (f_\theta(x_i))_{i \le n}$, $J_\theta = D_\theta F_\theta$, and $K_\theta = \frac{1}{n} J_\theta J_\theta^\top \in \mathbb{R}^{n \times n}$ the empirical tangent kernel (Jacot et al., 2018); define $\tilde{F}_\vartheta$ and $\tilde{K}_\vartheta$ in the same way for the TTN. Under full-batch gradient flow on the empirical square loss, $\dot{F}_\theta = -K_\theta(F_\theta - y)$, hence

$$\frac{d}{dt}(F_\theta - \tilde{F}_\vartheta) = -K_\theta(F_\theta - \tilde{F}_\vartheta) - (K_\theta - \tilde{K}_\vartheta)(\tilde{F}_\vartheta - y).$$

Function-space agreement is therefore plausible when the two models have close predictions and close residual-weighted tangent kernels (Surrogate note, 2026, Section 4.3). The two Jacobians need not have the same number of columns; their kernels are both $n \times n$.

Outline. Section 2 defines tensors, tensor networks and TTNs (Sections 2.1–2.2), their edge matricizations and hierarchical ranks (Section 2.3), the HSVD (Section 2.4) and the relevant notions of rank (Section 2.5). It then shows that deep polynomial networks are weight-tied TTNs and defines hierarchical features and learning capacity (Section 2.6), and constructs the TTN surrogate of a smooth DNN (Section 2.7). Section 3 derives the gradient flow of a depth-two TTN student, rewrites it as an evolution of the represented tensor, and solves it on the manifold of balanced parameters. Section 4 studies these balanced dynamics without assuming that the student is aligned with the teacher: it solves them for students of bond rank one, classifies the saddles by their linear rates, and derives the time scales of alignment, amplification and plateaus. Section 5 constructs observables for comparing the training dynamics of TTN and DNN students of a TTN teacher. Section 7 lists the experiments suggested in the text.

2 Theoretical background

We define tree tensor networks and review some of their important mathematical properties. Throughout, vector spaces are finite-dimensional real Euclidean spaces $\mathbb{R}^d$ with their standard basis

and inner product, and we use the resulting identification $\mathbb{R}^d \cong (\mathbb{R}^d)^*$ freely (Furman et al., 2026, Appendix A). General references are Kolda and Bader (2009), Hackbusch (2012) and Lim (2021). We write $[n] = \{1, \dots, n\}$.

2.1 Tensors, multilinear maps and matricizations

Definition 2.1 (Tensor)**.** Let $d_1, \dots, d_n$ be positive integers. An *order-n tensor* is an element $T \in \mathbb{R}^{d_1} \otimes \cdots \otimes \mathbb{R}^{d_n}$. Expanding $T = \sum T_{i_1 \cdots i_n}\, e_{i_1} \otimes \cdots \otimes e_{i_n}$ in the standard bases identifies T with the array $(T_{i_1 \cdots i_n}) \in \mathbb{R}^{d_1 \times \cdots \times d_n}$. The index i_k, and the space $\mathbb{R}^{d_k}$, is the k-th *mode*, or *leg*, of T. We write $\langle S, T \rangle = \sum_{i_1, \dots, i_n} S_{i_1 \cdots i_n} T_{i_1 \cdots i_n}$ and $\|T\|_F = \langle T, T \rangle^{1/2}$.

Definition 2.2 (Multilinear map)**.** Let $V_1, \dots, V_p, W$ be vector spaces. A map $F : V_1 \times \cdots \times V_p \to W$ is *p-linear* if it is linear in each argument when the others are fixed. By the universal property of the tensor product, p-linear maps $V_1 \times \cdots \times V_p \to W$ are in one-to-one correspondence with linear maps $V_1 \otimes \cdots \otimes V_p \to W$.

Proposition 2.3 (Tensors as multilinear maps; Furman et al., 2026, Proposition A.1)**.** Let $T \in \bigotimes_{k=1}^{n} \mathbb{R}^{d_k}$, and let $I \sqcup O = [n]$ be a partition of the modes into inputs and outputs. The formula

$$F_T\big((v_k)_{k \in I}\big)_{(i_o)_{o \in O}} = \sum_{(i_k)_{k \in I}} T_{i_1 \cdots i_n} \prod_{k \in I} (v_k)_{i_k} \tag{2.1}$$

defines a multilinear map $F_T : \prod_{k \in I} \mathbb{R}^{d_k} \to \bigotimes_{o \in O} \mathbb{R}^{d_o}$, and $T \mapsto F_T$ is a linear isomorphism onto the space of such maps. Equivalently, T is a linear map from $\bigotimes_{k \in I} \mathbb{R}^{d_k}$ to $\bigotimes_{o \in O} \mathbb{R}^{d_o}$.

This is the precise sense in which "a tensor is a multilinear map": it maps a tensor product of vector spaces linearly (equivalently, a Cartesian product of vector spaces multilinearly) to another tensor product of vector spaces. The split of the modes into inputs and outputs is a choice, not a property of T.

Example 2.4. An order-3 tensor $W \in \mathbb{R}^{d_1} \otimes \mathbb{R}^{d_2} \otimes \mathbb{R}^{d_3}$ with inputs $\{1, 2\}$ and output $\{3\}$ is the bilinear map $B_W(u, v)_k = \sum_{i,j} W_{ijk}\, u_i v_j$: a vertex with two input legs and one output leg. With input $\{1\}$ and outputs $\{2, 3\}$, the same W is a linear map $\mathbb{R}^{d_1} \to \mathbb{R}^{d_2} \otimes \mathbb{R}^{d_3}$. For $d_1 = d_2 = d_3 = d$, the diagonal tensor $\Delta_{ijk} = \mathbf{1}[i = j = k]$ gives the Hadamard (coordinatewise) product $\Delta(u, v) = u \odot v$, which is linear in u and in v but not jointly linear.

Definition 2.5 (Matricization)**.** Let T be an order-n tensor, $\emptyset \neq A \subsetneq [n]$ with complement A^c, and $m_A = \prod_{k \in A} d_k$. Bipartition the indices of T into the groups A and A^c and flatten each group, using the lexicographic order on multi-indices. The result is the *A-matricization* $T^{\langle A \rangle} \in \mathbb{R}^{m_A \times m_{A^c}}$, with entries $\big(T^{\langle A \rangle}\big)_{(i_k)_{k \in A}, (i_k)_{k \in A^c}} = T_{i_1 \cdots i_n}$. As a linear map, $T^{\langle A \rangle}$ is F_T with inputs A^c and outputs A. Its rank $\operatorname{rank}_A(T) := \operatorname{rank} T^{\langle A \rangle}$ does not depend on the orderings, and $\operatorname{rank}_A(T) = \operatorname{rank}_{A^c}(T)$. For $A = \{k\}$ one obtains the mode-k matricization $T_{(k)}$ (Kolda and Bader, 2009).

2.2 Tensor networks and tree tensor networks

Definition 2.6 (Tensor network)**.** A *tensor network* consists of

(i) a finite graph with vertex set V and edge set $E = E_{\text{int}} \sqcup E_{\text{ext}}$, where each *internal edge* or *bond* $e \in E_{\text{int}}$ joins two distinct vertices, and each *external edge* or *open leg* $a \in E_{\text{ext}}$ is attached to a single vertex;

(ii) a dimension for each edge: a *bond dimension* D_e for $e \in E_{\text{int}}$, and a leg dimension d_a for $a \in E_{\text{ext}}$;

(iii) for each vertex v, a *core* C_v: a tensor with one mode for each edge incident to v, of the corresponding dimension.

The tensor *represented* by the network is $T \in \bigotimes_{a \in E_{\text{ext}}} \mathbb{R}^{d_a}$, obtained by contracting all bonds:

$$T_{(i_a)_{a \in E_{\text{ext}}}} = \sum_{(i_e)_{e \in E_{\text{int}}}} \prod_{v \in V} (C_v)_{(i_e)_{e \ni v}}, \tag{2.2}$$

where each bond index i_e ranges over $[D_e]$, and $(i_e)_{e \ni v}$ collects the indices of all edges, bonds and legs, incident to v. We write $\vartheta = (C_v)_{v \in V}$ for the parameters and $T(\vartheta)$ for the represented tensor.

Informally: vertices carry tensors and edges are indices. Each core is linear in each of its legs; an open leg is a free index of the represented tensor; and joining two legs by a bond means summing over the shared index, which composes the corresponding multilinear maps (Furman et al., 2026, Proposition A.4). For instance, the product AB of two matrices is the network with two vertices and one bond, and a vertex with two input legs and one output leg is a bilinear map (Example 2.4). The order in which the sums in (2.2) are performed is irrelevant.

Definition 2.7 (Tree tensor network)**.** A *tree tensor network* (TTN) is a tensor network whose graph (V, E_{int}) is a tree. We always assume that every leaf of this tree (a vertex of degree one in (V, E_{int})) carries at least one open leg.

Choosing a root $\rho \in V$ and orienting every bond towards ρ turns a TTN into a feedforward computation: by Proposition 2.3, each non-root core C_v is a multilinear map from its child bonds and its open legs to its parent bond, and T is computed by composing these maps from the leaves to the root (Furman et al., 2026, Definition 3.1 and Appendix A.3). The root and the orientation are a presentation choice, since (2.2) does not depend on them. Figure 1 shows three examples: a chain is a DLN $T = W_L \cdots W_1$, and a star is a Tucker decomposition. A binary tree with one leg per leaf is the hierarchical Tucker format (Hackbusch and Kühn, 2009; Grasedyck, 2010), and a path with one leg per vertex is a tensor train, or matrix product state (Oseledets, 2011). TTNs originate in quantum many-body physics (Shi et al., 2006).

A TTN defines a model of functions once its legs receive inputs. If the input x splits into blocks $(x^{(a)})_{a \in E_{\text{ext}}}$ and each leg receives a fixed embedding $\xi_a(x^{(a)}) \in \mathbb{R}^{d_a}$ of its block (for example $\xi(z) = (1, z)$, or a one-hot encoding), the TTN computes

$$g_\vartheta(x) = \Big\langle T(\vartheta), \bigotimes_{a \in E_{\text{ext}}} \xi_a(x^{(a)}) \Big\rangle, \tag{2.3}$$

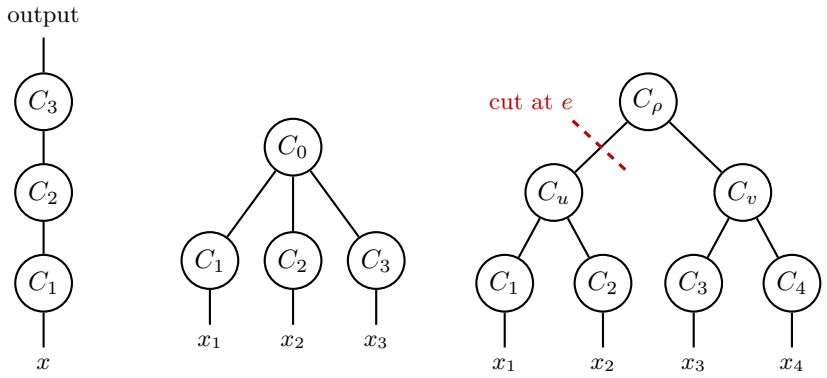
which is multilinear in the embedded blocks and nonlinear in x (Stoudenmire and Schwab, 2016). We will meet two situations: *physical* TTNs, whose legs receive distinct blocks of the input, and *computational* TTNs, whose legs all receive copies of the same input (Section 2.6).

2.3 Edge cuts, bond dimensions and hierarchical ranks

The key structural property of a tree is that removing any edge disconnects it. Each bond therefore defines a bipartition of the open legs, hence a matricization of the represented tensor.

Lemma 2.8 (Edge cuts)**.** Let T be the tensor represented by a TTN and let $e \in E_{\text{int}}$. Removing e splits the tree into two subtrees with vertex sets V_e and V_e^c. Let $A_e \subseteq E_{\text{ext}}$ be the set of open legs attached to V_e, and $A_e^c = E_{\text{ext}} \setminus A_e$.

(i) A_e and A_e^c are nonempty, so e defines the matricization $M_e := T^{\langle A_e \rangle} \in \mathbb{R}^{m_e \times m_e^c}$, where $m_e = \prod_{a \in A_e} d_a$ and $m_e^c = \prod_{a \in A_e^c} d_a$.



(a) chain: DLN $T = W_3 W_2 W_1$

(b) star: Tucker decomposition

(c) binary tree: $A_e = \{1, 2\}$, $M_e = T^{\langle\{1,2\}\rangle} = F_e H_e^\top$

Figure 1: Examples of tree tensor networks. Circles are cores, lines between circles are bonds, and dangling lines are open legs. (a) A chain is a deep linear network. (b) A star is a Tucker decomposition. (c) In a tree, cutting the bond e splits the open legs into $A_e = \{1, 2\}$ and its complement; the corresponding matricization factors through the bond, $M_e = F_e H_e^\top$ (Lemma 2.8), so its rank is at most the bond dimension D_e.

(ii) $M_e = F_e H_e^\top$, where $F_e \in \mathbb{R}^{m_e \times D_e}$ (resp. $H_e \in \mathbb{R}^{m_e^c \times D_e}$) is the contraction of the cores in V_e (resp. V_e^c) with the index of the bond e left open.
(iii) $\operatorname{rank} M_e \le D_e$.

Proof. (i) The subtree on V_e is either a single vertex, which then has degree one in the tree, or it has at least two leaves, at most one of which is the endpoint of e. In both cases V_e contains a leaf of the original tree, which carries an open leg. The same argument applies to V_e^c. (ii) Since e is the only bond between V_e and V_e^c, the sum (2.2) factorizes as a sum over $i_e \in [D_e]$ of the product of the contraction over V_e and the contraction over V_e^c. (iii) follows from (ii). □

Definition 2.9 (Bond dimension and hierarchical rank)**.** The dimension D_e of the index along which the two subtree matrices F_e and H_e are contracted is the *bond dimension* of e. The family

$$(M_e, r_e)_{e \in E_{\text{int}}}, \qquad r_e := \operatorname{rank} M_e,$$

collects the edge matricizations of T and their ranks, and the tuple $r(T) := (r_e)_{e \in E_{\text{int}}}$ is the *hierarchical (Tucker) rank* of T with respect to the tree. By Lemma 2.8, $r_e \le D_e$: the rank across a cut is upper bounded by the bond dimension.

The matrices M_e and ranks r_e depend only on the represented tensor T and on which legs lie on each side of each bond, not on the cores. Cutting an open leg $a \in E_{\text{ext}}$ instead of a bond gives the mode matricization $T^{\langle\{a\}\rangle}$.

The bond dimensions describe exactly what a TTN can represent.

Proposition 2.10 (Realizability; Grasedyck and Hackbusch, 2011; Furman et al., 2026, Proposition B.1)**.** Fix a tree with open legs. A tensor T is represented by a TTN on this tree with bond dimensions $(D_e)_{e \in E_{\text{int}}}$ if and only if $r_e(T) \le D_e$ for every $e \in E_{\text{int}}$. In that case one can take $D_e = r_e(T)$.

Sketch. Necessity is Lemma 2.8(iii). For sufficiency, the column spaces of the matrices M_e are nested along the tree (Proposition 2.14 below). Choosing a basis of each column space and expressing

it in the bases of the children defines cores with $D_e = r_e$ that represent T (Furman et al., 2026, Proposition B.1); padding the cores with zeros gives any larger bond dimensions. □

Hence the model class of a TTN architecture is the set $\{T : r_e(T) \leq D_e \ \forall e \in E_{\text{int}}\}$. Since the set of matrices of rank at most r is closed, this set is closed, and best approximations within it exist; this is not the case for the CP rank (Section 2.5).

Remark 2.11 (Why trees: canonical matricizations)**.** In a TTN, every bond e determines exactly one bipartition $(A_e \mid A_e^c)$ of the open legs, and its bond dimension bounds the rank across that bipartition. The edge matricizations are therefore canonically attached to the network. This is not the case for tensor networks with cycles. There, removing a single edge does not disconnect the graph; a bipartition of the open legs is separated only by a set C of edges, the rank across it is bounded by $\prod_{e\in C} D_e$, and a given edge belongs to the separating sets of many bipartitions. The correspondence between edges and matricizations is lost, and with it several properties used below: no characterization of the model class by ranks of matricizations such as Proposition 2.10 is available in general, sets of tensors representable with bounded bond dimensions need not be closed (Landsberg et al., 2012), and canonical (minimum-norm) parameterizations are no longer unique up to gauge (Furman et al., 2026, Remark F.7).

The parameterization of a TTN is redundant: an invertible change of basis on a bond can be absorbed by the two adjacent cores.

Proposition 2.12 (Gauge freedom)**.** Let $e = \{u, v\} \in E_{\text{int}}$ and $X \in \mathrm{GL}(D_e)$. For a core C incident to e, let $C \times_e X$ denote the core obtained by multiplying the mode of C corresponding to e by X, that is $(C \times_e X)_{\ldots j \ldots} = \sum_i C_{\ldots i \ldots} X_{ij}$. Replacing C_u by $C_u \times_e X$ and C_v by $C_v \times_e X^{-\top}$ leaves T unchanged. Consequently the *gauge group* $\mathcal{G} = \prod_{e\in E_{\text{int}}} \mathrm{GL}(D_e)$ acts on the parameters and preserves T. Every quantity defined from T alone, in particular M_e, r_e, and the singular values and singular subspaces of M_e, is gauge invariant, whereas individual core entries are not. If $D_e = r_e(T)$ for every e (a minimal representation), then any two parameterizations of T with these bond dimensions differ by a unique gauge transformation.

Proof. If V_e contains u, the substitution maps F_e to $F_e X$ and H_e to $H_e X^{-\top}$, so $M_e = F_e H_e^\top$ is unchanged; since M_e is a rearrangement of T, so is T. For the uniqueness statement for minimal representations see Hackbusch (2012) and Uschmajew and Vandereycken (2013). □

2.4 The hierarchical singular value decomposition

Since every bond defines a canonical matricization, we can take its singular value decomposition.

Definition 2.13 (SVD at a bond; hierarchical SVD)**.** For $e \in E_{\text{int}}$, let

$$M_e = U_e S_e V_e^\top$$

be a compact SVD: $U_e \in \mathbb{R}^{m_e \times r_e}$ and $V_e \in \mathbb{R}^{m_e^c \times r_e}$ have orthonormal columns, and $S_e = \mathrm{diag}(\sigma_{e,1}, \ldots, \sigma_{e,r_e})$ with $\sigma_{e,1} \geq \cdots \geq \sigma_{e,r_e} > 0$. Reshaping the j-th columns of U_e and V_e gives tensors $u_{e,j}$ over the legs in A_e and $v_{e,j}$ over the legs in A_e^c, and, up to the order of the legs,

$$T = \sum_{j=1}^{r_e} \sigma_{e,j}\, u_{e,j} \otimes v_{e,j}. \tag{2.4}$$

The $\sigma_{e,j}$ are the *hierarchical singular values* of T at e, and $u_{e,j}$ (resp. $v_{e,j}$) its left (resp. right) singular vectors, which describe the subtree on one side of e (resp. its environment). The *hierarchical SVD*

(HSVD) of T is the family $(U_e, S_e, V_e)_{e \in E_{\text{int}}}$ of the SVDs across all bonds of the TTN (Grasedyck, 2010).

The HSVD is more than a list of unrelated SVDs: the singular vectors at different bonds are nested and fit together into a TTN.

Proposition 2.14 (Nestedness; the HSVD is a TTN)**.** Root the tree at a vertex ρ and, for each bond e, let A_e be the set of legs below e. Let $u \neq \rho$ be a vertex with parent bond e, child bonds $e_1, \dots, e_k$, and open legs of total dimension n_u ($n_u = 1$ if u has no open leg). Then, after reordering the legs,

$$\operatorname{col}(M_e) \subseteq \mathbb{R}^{n_u} \otimes \operatorname{col}(M_{e_1}) \otimes \cdots \otimes \operatorname{col}(M_{e_k}). \tag{2.5}$$

Consequently $B_u := (I_{n_u} \otimes U_{e_1} \otimes \cdots \otimes U_{e_k})^\top U_e$ satisfies $U_e = (I_{n_u} \otimes U_{e_1} \otimes \cdots \otimes U_{e_k})\, B_u$. Set $B_\rho := (I_{n_\rho} \otimes U_{e_1} \otimes \cdots \otimes U_{e_k})^\top \operatorname{vec}(T)$ at the root, where $e_1, \dots, e_k$ now denote the child bonds of ρ. The TTN with cores $(B_v)_{v \in V}$ and bond dimensions $(r_e)_{e \in E_{\text{int}}}$ represents T, and in it the contraction of the subtree below each bond e equals U_e; in particular $F_e = U_e$ has orthonormal columns.

Proof. Each column of M_e is a partial contraction of T against a fixed vector on the legs outside the subtree of e. Matricizing such a column with respect to the legs below a child bond e_j gives columns that are partial contractions of T against vectors on the legs outside the subtree of e_j, hence columns of M_{e_j}. A tensor whose matricizations along every child subtree have their columns in $\operatorname{col}(M_{e_j})$ lies in the right-hand side of (2.5) (Furman et al., 2026, proof of Proposition B.1). The identity for U_e follows because $I \otimes U_{e_1}U_{e_1}^\top \otimes \cdots \otimes U_{e_k}U_{e_k}^\top$ is the orthogonal projector onto that right-hand side. The representation statement follows by induction from the leaves, the root step using (2.5) for T itself. □

Remark 2.15 (Canonical and gauge invariant)**.** The HSVD depends only on T and on the tree, so it is gauge invariant (Proposition 2.12). It is unique up to the usual ambiguities of the SVD: the signs of singular vectors, and rotations within the eigenspaces of $M_e M_e^\top$ associated with repeated singular values. The singular values, and the singular subspaces of distinct singular values, are therefore canonical, and near a degeneracy one should compare whole singular subspaces rather than individual vectors. The orthonormal TTN of Proposition 2.14 is one canonical gauge. Another is the balanced, or minimum-norm, gauge of Furman et al. (2026, Appendix F), which is unique up to the orthogonal gauge group $\prod_e \mathrm{O}(D_e)$ and preserved by gradient flow; both carry the same gauge-invariant data.

Truncating the HSVD gives a quasi-optimal compression.

Proposition 2.16 (Truncation; Grasedyck, 2010)**.** Let $k_e \leq r_e$ for $e \in E_{\text{int}}$, and let P_e be the orthogonal projector onto the span of the k_e leading left singular vectors of M_e, acting on the legs in A_e. Enumerate the bonds $e_1, \dots, e_m$ so that every bond comes before the bonds below it. Then $T_k := P_{e_m} \cdots P_{e_1} T$ satisfies $r_e(T_k) \leq k_e$ for every e, and

$$\|T - T_k\|_F^2 \;\leq\; \sum_{e \in E_{\text{int}}} \sum_{j > k_e} \sigma_{e,j}^2 \;\leq\; |E_{\text{int}}| \cdot \min\big\{\|T - T'\|_F^2 :\; r_e(T') \leq k_e \;\forall e \in E_{\text{int}}\big\}.$$

Proof. A projection at a bond below e, or in a subtree disjoint from the subtree below e, acts on one side of the cut at e only, so it does not increase r_e; projections at bonds above e are applied before P_e. For orthogonal projectors $\pi_1, \dots, \pi_m$, the decomposition $A - \pi_m B = (I - \pi_m)A + \pi_m(A - B)$ into orthogonal terms gives, by induction, $\|A - \pi_m \cdots \pi_1 A\|^2 \leq \sum_i \|(I - \pi_i)A\|^2$, and $\|(I - P_e)T\|_F^2 = \sum_{j > k_e} \sigma_{e,j}^2$ by the Eckart–Young theorem (Eckart and Young, 1936). Finally, for every T' with $r_e(T') \leq k_e$, Eckart–Young gives $\sum_{j > k_e} \sigma_{e,j}^2 \leq \|M_e - T'^{\langle A_e \rangle}\|_F^2 = \|T - T'\|_F^2$. The minimum exists because the constraint set is closed. □

If two bonds define the same bipartition, as the two child bonds of a binary root do, projecting at one of them suffices; this gives the factor $2d-3$ of Grasedyck (2010) for binary trees with d leaves.

Remark 2.17 (Stability)**.** By Weyl's inequality, $|\sigma_{e,j}(T) - \sigma_{e,j}(T')| \le \|M_e(T) - M_e(T')\|_{\mathrm{op}} \le \|T - T'\|_F$: hierarchical singular values are 1-Lipschitz functions of the tensor. Singular subspaces are stable only relative to the gap that separates their singular values from the rest of the spectrum (Davis and Kahan, 1970; Wedin, 1972). These facts matter whenever the HSVDs of two models, or of one model at two times, are compared.

2.5 Notions of rank

Tensors admit several notions of rank, which coincide for matrices but not for higher-order tensors (Kolda and Bader, 2009; Lim, 2021).

Definition 2.18 (CP rank)**.** The *CP rank* of T, $\mathrm{rank}_{\mathrm{CP}}(T)$, is the smallest number R of rank-one tensors into which T can be decomposed: $T = \sum_{s=1}^{R} a_s^{(1)} \otimes \cdots \otimes a_s^{(n)}$ (Hitchcock, 1927).

The CP rank is challenging to compute in practice: deciding whether $\mathrm{rank}_{\mathrm{CP}}(T) \le R$ is NP-hard (Håstad, 1990; Hillar and Lim, 2013). Moreover, for $n \ge 3$ and $R \ge 2$ the set of tensors of CP rank at most R is in general not closed, so a best approximation of low CP rank may fail to exist (de Silva and Lim, 2008).

Definition 2.19 (Mode rank and Tucker rank)**.** The *mode-k rank* of T is $r_k(T) = \mathrm{rank}\, T_{(k)}$, the rank of the matricization obtained by separating index k from the others. The *Tucker* (or multilinear) *rank* is the tuple $(r_1(T), \dots, r_n(T))$ of all mode ranks. It is the smallest core size in a Tucker decomposition $T = \sum_{j_1,\dots,j_n} \Omega_{j_1 \cdots j_n}\, u_{j_1}^{(1)} \otimes \cdots \otimes u_{j_n}^{(n)}$ (Tucker, 1966; De Lathauwer et al., 2000a), i.e. a TTN on a star (Figure 1b).

For a TTN, the *hierarchical Tucker rank* is the tuple $(r_e)_{e \in E_{\mathrm{int}}}$ of Definition 2.9 (Hackbusch and Kühn, 2009; Grasedyck, 2010). Mode ranks and hierarchical ranks are ranks of matrices: unlike the CP rank, they can be computed by SVDs, up to a numerical tolerance.

Proposition 2.20 (Comparison of ranks)**.**

(i) For every bipartition A, $\mathrm{rank}_A(T) \le \mathrm{rank}_{\mathrm{CP}}(T)$. In particular $r_k(T) \le \mathrm{rank}_{\mathrm{CP}}(T)$, and $r_e(T) \le \mathrm{rank}_{\mathrm{CP}}(T)$ for every tree.

(ii) In a rooted TTN, if a vertex u has parent bond e, child bonds $e_1, \dots, e_k$ and open legs of total dimension n_u, then $r_e \le n_u \prod_{j=1}^{k} r_{e_j}$.

(iii) A tensor of CP rank R is represented on every tree, with every assignment of legs to vertices, by a TTN whose bond dimensions all equal R and whose cores are diagonal in the bond indices.

Proof. (i) The A-matricization of a rank-one tensor has rank one, and the matrix rank is subadditive. (ii) Take dimensions in (2.5). (iii) Given $T = \sum_{s=1}^{R} \bigotimes_{a \in E_{\mathrm{ext}}} w_s^{(a)}$, set $(C_v)_{(i_a)_a,(j_e)_e} = \sum_{s=1}^{R} \prod_{e \ni v} \mathbf{1}[j_e = s] \prod_a (w_s^{(a)})_{i_a}$, where a ranges over the open legs at v and e over the bonds at v. Contracting forces all bond indices to agree, because the tree is connected (Diagonal TTN note, 2026, Section 2.4). □

The converse of (i) fails dramatically: TTNs with polynomially many parameters represent tensors whose CP rank is exponential in the number of leaves, for almost every choice of cores (Cohen et al., 2016). This is one precise sense in which hierarchical (deep) factorizations are more expressive than shallow ones.

Remark 2.21 (Ranks and critical points)**.** Furman et al. (2026) use a parameter-level notion: ϑ has *full Tucker rank* if, for every bond e and each of its endpoints v, the matricization of C_v that separates the mode of e has rank D_e. In general $r_e(T(\vartheta))$ is at most the ranks of these two matricizations, and equality holds at balanced (minimum-norm) parameters (Furman et al., 2026, Proposition D.2). For a TTN with free cores trained on the Frobenius loss $\frac{1}{2}\|T(\vartheta) - T^\star\|_F^2$ (their setting, in which inputs are drawn uniformly among the entries of T, i.e. one-hot inputs) towards a realizable target $T^\star$, every critical point that is not a global minimum is rank deficient (Furman et al., 2026, Theorem 5.2); at such a point the network behaves as a TTN with smaller bond dimensions. For the parity of n bits ($n \geq 4$ a power of two), realized on a balanced binary tree with bond dimension 2, the rank-one point whose cores all have entries $\frac{1}{2}$, which is a best rank-one approximation, is a saddle of order at least $n/2$ (Furman et al., 2026, Proposition 5.3). Hierarchical ranks are therefore natural dynamical variables, and saddle-to-saddle dynamics should appear as successive increases of the rank profile. Whether these results extend to other losses, data distributions or constrained cores is open.

2.6 Deep polynomial networks as tree tensor networks

We now show that deep networks with polynomial activations are TTNs with tied weights, evaluated on copies of the input.

Definition 2.22 (Deep polynomial network)**.** A feedforward network with L hidden layers of widths $d_1, \dots, d_L$, acting on inputs $x \in \mathbb{R}^{d_0}$, is

$$h_0 = x, \qquad z_\ell = W_\ell h_{\ell-1} + b_\ell, \qquad h_\ell = \sigma_\ell(z_\ell) \quad (1 \leq \ell \leq L), \qquad f_\theta(x) = c^\top h_L + b_{\text{out}}, \tag{2.6}$$

with $W_\ell \in \mathbb{R}^{d_\ell \times d_{\ell-1}}$, $b_\ell \in \mathbb{R}^{d_\ell}$, $c \in \mathbb{R}^{d_L}$ and $b_{\text{out}} \in \mathbb{R}$ collected in θ, and activations acting coordinatewise, $\sigma_\ell(z)_i = \sigma_{\ell,i}(z_i)$. It is a *deep polynomial network* (DPN) if every $\sigma_{\ell,i}$ is a polynomial of degree at most K. It is a *deep monomial network* (DMN) if moreover $\sigma_{\ell,i}(z) = z^K$ and all biases vanish ($b_\ell = 0$, $b_{\text{out}} = 0$). A DMN with $K = 2$ is a *deep quadratic network* (DQN).

Allowing the polynomial to depend on the neuron covers the Taylor surrogates of Section 2.7. Since degrees multiply under composition, a DPN is a polynomial in x of degree at most K^L.

2.6.1 Monomial activations: tied TTNs on copies of the input

Proposition 2.23 (A monomial layer is one TTN vertex)**.** Let $\sigma(z) = z^K$ and $W \in \mathbb{R}^{d' \times d}$. Then $\sigma(Wh) = A(h, \dots, h)$, where A is the symmetric K-linear map

$$A_{i\, j_1 \cdots j_K} = \prod_{s=1}^{K} W_{i j_s}, \qquad \text{i.e.} \qquad A(u_1, \dots, u_K) = (Wu_1) \odot \cdots \odot (Wu_K).$$

In particular, for $K = 2$,

$$\sigma(Wx)_i = \Big(\sum_j W_{ij} x_j\Big)^2 = \sum_{j,k} W_{ij} x_j \, W_{ik} x_k = A(x, x)_i. \tag{2.7}$$

The proof is the expansion of the product. Graphically, (2.7) is a vertex with one output leg i and two input legs; equivalently, it is the Hadamard vertex Δ of Example 2.4 whose two input legs carry the same (tied) weight matrix W, applied to two copies of x. Composing layers gives a tree.

Proposition 2.24 (DMNs are tied TTNs)**.** Let f_θ be a DMN with exponent $K \geq 2$ and L layers, and let $N = K^L$. Let $\mathcal{T}_{K,L}$ be the complete K-ary rooted tree of depth L in which every non-root vertex at height ℓ carries the core A_ℓ of Proposition 2.23 built from W_ℓ, the root (height L) carries the K-linear form $c^\top A_L$, and the vertices at height 1 carry the N open legs. Let $T(\theta) \in (\mathbb{R}^{d_0})^{\otimes N}$ be the represented tensor. Then

$$f_\theta(x) = \langle T(\theta),\, x^{\otimes N} \rangle \qquad \text{for all } x \in \mathbb{R}^{d_0},$$

and the bonds between heights ℓ and $\ell + 1$ have dimension d_ℓ.

Proof. Let $\Phi_\ell : (\mathbb{R}^{d_0})^{K^\ell} \to \mathbb{R}^{d_\ell}$ be the contraction of a subtree of height ℓ: $\Phi_0 = \mathrm{id}$ and $\Phi_\ell(y_1, \ldots, y_{K^\ell}) = A_\ell(\Phi_{\ell-1}(y^{(1)}), \ldots, \Phi_{\ell-1}(y^{(K)}))$, where $y^{(s)}$ is the s-th block of $K^{\ell-1}$ consecutive arguments. Since $A_\ell(h, \ldots, h) = \sigma(W_\ell h)$, induction gives $\Phi_\ell(x, \ldots, x) = h_\ell(x)$, hence $f_\theta(x) = c^\top \Phi_L(x, \ldots, x) = \langle T(\theta), x^{\otimes N} \rangle$. □

Remark 2.25 (Copies, not copying)**.** The map $x \mapsto x^{\otimes K}$ is not linear, so no tensor implements "copying" a vector. A DMN is therefore not a tensor network applied to x: it is the restriction of the N-linear map $T(\theta)$ to the diagonal $(x, \ldots, x)$. The open legs are formal copies of the same input, and a cut of $\mathcal{T}_{K,L}$ separates copies, not input variables. The same identification, for networks with bilinear layers, is described as a TTN with a copying operation by Pearce et al. (2024) and Dooms et al. (2025) (see Furman et al., 2026, Section 2).

Remark 2.26 (Hidden-unit symmetries are gauge transformations)**.** Permuting the hidden units of layer ℓ by a permutation matrix P ($W_\ell \mapsto PW_\ell$, $W_{\ell+1} \mapsto W_{\ell+1}P^\top$) does not change f_θ, and neither does rescaling them, for monomial activations, by an invertible diagonal matrix Λ ($W_\ell \mapsto \Lambda W_\ell$, $W_{\ell+1} \mapsto W_{\ell+1}\Lambda^{-K}$, with c in place of W_{L+1}). On the tied TTN these symmetries act as gauge transformations, by P and by Λ^K respectively, on every bond of height ℓ (Proposition 2.12). Hence $T(\theta)$ and its HSVD are invariant under them, while the weights are not. This is the advantage of the TTN over the DMN parameterization: the HSVD gives a more canonical, gauge-independent description of internal structure. The idea is then to use the TTN representing a DMN to interpret its internal structure, reading the HSVD as the features that the network learns. Section 2.6.3 makes this precise, and Remark 2.39 discusses its limits.

2.6.2 General polynomial activations: homogenization

Biases and lower-order terms break homogeneity. They are absorbed by adding a constant coordinate.

Definition 2.27 (Homogenization)**.** Let $p(z) = \sum_{k=0}^{K} a_k z^k$ be a polynomial of degree at most K. Its degree-K *homogenization* is

$$p_h(z_0, z) := \sum_{k=0}^{K} a_k\, z_0^{K-k} z^k,$$

which equals $z_0^K\, p(z/z_0)$ for $z_0 \neq 0$. It is homogeneous of degree K, and $p_h(1, z) = p(z)$. The construction generalizes to multivariate polynomial maps: if $P : \mathbb{R}^n \to \mathbb{R}^m$ has degree at most K, $P(z) = \sum_{|\alpha| \leq K} c_\alpha z^\alpha$ with $c_\alpha \in \mathbb{R}^m$, then $P_h(z_0, z) = \sum_{|\alpha| \leq K} c_\alpha\, z_0^{K-|\alpha|} z^\alpha$ is a homogeneous polynomial map on $\mathbb{R}^{1+n}$ with $P_h(1, z) = P(z)$.

A homogeneous polynomial can in turn be identified with a symmetric multilinear map.

Lemma 2.28 (Polarization)**.** Let $Q : V \to W$ be a homogeneous polynomial map of degree K between finite-dimensional real vector spaces. There is a unique symmetric K-linear map $S_Q : V^K \to W$ such that $S_Q(y, \dots, y) = Q(y)$ for all y, namely

$$S_Q(y_1, \dots, y_K) = \frac{1}{K!} \sum_{\emptyset \neq J \subseteq [K]} (-1)^{K-|J|} Q\Big(\sum_{j \in J} y_j\Big).$$

Conversely, every symmetric K-linear map S defines the homogeneous polynomial $y \mapsto S(y, \dots, y)$.

This is the classical polarization identity. Combining the two steps, a polynomial of degree at most K is a homogeneous polynomial of degree K in one more variable, which is a symmetric K-linear map on the augmented space. For a polynomial activation, this map is a vertex with one output leg and K input legs, which is why a DPN has a tree structure.

We write $H_\ell := \mathbb{R} \oplus \mathbb{R}^{d_\ell} \cong \mathbb{R}^{1+d_\ell}$, with coordinates indexed from 0, and $\bar{h} := (1, h) \in H_\ell$ for $h \in \mathbb{R}^{d_\ell}$.

Proposition 2.29 (A polynomial layer is one TTN vertex; Local TN note, 2026, Section 3)**.** Let $P_\ell(h) = \sigma_\ell(W_\ell h + b_\ell)$ with polynomial activations $\sigma_{\ell,i}(z) = \sum_{s=0}^K \alpha_{\ell,i,s}(z - m_{\ell,i})^s$, for arbitrary expansion points $m_{\ell,i}$. Then $P_\ell(h) = \sum_{r=0}^K A_\ell^{(r)}[h^{\otimes r}]$, where $A_\ell^{(r)} : (\mathbb{R}^{d_{\ell-1}})^r \to \mathbb{R}^{d_\ell}$ is the symmetric r-linear map with components

$$A_{\ell;i}^{(r)} = \sum_{s=r}^K \binom{s}{r} \alpha_{\ell,i,s}\, q_{\ell,i}^{s-r}\, w_{\ell,i}^{\otimes r}, \qquad q_{\ell,i} = b_{\ell,i} - m_{\ell,i}, \tag{2.8}$$

and $w_{\ell,i}^\top$ is the i-th row of W_ℓ. Define $G_\ell : H_{\ell-1}^K \to H_\ell$ by, for $u_s = (a_s, v_s) \in \mathbb{R} \oplus \mathbb{R}^{d_{\ell-1}}$,

$$G_\ell(u_1, \dots, u_K)_0 = \prod_{s=1}^K a_s, \qquad G_\ell(u_1, \dots, u_K)_{1:d_\ell} = \sum_{r=0}^K \binom{K}{r}^{-1} \sum_{\substack{S \subseteq [K] \\ |S|=r}} A_\ell^{(r)}[v_s : s \in S] \prod_{s \notin S} a_s. \tag{2.9}$$

Then G_ℓ is symmetric and K-linear, it is the polarization of the homogenized layer map, and

$$G_\ell(\bar{h}, \dots, \bar{h}) = \overline{P_\ell(h)} = (1, P_\ell(h)) \qquad \text{for all } h \in \mathbb{R}^{d_{\ell-1}}. \tag{2.10}$$

In coordinates, an entry of G_ℓ with r nonzero input indices equals the corresponding entry of $A_\ell^{(r)}$ divided by $\binom{K}{r}$, and the only nonzero entry of the constant output slice is $(G_\ell)_{0;0\cdots0} = 1$.

Proof. Equation (2.8) follows from the binomial expansion of $(w_{\ell,i}^\top h + q_{\ell,i})^s$. Each term of (2.9) is linear in each u_s, and permuting the u_s permutes the subsets S, so G_ℓ is symmetric and K-linear. With $u_s = \bar{h}$ for all s, the inner sum consists of $\binom{K}{r}$ equal terms $A_\ell^{(r)}[h^{\otimes r}]$, which gives (2.10). With $u_s = (a, v)$ for all s, G_ℓ returns $(a^K, \sum_r a^{K-r} A_\ell^{(r)}[v^{\otimes r}])$, the homogenization of $h \mapsto (1, P_\ell(h))$; by Lemma 2.28, G_ℓ is its polarization. □

The factors $\binom{K}{r}^{-1}$ prevent lower-order terms from being counted several times. Equation (2.8) also shows that the cores are constrained: all orders r reuse the same weight row $w_{\ell,i}$, so G_ℓ is not a free tensor.

Theorem 2.30 (Exact tied-TTN representation of a DPN; Local TN note, 2026, Proposition 1; Surrogate note, 2026, Proposition 1)**.** Let f_θ be a DPN with L layers and activations of degree at most $K \geq 2$, and let $N = K^L$. Consider the complete K-ary tree $\mathcal{T}_{K,L}$ in which every non-root

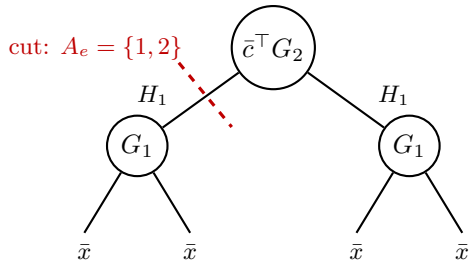


Figure 2: The order-two surrogate of a network with two hidden layers, as a weight-tied binary TTN (Theorem 2.30 and Example 2.32, $K = L = 2$). Both lower vertices carry the same core G_1; the $N = 4$ open legs all receive the augmented input $\bar{x} = (1, x) \in H_0$; the bonds carry augmented hidden states $\bar{h}_1 \in H_1 = \mathbb{R} \oplus \mathbb{R}^{d_1}$; and the root carries $\bar{c}^\top G_2$, i.e. its output is contracted with $\bar{c} = (b_{\text{out}}, c)$. The cut shown defines M_e with $A_e = \{1, 2\}$; the other bond of height one gives the complementary bipartition and the same singular values (Proposition 2.33).

vertex at height ℓ carries the core G_ℓ of Proposition 2.29, the root carries the K-linear form $\bar{c}^\top G_L$, i.e. its output is contracted with $\bar{c} = (b_{\text{out}}, c) \in H_L$, and the N open legs, attached to the vertices of height 1, all receive the augmented input $\bar{x} = (1, x) \in H_0$. Let $T(\theta) \in H_0^{\otimes N}$ be the represented tensor. Then

$$f_\theta(x) = \langle T(\theta),\, \bar{x}^{\otimes N} \rangle \qquad \text{for all } x \in \mathbb{R}^{d_0} \text{ and all } \theta, \tag{2.11}$$

and the bonds between heights ℓ and $\ell + 1$ have dimension $d_\ell + 1$. The map $\theta \mapsto T(\theta)$ is polynomial.

Proof. As in the proof of Proposition 2.24, let $\Phi_\ell : H_0^{K^\ell} \to H_\ell$ be the contraction of a subtree of height ℓ: $\Phi_0 = \mathrm{id}$ and $\Phi_\ell(y_1, \dots, y_{K^\ell}) = G_\ell(\Phi_{\ell-1}(y^{(1)}), \dots, \Phi_{\ell-1}(y^{(K)}))$. By (2.10) and induction, $\Phi_\ell(\bar{x}, \dots, \bar{x}) = \bar{h}_\ell(x)$. Hence $\langle T(\theta), \bar{x}^{\otimes N} \rangle = \bar{c}^\top \Phi_L(\bar{x}, \dots, \bar{x}) = b_{\text{out}} + c^\top h_L(x) = f_\theta(x)$. The identity is algebraic, so it holds for every θ. □

So a DPN is a weight-tied TTN applied to copies of the input $(1, x)$, the extra coordinate 1 being induced by the homogenization; the bond dimension $d_\ell + 1$ counts the hidden units of layer ℓ plus the constant. Figure 2 shows the case $K = L = 2$. Since (2.11) holds identically in θ, it can be differentiated: training the DPN in its own parameters is the same as training the tied TTN through the map $\theta \mapsto ((G_\ell(\theta))_\ell, \bar{c})$ (Remark 2.52).

Remark 2.31 (What the representation is, and is not)**.**

(i) It is a *constrained* TTN: the cores are tied across each level, their entries satisfy the algebraic relations (2.8), and the constant slice is fixed.

(ii) It is evaluated on the *diagonal*: the leaves are formal copies of the whole input (Remark 2.25).

(iii) The $N = K^L$ leaves are never materialized in computations: the recurrence (2.6) evaluates the same function at a cost linear in the depth, by sharing repeated subcomputations, i.e. as a circuit (a directed acyclic graph) rather than a tree.

(iv) For $K > 2$, every K-ary vertex can be refined exactly into binary vertices, with new bond dimensions given by matricization ranks of G_ℓ.

(v) The multilinear extension is not unique. Since $\langle T, \bar{x}^{\otimes N} \rangle = \langle \mathrm{Sym}\, T, \bar{x}^{\otimes N} \rangle$, where Sym averages over permutations of the N legs, f_θ determines only the symmetric part $\mathrm{Sym}\, T(\theta)$. It determines it uniquely: a homogeneous polynomial on H_0 is determined by its values on the hyperplane $\{y_0 = 1\}$, and a symmetric tensor by its homogeneous polynomial (Lemma 2.28). The tensor $T(\theta)$ is the particular extension induced by the architecture.

Example 2.32 (Two layers, quadratic activations; Local TN note, 2026, Section 4)**.** For $K = 2$, write each layer polynomial as $P_\ell(h) = a_\ell + A_\ell h + B_\ell(h,h)$ with $a_\ell \in \mathbb{R}^{d_\ell}$, A_ℓ linear and B_ℓ symmetric bilinear (these are $A_\ell^{(0)}, A_\ell^{(1)}, A_\ell^{(2)}$ in Proposition 2.29). For the order-two Taylor polynomial of σ_ℓ at m_ℓ (Section 2.7), with $\gamma_\ell^{(r)} = \sigma_\ell^{(r)}(m_\ell)$ taken coordinatewise and $q_\ell = b_\ell - m_\ell$,

$$a_\ell = \gamma_\ell^{(0)} + \gamma_\ell^{(1)} \odot q_\ell + \tfrac{1}{2}\gamma_\ell^{(2)} \odot q_\ell \odot q_\ell, \qquad A_\ell u = \gamma_\ell^{(1)} \odot W_\ell u + \gamma_\ell^{(2)} \odot q_\ell \odot W_\ell u,$$
$$B_\ell(u,v) = \tfrac{1}{2}\gamma_\ell^{(2)} \odot W_\ell u \odot W_\ell v.$$

The core (2.9) is $G_\ell\big((\alpha,u),(\beta,v)\big) = \big(\alpha\beta,\ a_\ell\,\alpha\beta + \tfrac{1}{2}A_\ell(\alpha v + \beta u) + B_\ell(u,v)\big)$. For an output coordinate $i \geq 1$, its matrix in the two input indices is

$$(G_\ell)_{i;\cdot\cdot} = \begin{pmatrix} (a_\ell)_i & \tfrac{1}{2}(A_\ell)_{i,:} \\ \tfrac{1}{2}(A_\ell)_{i,:}^\top & (B_\ell)_{i;:,:} \end{pmatrix}.$$

The network is $f_\theta(x) = \bar{c}^\top G_2\big(G_1(\bar{x},\bar{x}), G_1(\bar{x},\bar{x})\big)$, a polynomial of degree four (not two) in x, with coefficient tensor $T_{j_1j_2j_3j_4} = \sum_{r,a,b} \bar{c}_r\,(G_2)_{r;ab}\,(G_1)_{a;j_1j_2}\,(G_1)_{b;j_3j_4}$. As a scalar check ($d_0 = d_1 = d_2 = 1$), for the SiLU activation $\sigma(z) = z/(1+e^{-z})$ with unit weights, zero biases, centres 0 and unit readout, $p(z) = z/2 + z^2/4$ and $f(x) = p(p(x)) = x/4 + 3x^2/16 + x^3/16 + x^4/64$.

2.6.3 Hierarchical features and learning capacity

The representation allows us to define a notion of hierarchical features of a DPN, from the HSVD of its TTN, and a measure of learning capacity, from its hierarchical ranks. For the tied tree, the HSVD reduces to one SVD per layer.

Proposition 2.33 (One SVD per layer)**.** In the setting of Theorem 2.30, $T(\theta)$ is invariant under the permutations of its N legs induced by automorphisms of $\mathcal{T}_{K,L}$. Consequently, for two bonds e, e' at the same height, $M_{e'}$ is obtained from M_e by permuting rows and columns: the two bonds have the same hierarchical singular values, and their singular vectors correspond under the relabelling of the leaves.

Proof. The cores G_ℓ and $\bar{c}^\top G_L$ are symmetric in their K inputs, so the contraction is invariant under exchanging the children of any vertex. These exchanges generate the automorphism group of $\mathcal{T}_{K,L}$, which acts transitively on the vertices of each height. □

We may therefore index the HSVD of a DPN by height: for $\ell = 1, \ldots, L-1$, it consists of singular values $\sigma_{\ell,1} \geq \cdots \geq \sigma_{\ell,r_\ell} > 0$ and singular vectors $u_{\ell,j} \in H_0^{\otimes K^\ell}$, $v_{\ell,j} \in H_0^{\otimes (N - K^\ell)}$. (A network with one hidden layer has no bond; its cuts are mode matricizations.)

Definition 2.34 (Hierarchical features)**.** The *hierarchical features* of the DPN f_θ at height ℓ are the pairs $(\sigma_{\ell,j}, \varphi_{\ell,j})$, $j \leq r_\ell$, where

$$\varphi_{\ell,j}(x) := \big\langle u_{\ell,j},\ \bar{x}^{\otimes K^\ell} \big\rangle$$

is a polynomial of degree at most K^ℓ. The complementary functions are $\psi_{\ell,j}(x) := \langle v_{\ell,j}, \bar{x}^{\otimes (N-K^\ell)} \rangle$.

Proposition 2.35 (What hierarchical features are)**.** For every $\ell \in \{1, \ldots, L-1\}$:

(i) $f_\theta(x) = \sum_{j=1}^{r_\ell} \sigma_{\ell,j}\,\varphi_{\ell,j}(x)\,\psi_{\ell,j}(x)$ for all x;

(ii) there is a matrix $Q_\ell \in \mathbb{R}^{(d_\ell+1)\times r_\ell}$ such that $\varphi_{\ell,j}(x) = \sum_i (Q_\ell)_{ij}\,\bar{h}_{\ell,i}(x)$: the hierarchical features at height ℓ are linear readouts of the augmented activations $\bar{h}_\ell = (1, h_\ell)$ of layer ℓ;

(iii) $r_\ell \le d_\ell + 1$.

Proof. (i) Contract (2.4) with $\bar{x}^{\otimes N}$. (ii) By Lemma 2.8, $M_e = F_e H_e^\top$, so $\mathrm{col}(U_e) = \mathrm{col}(M_e) \subseteq \mathrm{col}(F_e)$, i.e. $U_e = F_e Q_\ell$ for some matrix Q_ℓ. The i-th column of F_e is the i-th coordinate of the subtree contraction Φ_ℓ, and $\Phi_\ell(\bar{x}, \dots, \bar{x}) = \bar{h}_\ell(x)$ (proof of Theorem 2.30). (iii) is Lemma 2.8(iii). □

The HSVD at height ℓ thus selects, among the directions of the activation space of layer ℓ, those that the rest of the network uses, ranks them by $\sigma_{\ell,j}$, and is invariant under the hidden-unit symmetries of Remark 2.26. Two cautions apply. The orthonormality of the $u_{\ell,j}$ refers to the Frobenius inner product of coefficient tensors, not to $L^2(\nu_x)$. And $\varphi_{\ell,j}$ and $\psi_{\ell,j}$ are functions of the same input, so (i) is not a decomposition of f_θ into functions of independent variables (Remark 2.39).

Definition 2.36 (Learning capacity)**.** The *learning capacity* of a DPN at θ is the hierarchical rank profile of its TTN, $r(\theta) = (r_e(T(\theta)))_{e \in E_{\mathrm{int}}}$, that is $(r_\ell(\theta))_{\ell=1}^{L-1}$. It is bounded by the bond dimensions, $r_\ell \le D_\ell = d_\ell + 1$, which form the maximal capacity of the architecture.

By Proposition 2.10, a free TTN with bond dimensions D represents exactly the tensors with $r \le D$, so $r(\theta)$ measures how much of the available capacity the model uses. Remark 2.21 further suggests that learning proceeds by increasing ranks, although its results are proved for free TTNs on the Frobenius loss, not for the tied TTN of a DPN, whose loss depends on $T(\theta)$ only through $\mathrm{Sym}\, T(\theta)$ (Remark 2.31(v)). A tuple of ranks raises a natural question.

Question 2.37 (A global notion of learning capacity)**.** What would be a reasonable global notion of learning capacity? The minimum, the sum, or the maximum of the ranks? Or some other complexity measure using the structure of the TTN?

Remark 2.38 (Some candidates)**.**

(a) $\min_e r_e$ and $\max_e r_e$ measure the narrowest interface and the cost of contraction, respectively, but ignore where the ranks sit in the tree.

(b) $\sum_e \log r_e$ is the logarithm of the product of the ranks; $\log r_e$ bounds the entropy of the normalized squared singular values $\sigma_{e,j}^2 / \sum_i \sigma_{e,i}^2$ at e (the entanglement entropy of quantum physics).

(c) For a feasible profile r, the dimension of the set of tensors whose rank profile is exactly r,

$$\dim(r) = \sum_{v \in V} n_v \prod_{e \ni v} r_e \; - \sum_{e \in E_{\mathrm{int}}} r_e^2,$$

where n_v is the product of the dimensions of the open legs at v, counts the parameters of a minimal TTN modulo the gauge group. It weights each rank by its position in the tree, and reduces to $r(m+n-r)$ for $m \times n$ matrices (Holtz et al., 2012; Uschmajew and Vandereycken, 2013).

(d) Exact ranks are unstable, since a generic perturbation makes them maximal. Continuous surrogates, such as the participation ratio $(\sum_j \sigma_{e,j}^2)^2 / \sum_j \sigma_{e,j}^4$ (Diagonal TTN note, 2026, Section 9.4) or the entropy-based effective rank (Roy and Vetterli, 2007), can be aggregated as in (a)–(c).

Which of these, if any, predicts learning behaviour is an empirical question.

Remark 2.39 (Which tensor carries the features?)**.** Two caveats, emphasized by Local TN note (2026, Section 8) and Surrogate note (2026, Sections 3.4 and 7.4), qualify the identification of the HSVD of $T(\theta)$ with the features learned by the network.

First, the HSVD of $T(\theta)$ is invariant under gauge transformations, hence under hidden-unit symmetries, but it is not an invariant of the function f_θ. By Remark 2.31(v), f_θ determines only

$\operatorname{Sym} T(\theta)$, and different networks computing the same function can have different hierarchical ranks (Example 2.40). The HSVD is an invariant of the computation, not of the input–output map.

Second, cuts of the unrolled tree separate copies of the input, not input variables. When the input has independent blocks $x = (x_S, x_{S^c})$, with $\nu_x = \mu_S \otimes \mu_{S^c}$, an architecture-independent alternative is the functional SVD across a physical cut: every $g \in L^2(\nu_x)$ has a Schmidt decomposition

$$g(x_S, x_{S^c}) = \sum_{q \geq 1} s_q(g)\, \phi_q^S(x_S)\, \phi_q^{S^c}(x_{S^c}),$$

with $(\phi_q^S)_q$ orthonormal in $L^2(\mu_S)$ and $(\phi_q^{S^c})_q$ orthonormal in $L^2(\mu_{S^c})$. Its singular values and subspaces depend only on g and ν_x, so they can be compared between a teacher, a DNN, its polynomial surrogate and any TTN. Organizing such cuts along a tree of input blocks gives a physical-block TTN and HSVD, whose ranks are not bounded by the widths d_ℓ. In practice, they can be estimated from the crossed-grid matrix $\big[g(u_a, v_b)/\sqrt{n_L n_R}\big]_{a \leq n_L, b \leq n_R}$ built from independent samples $u_a \sim \mu_S$ and $v_b \sim \mu_{S^c}$ (Local TN note, 2026, Section 8.4). The computational HSVD of Definition 2.34 and the physical one answer different questions: the first describes how the network computes, the second what it computes.

Example 2.40 (Same function, different hierarchical ranks)**.** Take $d_0 = 2$ and two DQNs with two layers (Definition 2.22, $K = L = 2$) computing $f(x) = x_1^2 x_2^2$. Network P has $W_1 = \left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$, so that $h_1 = ((x_1 + x_2)^2, (x_1 - x_2)^2)$, then $W_2 = (\frac{1}{4}, -\frac{1}{4})$ and $c = 1$, so that $z_2 = x_1 x_2$ and $f = z_2^2$. Network Q has $W_1 = I_2$, so that $h_1 = (x_1^2, x_2^2)$, then $W_2 = \left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$ and $c = (\frac{1}{4}, -\frac{1}{4})$. With the bilinear forms $Z(u, v) = \frac{1}{2}(u_1 v_2 + u_2 v_1)$, $Z_a(u, v) = u_1 v_1$ and $Z_b(u, v) = u_2 v_2$, Proposition 2.24 gives

$$\begin{aligned} T_P(y_1, y_2, y_3, y_4) &= Z(y_1, y_2)\, Z(y_3, y_4), \\ T_Q(y_1, y_2, y_3, y_4) &= \tfrac{1}{2}\big[Z_a(y_1, y_2) Z_b(y_3, y_4) + Z_b(y_1, y_2) Z_a(y_3, y_4)\big]. \end{aligned}$$

Both restrict to $x_1^2 x_2^2$ on the diagonal, but the hierarchical rank of the bond at height one is $r_1 = 1$ for P and $r_1 = 2$ for Q. By contrast, for independent inputs x_1 and x_2, the functional rank of f across the physical cut $x_1 \mid x_2$ is 1, whichever network computes it.

Question 2.41 (Interpretability)**.** If we use the TTN as a surrogate model and its HSVD to interpret it, do the hierarchical singular values and vectors correspond to some meaningful structure? For example, if we use a toy model that we can interpret, do the features of the TTN correspond to the known features? Modular arithmetic would be an interesting example to investigate.

The following computation shows what to expect in the modular arithmetic case, in which the mechanism learned by small networks is known.

Example 2.42 (Modular addition)**.** Let p be an odd prime, encode inputs $a, b \in \mathbb{Z}_p$ as one-hot vectors, and let a model output logits $o(a, b) \in \mathbb{R}^p$, as in the grokking setting of Power et al. (2022). Since every function of (a, b) is bilinear in the one-hot encodings, the model is exactly a tensor $L \in \mathbb{R}^{p \times p \times p}$, $L_{abc} = o(a, b)_c$: a TTN on the physical tree with leaves a, b and output c. Its HSVD across the cut that separates a is the SVD of $L^{\langle\{a\}\rangle} \in \mathbb{R}^{p \times p^2}$. This is a function-level (physical) HSVD: it depends only on the logits, not on the architecture that computes them (Remark 2.39).

(i) The target $T^\star_{abc} = \mathbf{1}[c \equiv a + b]$ has a maximally degenerate spectrum. The rows of $T^{\star\langle\{a\}\rangle}$ are orthogonal and each contains p ones, so $T^{\star\langle\{a\}\rangle}\big(T^{\star\langle\{a\}\rangle}\big)^\top = p\, I_p$ and all p singular values equal $\sqrt{p}$. The SVD alone therefore singles out no basis, in particular not the Fourier basis. The same holds for the two other cuts.

(ii) Nanda et al. (2023) found that one-layer transformers trained on modular addition implement a Fourier algorithm whose logits are approximately $L_{abc} = \sum_{k\in\mathcal{F}} \lambda_k \cos\big(2\pi k(a+b-c)/p\big)$, for a small set $\mathcal{F} \subseteq \{1,\dots,(p-1)/2\}$ of key frequencies. If the $\lambda_k > 0$ are distinct, then $L^{\langle\{a\}\rangle}$ has rank $2|\mathcal{F}|$ and singular values $\lambda_k p^{3/2}/2$, each with multiplicity two, and the corresponding left singular subspaces are $\mathrm{span}\{(\cos(2\pi ka/p))_a, (\sin(2\pi ka/p))_a\}$. Indeed, with $\omega_k = 2\pi k/p$,

$$\cos\big(\omega_k(a+b-c)\big) = \cos(\omega_k a)\cos\big(\omega_k(b-c)\big) - \sin(\omega_k a)\sin\big(\omega_k(b-c)\big),$$

and these vectors are mutually orthogonal across k and between cos and sin, with squared norms $p/2$ as functions of a and $p^2/2$ as functions of (b,c).

For this idealized mechanism, the physical HSVD thus recovers exactly the known features, as two-dimensional frequency subspaces ranked by their strengths λ_k; the rotation within each subspace reflects the translation symmetry of $\mathbb{Z}_p$. Question 2.41 asks more: whether the HSVD of the computational TTN of a trained network (Section 2.6.3), which describes how the network computes, also recovers these subspaces. A natural test case is an MLP with quadratic activations acting on the concatenated one-hot inputs, for which an analytic Fourier solution is known in the case of one hidden layer (Gromov, 2023). With one hidden layer, the computational cuts are the mode matricizations of the root core; with more layers, Definition 2.34 applies.

2.7 Surrogate model of a deep neural network

Consider a general DNN (2.6) with smooth activation functions, more precisely of class C^{k+1} on intervals that contain all the pre-activations of the network and of its surrogate. The idea is to Taylor expand the DNN into a DPN around a training checkpoint, and to represent this DPN as a TTN.

Definition 2.43 (Layerwise Taylor surrogate; Local TN note, 2026, Section 2.2)**.** Let $\theta_j = \theta(t_j)$ be a checkpoint, $\mathcal{C}$ a finite calibration set of inputs, and $k \geq 1$. For every layer ℓ and neuron i, let the *centre*

$$m_{\ell,j,i} := \frac{1}{|\mathcal{C}|}\sum_{x\in\mathcal{C}} z_{\ell,i}(\theta_j, x)$$

be the average pre-activation of the neuron over the calibration set at the checkpoint, and let

$$p^{(k)}_{\ell,j,i}(z) := \sum_{r=0}^{k} \frac{\sigma^{(r)}_{\ell,i}(m_{\ell,j,i})}{r!}\,(z - m_{\ell,j,i})^r$$

be the order-k Taylor polynomial of $\sigma_{\ell,i}$ at that centre. The *order-k surrogate at checkpoint j* is the DPN obtained by replacing every activation by its Taylor polynomial, while using the surrogate's own hidden states:

$$\hat{h}_0 = x, \qquad \hat{z}_\ell = W_\ell \hat{h}_{\ell-1} + b_\ell, \qquad \hat{h}_\ell = p^{(k)}_{\ell,j}(\hat{z}_\ell), \qquad \hat{f}_j(\theta, x) = c^\top \hat{h}_L + b_{\mathrm{out}}.$$

The centres and the Taylor coefficients are frozen, and θ remains trainable.

Definition 2.44 (TTN surrogate)**.** For $k \geq 2$, the *TTN surrogate* of the DNN at checkpoint j is the tied TTN of Theorem 2.30, with $K = k$, that represents the surrogate: $\hat{f}_j(\theta, x) = \langle T_j(\theta), \bar{x}^{\otimes k^L}\rangle$. We say that this TTN is a surrogate model of the DNN at the training point θ_j.

Remark 2.45 (On the definition)**.**

(i) The surrogate must use its own hidden states: substituting the true hidden states of the DNN at every layer would define an error diagnostic, not a trainable model.

(ii) Its input degree is k^L, not k. Layerwise Taylorization is not the Taylor expansion of the predictor in θ (Bai et al., 2020), since composition keeps cross-layer terms of all orders. For $k = 1$, every activation is replaced by its tangent line and the surrogate is a deep affine network, i.e. a DLN with biases, not the linearized (NTK) model; its TTN degenerates into a chain. The family of surrogates therefore starts from the model whose dynamics is understood.

(iii) The centre lives in pre-activation space, since σ is expanded in its argument. Expanding around the mean pre-activation over a calibration set places the expansion at a typical value over the input distribution; the mean minimizes the mean squared distance to the centre, which controls the order-one remainder, but it is a heuristic and not an optimal choice in general.

(iv) Choosing instead the per-example centre $m_{\ell,j}(x) = z_\ell(\theta_j, x)$ makes the surrogate agree with the DNN and its θ-derivatives up to order k at θ_j, so that $\hat{f}_j(\theta_j + \Delta, \cdot) - f_{\theta_j+\Delta} = O(\|\Delta\|^{k+1})$ under suitable bounds. Its coefficients, however, depend on x, so it is not one polynomial in x and has no single TTN. It is a useful control, which separates the error due to the spread of pre-activations across inputs from the error due to training (Local TN note, 2026, Section 2.3).

(v) Layerwise Taylorization of trained networks also underlies NN2Poly (Morala et al., 2025); for higher-order dynamical approximations in parameter space, see Bai et al. (2020) and Huang and Yau (2020).

(vi) In automatic differentiation, the centres and the derivatives $\sigma^{(r)}(m)$ must be detached from the computational graph, while the cores G_ℓ, recomputed from the current θ, must remain differentiable.

We would like to know how good an approximation of the DNN such a TTN is. Since training follows gradients, the relevant errors are on derivatives as well as values.

Lemma 2.46 (Taylor remainders)**.** If σ is C^{k+1} on the segment between m and z, and M_{k+1} bounds $|\sigma^{(k+1)}|$ there, then

$$|\sigma(z) - p^{(k)}(z)| \le \frac{M_{k+1}}{(k+1)!}\,|z-m|^{k+1}, \qquad |\sigma'(z) - (p^{(k)})'(z)| \le \frac{M_{k+1}}{k!}\,|z-m|^{k}.$$

Proof. Use the Lagrange form of the remainder; for the second inequality, note that $(p^{(k)})'$ is the order-$(k-1)$ Taylor polynomial of σ' at m. □

Proposition 2.47 (Loss-gradient defect; Local TN note, 2026, Proposition 2)**.** Let $\mathcal{L}(\theta) = \frac{1}{2}\mathbb{E}_\nu[(f_\theta(x) - y)^2]$ and $\widehat{\mathcal{L}}_j(\theta) = \frac{1}{2}\mathbb{E}_\nu[(\hat{f}_j(\theta, x) - y)^2]$, and define at a common parameter θ

$$\varepsilon_{0,j} = \|f_\theta - \hat{f}_j(\theta,\cdot)\|_{L^2(\nu_x)}, \qquad \varepsilon_{1,j} = \|\nabla_\theta f_\theta - \nabla_\theta \hat{f}_j(\theta,\cdot)\|_{L^2(\nu_x)},$$
$$R_y = \|f_\theta - y\|_{L^2(\nu)}, \qquad B_j = \|\nabla_\theta \hat{f}_j(\theta,\cdot)\|_{L^2(\nu_x)},$$

where $\|v\|_{L^2} = \mathbb{E}[\|v\|^2]^{1/2}$ for vector-valued v. Whenever differentiation under the expectation is valid,

$$\|\nabla\mathcal{L}(\theta) - \nabla\widehat{\mathcal{L}}_j(\theta)\| \le R_y\,\varepsilon_{1,j} + B_j\,\varepsilon_{0,j} =: \beta_j(\theta).$$

Proof. $\nabla\mathcal{L} - \nabla\widehat{\mathcal{L}}_j = \mathbb{E}\big[(f-y)(\nabla f - \nabla\hat{f}_j) + (f - \hat{f}_j)\nabla\hat{f}_j\big]$; apply the triangle and Cauchy–Schwarz inequalities. □

Two functions can thus be close while their parameter derivatives, and therefore their training updates, are different: prediction agreement alone is not sufficient.

Proposition 2.48 (Trajectory comparison; Local TN note, 2026, Section 6)**.** Let θ and $\tilde{\theta}$ solve $\dot{\theta} = -\nabla\mathcal{L}(\theta)$ and $\dot{\tilde{\theta}} = -\nabla\widehat{\mathcal{L}}_j(\tilde{\theta})$ on $[t_j, t_j + h]$, and assume that both stay in a set $\mathcal{U}$ on which $\nabla\mathcal{L}$ is κ_j-Lipschitz and $\|\nabla\mathcal{L} - \nabla\widehat{\mathcal{L}}_j\| \le \beta_j$. Then $E(s) := \|\theta(t_j + s) - \tilde{\theta}(t_j + s)\|$ satisfies

$$E(s) \le e^{\kappa_j s} E(0) + \beta_j \, \frac{e^{\kappa_j s} - 1}{\kappa_j}, \qquad 0 \le s \le h,$$

the fraction being s if $\kappa_j = 0$. For gradient descent with steps η_n, $E_{n+1} \le (1 + \eta_n\kappa_j)E_n + \eta_n\beta_j$.

Proof. The upper Dini derivative satisfies $D^+E \le \|\nabla\mathcal{L}(\theta) - \nabla\mathcal{L}(\tilde{\theta})\| + \|\nabla\mathcal{L}(\tilde{\theta}) - \nabla\widehat{\mathcal{L}}_j(\tilde{\theta})\| \le \kappa_j E + \beta_j$; integrate (Grönwall). The discrete case is identical. □

These statements make the quality of the approximation precise: the local surrogate is faithful over windows on which the loss-gradient defect β_j is small compared with the gradient itself. They are conditional, and their constants can grow with depth, width and parameter norms; Local TN note (2026, Appendix A) propagates the remainders of Lemma 2.46 through the layers into $\varepsilon_{0,j}$ and $\varepsilon_{1,j}$.

Remark 2.49 (Fixed centres have an error floor)**.** With input-independent centres, the radius $|z - m|$ in Lemma 2.46 includes the spread $S_{\ell,j} = \sup_x \|z_\ell(\theta_j, x) - m_{\ell,j}\|_\infty$ of the pre-activations around the centres, which does not shrink with the time window (Local TN note, 2026, Section 6.3). The gradient defect $\|\nabla\mathcal{L}(\theta_j) - \nabla\widehat{\mathcal{L}}_j(\theta_j)\|$ at the checkpoint is then generally nonzero, and since both trajectories start at θ_j, $E(s) = s\,\|\nabla\mathcal{L}(\theta_j) - \nabla\widehat{\mathcal{L}}_j(\theta_j)\| + o(s)$: the error grows linearly from the start of the window, and Proposition 2.48 gives a matching upper bound. If the fixed-centre surrogate fails while the per-example surrogate of Remark 2.45(iv) succeeds, the problem is the spread of activations across inputs, not the tensor representation. Chebyshev or distribution-weighted polynomial fits over the observed range of pre-activations are then natural, non-Taylor alternatives (Surrogate note, 2026, Section 8.2).

We would also like to know whether the HSVD of the TTN can be used to extract the structure that the DPN has learned, and to predict its training dynamics. The following identity connects the two.

Proposition 2.50 (Dynamics of hierarchical singular values)**.** Let $N = k^L$. Along the gradient flow $\dot{\theta} = -\nabla\widehat{\mathcal{L}}_j(\theta)$ of the surrogate, the tensor $T(t) := T_j(\theta(t))$ evolves as

$$\dot{T} = -\mathcal{K}_\theta[R_\theta], \qquad \mathcal{K}_\theta := DT_j(\theta)\,DT_j(\theta)^*, \qquad R_\theta := \mathbb{E}_\nu[(\hat{f}_j(\theta, x) - y)\,\bar{x}^{\otimes N}],$$

where $\mathcal{K}_\theta$ is a positive semidefinite operator on $H_0^{\otimes N}$ and R_θ is the residual-weighted moment tensor. If $\sigma_{e,i}$ is a simple nonzero singular value of $M_e(t)$, then it is differentiable and

$$\dot{\sigma}_{e,i} = u_{e,i}^\top \dot{M}_e \, v_{e,i} = -\langle Y_{e,i},\, \mathcal{K}_\theta[R_\theta]\rangle,$$

where $Y_{e,i}$ is the tensor whose A_e-matricization is $u_{e,i}v_{e,i}^\top$.

Proof. We have $\widehat{\mathcal{L}}_j = \mathcal{J} \circ T_j$ with $\mathcal{J}(T) = \frac{1}{2}\mathbb{E}_\nu[(\langle T, \bar{x}^{\otimes N}\rangle - y)^2]$, whose gradient is R. By the chain rule, $\dot{\theta} = -DT_j(\theta)^*[R_\theta]$ and $\dot{T} = DT_j(\theta)[\dot{\theta}]$. For a simple singular value with unit singular vectors, $\sigma = u^\top M v$ and $\dot{\sigma} = u^\top \dot{M} v + \sigma(\dot{u}^\top u + v^\top \dot{v}) = u^\top \dot{M} v$, because $\|u\| = \|v\| = 1$. □

The identity is exact but not closed: $\mathcal{K}_\theta$, R_θ and the singular vectors all evolve. Predicting the training dynamics from the HSVD amounts to finding regimes in which a few singular values

and subspaces approximately close under these equations (Surrogate note, 2026, Section 12), as the singular modes of a DLN do in the analysis of Saxe et al. (2014). When surrogates built at different checkpoints are compared, their HSVDs should be evaluated at the same parameter value, since otherwise a change of Taylor centres would be mistaken for learning (Local TN note, 2026, Section 8.5; Surrogate note, 2026, Section 9.1); Remark 2.17 bounds how much the spectrum can move.

A more direct approach: polynomial activations. Another, more direct approach is to replace the activation functions of the original network by polynomial activation functions. We then train the corresponding TTN, define some useful observables, such as the HSVD, and check whether their training dynamics is similar to that of the original DNN.

Definition 2.51 (Polynomial-activation proxy)**.** Given the architecture (2.6) with activation σ, choose polynomials q_ℓ of degree K, for example the Taylor polynomial of σ at a fixed point, or a least-squares fit of σ on an interval $[-a, a]$ that contains most pre-activations. The *polynomial proxy* f_θ^q is the DPN obtained by replacing σ_ℓ with q_ℓ, initialized and trained exactly like the DNN. By Theorem 2.30, $f_\theta^q(x) = \langle T^q(\theta), \bar{x}^{\otimes K^L} \rangle$, so the hierarchical singular values $\sigma_{\ell,j}(t)$ and ranks $r_\ell(t)$ of Definitions 2.34 and 2.36 can be tracked along its training. They are diagnostics of the TTN representation; they can be compared with the DNN through the HSVDs of its local TTN surrogates (Definition 2.44), or through physical HSVDs (Remark 2.39).

Remark 2.52 (Algebraic equivalence is not dynamical equivalence)**.** "Training the corresponding TTN" can mean two different things. (a) Training the DPN in its parameters θ is identical to training the tied TTN through the map $\Gamma : \theta \mapsto ((G_\ell(\theta))_\ell, \bar{c})$, because (2.11) holds identically in θ. (b) Training the cores g as free variables is a different dynamical system: if $g = \Gamma(\theta)$ and $\mathcal{J}(g)$ denotes the loss as a function of the cores, (a) induces $\dot{g} = -D\Gamma(\theta) D\Gamma(\theta)^\top \nabla_g \mathcal{J}(g)$, a preconditioned and possibly degenerate flow, whereas (b) is $\dot{g} = -\nabla_g \mathcal{J}(g)$. Moreover, free training breaks the ties and constraints of Remark 2.31(i). Surrogate note (2026, Section 4.1) proposes a ladder of relaxations that isolates each simplification: the tied TTN trained through θ; free tied cores with the induced metric; a block-diagonal approximation of that metric; Euclidean core gradients; untied cores; compressed ranks; and finally a different tree. For computations, one should use the recurrence (2.6) rather than the exponentially large tree.

Remark 2.53 (Choosing the degree)**.** A problem is to choose the degree of the nodes of the TTN. Several considerations bear on it.

(i) Approximation: on a bounded range of pre-activations, least-squares or Chebyshev approximations of a smooth activation improve rapidly with K, whereas Taylor polynomials are accurate only near their centre.

(ii) Stability: polynomials are unbounded and the degree K^L of the network grows exponentially with depth, so pre-activations that leave the fitted range are amplified. The fraction of pre-activations outside that range is a direct diagnostic (Surrogate note, 2026, Section 8.8).

(iii) Structure: $K = 2$ is the smallest nonlinear choice and gives binary trees. Networks with bilinear layers $(Wx) \odot (Vx)$, which are untied versions of quadratic activations, are competitive in practice and can be interpreted through their tensors (Pearce et al., 2024).

(iv) Cost: each core has order $K + 1$, while the bond dimensions $d_\ell + 1$ do not depend on K.

A pragmatic rule is to choose the smallest K for which the defects of Proposition 2.47 are below a tolerance at initialization and at a few checkpoints.

3 Training dynamics of TTNs

We derive the population gradient flow of a TTN student on the smallest balanced binary tree with two bilinear levels: four input blocks, two lower cores that each combine two blocks, and a root core that combines the messages of the lower cores. The purpose is twofold. First, we write the gradient flow explicitly, entry by entry. Second, we rewrite it as an evolution of the represented tensor, driven by the geometry of the data and by the differential of the TTN parameterization. This second form is an instance of the coefficient-space evolution of Proposition 2.50, and it generalizes directly to larger trees. On the manifold of balanced parameters, which gradient flow preserves, this evolution closes, and it decouples into scalar equations for the singular values when the student is aligned with the teacher (Section 3.6). In this smallest example, the two bonds cut the legs in the same way, so the dynamics are those of a matrix factorized into three factors, in a geometry set by the data (Remark 3.2).

3.1 A four-leaf, depth-two teacher–student model

Student. The input consists of four blocks $x^{(a)} \in \mathbb{R}^{d_a}$, $a \in [4]$, and we consider the balanced tree $((1,2),(3,4))$. To lighten notation, the legs receive the blocks themselves, i.e. $\xi_a(x^{(a)}) = x^{(a)}$ in (2.3). Since the data enter only through the inputs of the legs, every statement of this section holds for arbitrary embeddings, with $x^{(a)}$ replaced by $\xi_a(x^{(a)})$. We use the minimal parameterization with three cores, in which the two lower cores carry the legs directly; in Figure 1(c), this amounts to absorbing the leaf cores into C_u and C_v. The two lower cores and the root core are

$$\mathcal{A} \in \mathbb{R}^{D_L \times d_1 \times d_2}, \qquad \mathcal{B} \in \mathbb{R}^{D_R \times d_3 \times d_4}, \qquad C \in \mathbb{R}^{D_L \times D_R},$$

where D_L and D_R are the dimensions of the left and right bonds, and we write $\vartheta = (\mathcal{A}, \mathcal{B}, C)$. The lower cores send the messages

$$h^L_\alpha(x) = \sum_{i_1=1}^{d_1} \sum_{i_2=1}^{d_2} \mathcal{A}_{\alpha i_1 i_2}\, x^{(1)}_{i_1} x^{(2)}_{i_2}, \qquad \alpha \in [D_L], \tag{3.1}$$

$$h^R_\beta(x) = \sum_{i_3=1}^{d_3} \sum_{i_4=1}^{d_4} \mathcal{B}_{\beta i_3 i_4}\, x^{(3)}_{i_3} x^{(4)}_{i_4}, \qquad \beta \in [D_R], \tag{3.2}$$

through the two bonds, and the root contracts them:

$$g_\vartheta(x) = \sum_{\alpha=1}^{D_L} \sum_{\beta=1}^{D_R} C_{\alpha\beta}\, h^L_\alpha(x)\, h^R_\beta(x). \tag{3.3}$$

Equivalently, $g_\vartheta(x) = \langle T(\vartheta), x^{(1)} \otimes x^{(2)} \otimes x^{(3)} \otimes x^{(4)} \rangle$ as in (2.3), where the represented tensor $T(\vartheta) \in \mathbb{R}^{d_1 \times d_2 \times d_3 \times d_4}$ has entries

$$T(\vartheta)_{i_1 i_2 i_3 i_4} = \sum_{\alpha=1}^{D_L} \sum_{\beta=1}^{D_R} \mathcal{A}_{\alpha i_1 i_2}\, C_{\alpha\beta}\, \mathcal{B}_{\beta i_3 i_4}. \tag{3.4}$$

Matrix form. To expose the factorization across the central cut, let

$$z := x^{(1)} \otimes x^{(2)} \in \mathbb{R}^{d_{12}}, \qquad w := x^{(3)} \otimes x^{(4)} \in \mathbb{R}^{d_{34}}, \qquad d_{12} := d_1 d_2, \quad d_{34} := d_3 d_4,$$

and matricize the lower cores as $A \in \mathbb{R}^{d_{12} \times D_L}$ and $B \in \mathbb{R}^{d_{34} \times D_R}$, with $A_{(i_1,i_2),\alpha} := \mathcal{A}_{\alpha i_1 i_2}$ and $B_{(i_3,i_4),\beta} := \mathcal{B}_{\beta i_3 i_4}$. Pairs of indices are ordered lexicographically, as in Definition 2.5 and in the Kronecker products z and w. Then

$$h^L = A^\top z, \qquad h^R = B^\top w, \qquad g_\vartheta(x) = z^\top ACB^\top w, \tag{3.5}$$

and the $\{1,2\}$-matricization of the represented tensor is

$$M(\vartheta) := T(\vartheta)^{\langle\{1,2\}\rangle} = ACB^\top \in \mathbb{R}^{d_{12} \times d_{34}}. \tag{3.6}$$

This is the factorization $M_e = F_e H_e^\top$ of Lemma 2.8 at the left bond, with $F_e = A$ and $H_e = BC^\top$.

Teacher and loss. The teacher $f^\star(x) = \langle T^\star, x^{(1)} \otimes x^{(2)} \otimes x^{(3)} \otimes x^{(4)} \rangle$ acts on the same four blocks, and $M^\star := T^{\star\langle\{1,2\}\rangle}$ denotes its central matricization. It may be represented on the same tree with smaller bond dimensions $D_L^\star \le D_L$ and $D_R^\star \le D_R$, as $M^\star = A^\star C^\star B^{\star\top}$; overparameterization then means that the student has more bond coordinates than the teacher requires. The derivation below depends on the teacher only through $M^\star$. Let

$$\Delta := M(\vartheta) - M^\star, \qquad e_\vartheta(x) := g_\vartheta(x) - f^\star(x) = z^\top \Delta w, \tag{3.7}$$

and consider the population square loss

$$\mathcal{L}(\vartheta) := \tfrac{1}{2}\mathbb{E}\big[e_\vartheta(X)^2\big], \tag{3.8}$$

where $X = (X^{(1)}, \dots, X^{(4)})$ is a random input. Gradients are Euclidean: $\nabla_A \mathcal{L}$ is the gradient with respect to the entries of A, i.e. of $\mathcal{A}$, and similarly for B and C.

Remark 3.1 (Parameterization)**.** Equations (3.1)–(3.3) treat the two lower cores, of order three, and the root matrix as the trainable cores. If a lower core is itself factorized into leaf maps and an internal core, as in Figure 1(c), then A or B is an effective contracted core. The function is unchanged, but gradient flow in the additional factors is obtained by one more application of the chain rule, and it differs from gradient flow in A or B as free variables (compare Remark 2.52). Likewise, if the two lower cores are tied, which requires $d_1 = d_3$, $d_2 = d_4$ and $D_L = D_R$, the gradient with respect to the shared core is the sum of the gradients with respect to $\mathcal{A}$ and $\mathcal{B}$ computed below.

Remark 3.2 (One cut, three factors)**.** Both bonds of this tree induce the same bipartition $\{1,2\} \mid \{3,4\}$ of the legs. Hence $M(\vartheta)$ is the only edge matricization of $T(\vartheta)$, and by Proposition 2.10 the student represents exactly the tensors T with $\operatorname{rank} T^{\langle\{1,2\}\rangle} \le \min(D_L, D_R)$. As a function of the parameters, $M(\vartheta) = ACB^\top$ is a product of three matrices. For whitened inputs, the loss is therefore that of a three-layer linear network with weights $B^\top$, C and A (Proposition 3.3), so that its gradient flow is that of a DLN (Saxe et al., 2014), and its balance invariants (Proposition 3.6) are those of a DLN. What is specific to tensor networks enters through the data, whose second moments factor over the legs when the blocks are independent (Section 3.3), and, in larger trees or when the lower cores are themselves factorized (Remark 3.1), through nested cuts and additional factors.

3.2 Independent whitened inputs

Assume first that the four blocks are independent and whitened:

$$\mathbb{E}\big[X^{(a)} X^{(a)\top}\big] = I_{d_a} \quad (a \in [4]), \qquad X^{(1)}, \dots, X^{(4)} \text{ independent.} \tag{3.9}$$

Only these second moments enter below: the blocks need not be Gaussian, and they need not be centred. Since $zz^\top = (X^{(1)}X^{(1)\top}) \otimes (X^{(2)}X^{(2)\top})$, and similarly for w, independence gives

$$\mathbb{E}[zz^\top] = \mathbb{E}[X^{(1)}X^{(1)\top}] \otimes \mathbb{E}[X^{(2)}X^{(2)\top}] = I_{d_{12}}, \tag{3.10}$$

$$\mathbb{E}[ww^\top] = \mathbb{E}[X^{(3)}X^{(3)\top}] \otimes \mathbb{E}[X^{(4)}X^{(4)\top}] = I_{d_{34}}, \tag{3.11}$$

and z and w are independent.

Proposition 3.3 (Whitened population loss and gradient flow)**.** Under (3.9),

$$\mathcal{L}(\vartheta) = \tfrac{1}{2}\|ACB^\top - M^\star\|_F^2 = \tfrac{1}{2}\|\Delta\|_F^2. \tag{3.12}$$

Its gradients are

$$\nabla_A\mathcal{L} = \Delta BC^\top, \qquad \nabla_B\mathcal{L} = \Delta^\top AC, \qquad \nabla_C\mathcal{L} = A^\top\Delta B, \tag{3.13}$$

so that the population gradient flow is

$$\dot{A} = (M^\star - ACB^\top)BC^\top, \qquad \dot{B} = (M^\star - ACB^\top)^\top AC, \qquad \dot{C} = A^\top(M^\star - ACB^\top)B. \tag{3.14}$$

Proof. We first reduce the loss to a Frobenius norm. Since $e_\vartheta = z^\top\Delta w$ is a scalar, cyclicity of the trace and the independence of z and w give

$$2\mathcal{L} = \mathbb{E}[z^\top\Delta ww^\top\Delta^\top z] = \operatorname{tr}\left(\Delta\,\mathbb{E}[ww^\top]\,\Delta^\top\,\mathbb{E}[zz^\top]\right) = \operatorname{tr}(\Delta\Delta^\top) = \|\Delta\|_F^2. \tag{3.15}$$

For the gradients, we differentiate the prediction before taking the expectation. For a perturbation δA of A, (3.5) gives

$$D_A g_\vartheta(x)[\delta A] = z^\top(\delta A)CB^\top w = \langle z(CB^\top w)^\top, \delta A\rangle_F, \tag{3.16}$$

so the gradient of the per-example loss $\frac{1}{2}e_\vartheta(x)^2$ with respect to A is $e_\vartheta(x)\,z(CB^\top w)^\top$, and

$$\nabla_A\mathcal{L} = \mathbb{E}[e_\vartheta\, z(CB^\top w)^\top] = \mathbb{E}[e_\vartheta\, zw^\top]BC^\top. \tag{3.17}$$

By the independence of z and w, the residual-weighted moment matrix in this expression is

$$\mathbb{E}[e_\vartheta\, zw^\top] = \mathbb{E}[z(z^\top\Delta w)w^\top] = \mathbb{E}[zz^\top]\,\Delta\,\mathbb{E}[ww^\top] = \Delta, \tag{3.18}$$

hence $\nabla_A\mathcal{L} = \Delta BC^\top$. The other two gradients follow in the same way. For perturbations δB and δC,

$$D_B g_\vartheta(x)[\delta B] = z^\top AC(\delta B)^\top w = \langle w(C^\top A^\top z)^\top, \delta B\rangle_F, \tag{3.19}$$

$$D_C g_\vartheta(x)[\delta C] = z^\top A(\delta C)B^\top w = \langle (A^\top z)(B^\top w)^\top, \delta C\rangle_F, \tag{3.20}$$

so that

$$\nabla_B\mathcal{L} = \mathbb{E}[e_\vartheta\, wz^\top]AC = \Delta^\top AC, \tag{3.21}$$

$$\nabla_C\mathcal{L} = A^\top\mathbb{E}[e_\vartheta\, zw^\top]B = A^\top\Delta B. \tag{3.22}$$

Gradient flow is $\dot{A} = -\nabla_A\mathcal{L}$, $\dot{B} = -\nabla_B\mathcal{L}$ and $\dot{C} = -\nabla_C\mathcal{L}$, which gives (3.14). □

Index form. The same gradients can be written in the indices of the legs. Let

$$\Delta_{i_1 i_2 i_3 i_4} := T(\vartheta)_{i_1 i_2 i_3 i_4} - T^\star_{i_1 i_2 i_3 i_4} = \sum_{\alpha=1}^{D_L} \sum_{\beta=1}^{D_R} \mathcal{A}_{\alpha i_1 i_2}\, C_{\alpha\beta}\, \mathcal{B}_{\beta i_3 i_4} - T^\star_{i_1 i_2 i_3 i_4} \tag{3.23}$$

be the error tensor, whose $\{1,2\}$-matricization is Δ. Then

$$\frac{\partial \mathcal{L}}{\partial \mathcal{A}_{\alpha i_1 i_2}} = \sum_{\beta=1}^{D_R} \sum_{i_3=1}^{d_3} \sum_{i_4=1}^{d_4} \Delta_{i_1 i_2 i_3 i_4}\, C_{\alpha\beta}\, \mathcal{B}_{\beta i_3 i_4}, \tag{3.24}$$

$$\frac{\partial \mathcal{L}}{\partial \mathcal{B}_{\beta i_3 i_4}} = \sum_{\alpha=1}^{D_L} \sum_{i_1=1}^{d_1} \sum_{i_2=1}^{d_2} \Delta_{i_1 i_2 i_3 i_4}\, \mathcal{A}_{\alpha i_1 i_2}\, C_{\alpha\beta}, \tag{3.25}$$

$$\frac{\partial \mathcal{L}}{\partial C_{\alpha\beta}} = \sum_{i_1=1}^{d_1} \sum_{i_2=1}^{d_2} \sum_{i_3=1}^{d_3} \sum_{i_4=1}^{d_4} \Delta_{i_1 i_2 i_3 i_4}\, \mathcal{A}_{\alpha i_1 i_2}\, \mathcal{B}_{\beta i_3 i_4}. \tag{3.26}$$

Each core is thus updated by contracting the error tensor with all the other cores of the tree. In terms of the error $\mathcal{E} := T^\star - T(\vartheta) = -\Delta$, the gradient flow reads

$$\begin{aligned} \dot{\mathcal{A}}_{\alpha i_1 i_2} &= \sum_{\beta, i_3, i_4} \mathcal{E}_{i_1 i_2 i_3 i_4}\, C_{\alpha\beta}\, \mathcal{B}_{\beta i_3 i_4}, \\ \dot{\mathcal{B}}_{\beta i_3 i_4} &= \sum_{\alpha, i_1, i_2} \mathcal{E}_{i_1 i_2 i_3 i_4}\, \mathcal{A}_{\alpha i_1 i_2}\, C_{\alpha\beta}, \\ \dot{C}_{\alpha\beta} &= \sum_{i_1, i_2, i_3, i_4} \mathcal{E}_{i_1 i_2 i_3 i_4}\, \mathcal{A}_{\alpha i_1 i_2}\, \mathcal{B}_{\beta i_3 i_4}. \end{aligned} \tag{3.27}$$

Differential derivation. Once (3.12) is known, the three gradients also follow in one line. Since $dM = (dA)CB^\top + A(dC)B^\top + AC(dB)^\top$,

$$d\mathcal{L} = \langle \Delta, dM \rangle_F = \langle \Delta B C^\top, dA \rangle_F + \langle \Delta^\top A C, dB \rangle_F + \langle A^\top \Delta B, dC \rangle_F, \tag{3.28}$$

which is (3.13).

3.3 Independent, non-whitened inputs

We keep the independence of the four blocks but allow arbitrary second moments:

$$\Sigma_a := \mathbb{E}[X^{(a)} X^{(a)\top}] \succeq 0 \quad (a \in [4]), \qquad X^{(1)}, \dots, X^{(4)} \text{ independent.} \tag{3.29}$$

For a centred block, Σ_a is the covariance of $X^{(a)}$; in general, it is the Gram matrix of the inputs of leg a, as in (5.2). The Gram matrices of the two sides of the central cut are

$$\Sigma_{12} := \mathbb{E}[z z^\top] = \Sigma_1 \otimes \Sigma_2, \qquad \Sigma_{34} := \mathbb{E}[w w^\top] = \Sigma_3 \otimes \Sigma_4, \tag{3.30}$$

that is, $\Sigma_{\{1,2\}}$ and $\Sigma_{\{3,4\}}$ in the notation of (5.2). Neither needs to be invertible.

Proposition 3.4 (Data-weighted loss and gradient flow)**.** Under (3.29),

$$\mathcal{L}(\vartheta) = \tfrac{1}{2} \operatorname{tr}\left(\Delta^\top \Sigma_{12} \Delta \Sigma_{34}\right) = \tfrac{1}{2} \|\Sigma_{12}^{1/2} \Delta \Sigma_{34}^{1/2}\|_F^2. \tag{3.31}$$

As a function of M, the loss has gradient

$$\nabla_M \mathcal{L} = \Sigma_{12} \Delta \Sigma_{34}, \tag{3.32}$$

and its parameter gradients are

$$\nabla_A \mathcal{L} = \Sigma_{12} \Delta \Sigma_{34} B C^\top, \qquad \nabla_B \mathcal{L} = \Sigma_{34} \Delta^\top \Sigma_{12} A C, \qquad \nabla_C \mathcal{L} = A^\top \Sigma_{12} \Delta \Sigma_{34} B. \tag{3.33}$$

Equivalently, with the weighted error

$$\mathcal{E}_\Sigma := \Sigma_{12}(M^\star - M(\vartheta))\Sigma_{34}, \tag{3.34}$$

the gradient flow is

$$\dot{A} = \mathcal{E}_\Sigma B C^\top, \qquad \dot{B} = \mathcal{E}_\Sigma^\top A C, \qquad \dot{C} = A^\top \mathcal{E}_\Sigma B. \tag{3.35}$$

Proof. Only the two Gram matrices change in (3.15):

$$2\mathcal{L} = \mathbb{E}[z^\top \Delta w w^\top \Delta^\top z] = \operatorname{tr}\left(\Delta \Sigma_{34} \Delta^\top \Sigma_{12}\right) = \operatorname{tr}\left(\Delta^\top \Sigma_{12} \Delta \Sigma_{34}\right) = \|\Sigma_{12}^{1/2} \Delta \Sigma_{34}^{1/2}\|_F^2. \tag{3.36}$$

Differentiating this quadratic form in M gives (3.32), since Σ_{12} and Σ_{34} are symmetric. Equivalently, the residual-weighted moment matrix is now

$$\mathbb{E}[e_\vartheta\, z w^\top] = \mathbb{E}[z(z^\top \Delta w) w^\top] = \mathbb{E}[z z^\top]\, \Delta\, \mathbb{E}[w w^\top] = \Sigma_{12} \Delta \Sigma_{34}. \tag{3.37}$$

The first equalities in (3.17), (3.21) and (3.22) hold for any input law. Substituting (3.37) into them proves (3.33), and negating the gradients gives (3.35). □

The proof uses only the independence of z and w, i.e. of the two sides of the cut; the independence of the blocks within each side serves only to factor Σ_{12} and Σ_{34} as Kronecker products. Proposition 3.4 therefore holds whenever $(X^{(1)}, X^{(2)})$ and $(X^{(3)}, X^{(4)})$ are independent, with $\Sigma_{12} := \mathbb{E}[z z^\top]$ and $\Sigma_{34} := \mathbb{E}[w w^\top]$.

Index form. Under the product law (3.29), the gradients can also be written in the indices of the legs. Let

$$(\Delta_\Sigma)_{i_1 i_2 i_3 i_4} := \sum_{j_1, j_2, j_3, j_4} (\Sigma_1)_{i_1 j_1} (\Sigma_2)_{i_2 j_2} (\Sigma_3)_{i_3 j_3} (\Sigma_4)_{i_4 j_4}\, \Delta_{j_1 j_2 j_3 j_4} \tag{3.38}$$

be the weighted error tensor, whose $\{1,2\}$-matricization is $\Sigma_{12}\Delta\Sigma_{34}$. The gradients are the contractions (3.24)–(3.26) with Δ replaced by Δ_Σ. Non-whitened inputs therefore do not change the contraction pattern of the tree; they change the error tensor that is propagated through it.

Remark 3.5 (Whitened analysis coordinates versus parameter dynamics)**.** If Σ_{12} and Σ_{34} are invertible, the change of variables

$$\widetilde{A} := \Sigma_{12}^{1/2} A, \qquad \widetilde{B} := \Sigma_{34}^{1/2} B, \qquad \widetilde{M}^\star := \Sigma_{12}^{1/2} M^\star \Sigma_{34}^{1/2}$$

turns (3.31) into the whitened loss $\mathcal{L} = \widetilde{\mathcal{L}}(\widetilde{A}, \widetilde{B}, C) := \frac{1}{2}\|\widetilde{A} C \widetilde{B}^\top - \widetilde{M}^\star\|_F^2$. This is a Euclidean description of the geometry of function space. It does not turn gradient flow in (A, B, C) into gradient flow in $(\widetilde{A}, \widetilde{B}, C)$, because the change of variables also changes the metric on the parameters: by the chain rule, gradient flow in (A, B, C) gives

$$\dot{\widetilde{A}} = -\Sigma_{12} \nabla_{\widetilde{A}} \widetilde{\mathcal{L}}, \qquad \dot{\widetilde{B}} = -\Sigma_{34} \nabla_{\widetilde{B}} \widetilde{\mathcal{L}}, \qquad \dot{C} = -\nabla_C \widetilde{\mathcal{L}}.$$

The anisotropy of the inputs therefore acts as a preconditioner of the factor dynamics, not merely as a relabelling of the target tensor.

3.4 A compact residual-moment formulation

The preceding formulas reduce to a single statement, which remains valid when the blocks are dependent. For an arbitrary joint law of the four blocks with $\mathbb{E}\|z \otimes w\|^2 < \infty$, define the residual-weighted moment matrix

$$R_\Delta := \mathbb{E}\big[e_\vartheta(X)\, zw^\top\big] = \mathbb{E}\big[z(z^\top \Delta w)w^\top\big] \in \mathbb{R}^{d_{12}\times d_{34}}, \tag{3.39}$$

which is the $\{1,2\}$-matricization of the residual-weighted moment tensor $\mathbb{E}\big[e_\vartheta(X)\, X^{(1)} \otimes X^{(2)} \otimes X^{(3)} \otimes X^{(4)}\big]$, as in Proposition 2.50. It is the gradient of the loss with respect to the represented matricization,

$$\nabla_M \mathcal{L} = R_\Delta, \tag{3.40}$$

since $D_M\mathcal{L}[H] = \mathbb{E}[e_\vartheta\, z^\top H w] = \langle R_\Delta, H\rangle_F$ for every perturbation H of M. As the first equalities in (3.17), (3.21) and (3.22) hold for any input law, the parameter gradients are

$$\nabla_A \mathcal{L} = R_\Delta BC^\top, \qquad \nabla_B \mathcal{L} = R_\Delta^\top AC, \qquad \nabla_C \mathcal{L} = A^\top R_\Delta B. \tag{3.41}$$

The data regimes considered above are special cases:

$$R_\Delta = \begin{cases} \Delta, & \text{independent whitened blocks,} \\ \Sigma_{12}\Delta\Sigma_{34}, & z \text{ and } w \text{ independent,} \\ \mathbb{E}\big[z(z^\top \Delta w)w^\top\big], & \text{arbitrary joint law.} \end{cases} \tag{3.42}$$

In the second case, $\Sigma_{12} = \mathbb{E}[zz^\top]$ and $\Sigma_{34} = \mathbb{E}[ww^\top]$ factor as $\Sigma_1 \otimes \Sigma_2$ and $\Sigma_3 \otimes \Sigma_4$ when the blocks on each side are also independent, and the weighted error (3.34) is $\mathcal{E}_\Sigma = -R_\Delta$. When the two sides of the cut are dependent, R_Δ generally does not factor through the second moments of the two sides.

Vectorized form. We vectorize matrices row by row, i.e. in the lexicographic order of Definition 2.5, so that $\mathrm{vec}(M(\vartheta)) = \mathrm{vec}(T(\vartheta))$ and $\mathrm{vec}(zw^\top) = z \otimes w = x^{(1)} \otimes x^{(2)} \otimes x^{(3)} \otimes x^{(4)}$. Let

$$\Sigma_{[4]} := \mathbb{E}\big[(z \otimes w)(z \otimes w)^\top\big] \tag{3.43}$$

be the Gram matrix of the full product of the inputs of the legs, i.e. $\Sigma_{[N]}$ in (5.2) with $N = 4$. Since $e_\vartheta(x) = \langle \Delta, zw^\top\rangle_F = \mathrm{vec}(\Delta)^\top(z \otimes w)$,

$$\mathrm{vec}(R_\Delta) = \Sigma_{[4]}\,\mathrm{vec}(\Delta), \qquad \mathcal{L} = \tfrac{1}{2}\,\mathrm{vec}(\Delta)^\top \Sigma_{[4]}\,\mathrm{vec}(\Delta), \tag{3.44}$$

the second identity being (5.7) applied to $T(\vartheta) - T^\star$. If z and w are independent,

$$\Sigma_{[4]} = \Sigma_{12} \otimes \Sigma_{34}, \tag{3.45}$$

which equals $\Sigma_1 \otimes \Sigma_2 \otimes \Sigma_3 \otimes \Sigma_4$ under the product law, i.e. for independent blocks, and the identity for independent whitened blocks. If the four blocks are jointly Gaussian but dependent, $\Sigma_{[4]}$ is still explicit by Wick's theorem, but it is not, in general, a Kronecker product of the leaf Gram matrices Σ_a: it contains the cross-block contractions that are absent under the product law.

3.5 Differential, tangent-kernel and represented-tensor forms

Let $\Phi(A,B,C) := ACB^\top$ be the map from the parameters to the represented matricization. Its differential is

$$D\Phi(A,B,C)[\delta A,\delta B,\delta C] = (\delta A)CB^\top + AC(\delta B)^\top + A(\delta C)B^\top, \tag{3.46}$$

and, with the Frobenius inner product on each parameter block, its adjoint is

$$D\Phi(A,B,C)^*[X] = \big(XBC^\top,\ X^\top AC,\ A^\top XB\big), \tag{3.47}$$

whose three entries correspond to A, B and C. With $D\Phi$ evaluated at ϑ, the whole gradient computation is the chain rule

$$\nabla_\vartheta \mathcal{L} = D\Phi^*[R_\Delta], \qquad \dot\vartheta = -D\Phi^*[R_\Delta]. \tag{3.48}$$

Represented matricization. The induced evolution of $M = \Phi(A,B,C)$ is

$$\dot M = -D\Phi\, D\Phi^*[R_\Delta] = -\mathcal{K}_\vartheta[R_\Delta], \tag{3.49}$$

where

$$\mathcal{K}_\vartheta[X] := ACC^\top A^\top X + AA^\top XBB^\top + XBC^\top CB^\top \tag{3.50}$$

is the analogue for this TTN, in matricized form, of the operator $\mathcal{K}_\theta = DT_j\, DT_j^*$ of Proposition 2.50. It is positive semidefinite:

$$\langle X, \mathcal{K}_\vartheta[X]\rangle_F = \|XBC^\top\|_F^2 + \|X^\top AC\|_F^2 + \|A^\top XB\|_F^2 \geq 0. \tag{3.51}$$

It is also the tangent kernel of the TTN, expressed in coefficient space: since $\nabla_\vartheta g_\vartheta(x) = D\Phi^*[zw^\top]$, the empirical tangent kernel of Remark 1.4 on inputs $x_1,\dots,x_n$ has entries

$$(\tilde K_\vartheta)_{ij} = \tfrac{1}{n}\langle z_i w_i^\top, \mathcal{K}_\vartheta[z_j w_j^\top]\rangle_F,$$

where z_i and w_i are formed from x_i. Along the flow,

$$\frac{d}{dt}\mathcal{L} = -\|\nabla_A\mathcal{L}\|_F^2 - \|\nabla_B\mathcal{L}\|_F^2 - \|\nabla_C\mathcal{L}\|_F^2 = -\langle R_\Delta, \mathcal{K}_\vartheta[R_\Delta]\rangle_F \leq 0. \tag{3.52}$$

By (3.44), the population loss is a convex quadratic function of M for every input law. The dynamics of M is nevertheless nonlinear, and not closed: $\mathcal{K}_\vartheta$ depends on the factors A, B and C, not only on their product, and it changes along the trajectory.

Vectorized flow. Let $m := \mathrm{vec}(M(\vartheta)) = \mathrm{vec}(T(\vartheta))$ and $m^\star := \mathrm{vec}(M^\star) = \mathrm{vec}(T^\star)$, and let $J_\vartheta := D_\vartheta m$ be the Jacobian with respect to the entries of the three cores, so that $\mathrm{vec}(\mathcal{K}_\vartheta[X]) = J_\vartheta J_\vartheta^\top \mathrm{vec}(X)$. By (3.44),

$$\dot m = -J_\vartheta J_\vartheta^\top \Sigma_{[4]}(m - m^\star). \tag{3.53}$$

For independent whitened blocks this reduces to

$$\dot m = -J_\vartheta J_\vartheta^\top (m - m^\star), \tag{3.54}$$

and when z and w are independent, $\Sigma_{[4]} = \Sigma_{12} \otimes \Sigma_{34}$. Equation (3.53) separates three objects:

$$\underbrace{J_\vartheta J_\vartheta^\top}_{\text{evolving tangent geometry of the TTN}} \qquad \underbrace{\Sigma_{[4]}}_{\text{geometry of the data}} \qquad \underbrace{m - m^\star}_{\text{coefficient error}}.$$

Since $m = \mathrm{vec}(T(\vartheta))$, (3.53) is the depth-two instance of the coefficient-space evolution

$$\dot{T} = -DT(\vartheta)\, DT(\vartheta)^* \nabla_T \mathcal{L},$$

which holds for any TTN and which Proposition 2.50 states for the TTN surrogate of a DNN.

Proposition 3.6 (Balance invariants)**.** For any input law for which the preceding gradients exist, gradient flow preserves

$$A^\top A - CC^\top \qquad \text{and} \qquad B^\top B - C^\top C. \tag{3.55}$$

Proof. Write $R := R_\Delta$. By (3.41), $\dot{A} = -RBC^\top$, $\dot{B} = -R^\top AC$ and $\dot{C} = -A^\top RB$. Therefore

$$\begin{aligned} \tfrac{d}{dt}(A^\top A) &= -CB^\top R^\top A - A^\top RBC^\top = \tfrac{d}{dt}(CC^\top), \\ \tfrac{d}{dt}(B^\top B) &= -C^\top A^\top RB - B^\top R^\top AC = \tfrac{d}{dt}(C^\top C), \end{aligned}$$

and subtracting proves (3.55). □

These invariants are the conserved quantities Q_e of (7.1) for the left and right bonds. They express the gauge invariance of the loss (Proposition 2.12): the gauge group $\mathrm{GL}(D_L) \times \mathrm{GL}(D_R)$ acts by $(A, C, B) \mapsto (AX,\, X^{-1}CY^{-\top},\, BY)$ and leaves $M(\vartheta)$ unchanged. They show that the distribution of scale among adjacent cores is not washed out by gradient flow: an initialization is specified not only by an overall scale, but also by a balance profile across the levels of the tree (Section 7.2.2).

Remark 3.7 (Empirical loss)**.** For a sample $x_1, \ldots, x_n$ with noiseless labels $y_s = f^\star(x_s)$, let z_s and w_s be the Kronecker products formed from x_s, and $e_s := e_\vartheta(x_s) = z_s^\top \Delta w_s$. The empirical loss $\mathcal{L}_n(\vartheta) := \frac{1}{2n}\sum_{s=1}^n e_s^2$ has the parameter gradients (3.41) with R_Δ replaced by

$$\widehat{R}_\Delta := \frac{1}{n}\sum_{s=1}^n e_s\, z_s w_s^\top.$$

For fixed ϑ and independent samples, $\mathbb{E}[\widehat{R}_\Delta] = R_\Delta$, which equals $\Sigma_{12}\Delta\Sigma_{34}$ when z and w are independent. At finite n, however, $\widehat{R}_\Delta$ contains cross-mode fluctuations and need not equal its population value.

3.6 Gradient flow on the balanced manifold

By Proposition 3.6, the set of *balanced* parameters, on which both conserved quantities vanish,

$$A^\top A = CC^\top, \qquad B^\top B = C^\top C, \tag{3.56}$$

is invariant under gradient flow, for every input law (cf. Du et al., 2018). These are matrix identities, not equalities of norms. They are the stationarity conditions of the parameter norm $\|A\|_F^2 + \|B\|_F^2 + \|C\|_F^2$ along the orbits of the gauge group, i.e. they characterize the balanced, or minimum-norm, gauge of Remark 2.15. Gradient flow preserves balance, and imbalance alike: it does not drive an unbalanced initialization towards balance. On the balanced manifold, the dynamics simplifies in two steps. Balance alone closes the evolution (3.49) of the represented matricization; balance together with an alignment of the student with the teacher decouples it into one scalar equation per singular value. As Remark 3.2 leads one to expect, both steps are the depth-three instances of known results for deep linear networks (Saxe et al., 2014; Arora et al., 2018).

Lemma 3.8 (Balanced cores)**.** Let ϑ satisfy (3.56), and let $k := \operatorname{rank} C$. There are $S = \operatorname{diag}(s_1, \dots, s_k)$ with $s_i > 0$, and matrices $U \in \mathbb{R}^{d_{12} \times k}$, $V \in \mathbb{R}^{d_{34} \times k}$, $W_L \in \mathbb{R}^{D_L \times k}$ and $W_R \in \mathbb{R}^{D_R \times k}$ with orthonormal columns, such that

$$A = USW_L^\top, \qquad C = W_L S W_R^\top, \qquad B = VSW_R^\top. \tag{3.57}$$

Consequently, $M(\vartheta) = US^3V^\top$ is a compact SVD, so that the nonzero singular values of M are $\sigma_i = s_i^3$, $i \in [k]$, and

$$AA^\top = (MM^\top)^{1/3}, \qquad BB^\top = (M^\top M)^{1/3}, \tag{3.58}$$

where powers of positive semidefinite matrices are taken in their eigenbases.

Proof. Let $C = W_L S W_R^\top$ be a compact SVD. By (3.56), $A^\top A = CC^\top = W_L S^2 W_L^\top$. Hence $U := AW_L S^{-1}$ has orthonormal columns, and $\|A(I - W_L W_L^\top)\|_F^2 = \operatorname{tr}\left((I - W_L W_L^\top) A^\top A (I - W_L W_L^\top)\right) = 0$, so that $A = AW_L W_L^\top = USW_L^\top$. The same argument, applied to $B^\top B = C^\top C = W_R S^2 W_R^\top$, gives $B = VSW_R^\top$ with $V := BW_R S^{-1}$. Then $M = ACB^\top = US^3V^\top$, $MM^\top = US^6U^\top$ and $AA^\top = US^2U^\top = (MM^\top)^{1/3}$, and similarly for B. □

Balance thus spreads each singular value of M evenly over the three cores, $s_i = \sigma_i^{1/3}$, and the cores carry the singular vectors of the central cut: U and V are the left and right singular vectors of M (Definition 2.13), while W_L and W_R are orthonormal frames in the two bonds, on which the orthogonal gauge transformations act.

Proposition 3.9 (Closed dynamics of the represented matricization)**.** If ϑ is balanced, then for every $X \in \mathbb{R}^{d_{12} \times d_{34}}$,

$$\mathcal{K}_\vartheta[X] = (MM^\top)^{2/3} X + (MM^\top)^{1/3} X (M^\top M)^{1/3} + X (M^\top M)^{2/3}. \tag{3.59}$$

Consequently, along gradient flow from a balanced initialization and for any input law,

$$\dot{M} = -(MM^\top)^{2/3} R_\Delta - (MM^\top)^{1/3} R_\Delta (M^\top M)^{1/3} - R_\Delta (M^\top M)^{2/3}, \tag{3.60}$$

and

$$\frac{d}{dt}\mathcal{L} = -\|(MM^\top)^{1/3} R_\Delta\|_F^2 - \|(MM^\top)^{1/6} R_\Delta (M^\top M)^{1/6}\|_F^2 - \|R_\Delta (M^\top M)^{1/3}\|_F^2. \tag{3.61}$$

Proof. By (3.56), $ACC^\top A^\top = A(A^\top A)A^\top = (AA^\top)^2$ and $BC^\top CB^\top = B(B^\top B)B^\top = (BB^\top)^2$. Substituting these identities and (3.58) into (3.50) gives (3.59), and (3.49) then gives (3.60), since balance is preserved. In (3.52), the same identities give $\|\nabla_A \mathcal{L}\|_F^2 = \|R_\Delta B C^\top\|_F^2 = \|R_\Delta B B^\top\|_F^2$, $\|\nabla_B \mathcal{L}\|_F^2 = \|R_\Delta^\top A C\|_F^2 = \|AA^\top R_\Delta\|_F^2$ and $\|\nabla_C \mathcal{L}\|_F^2 = \operatorname{tr}\left(R_\Delta^\top AA^\top R_\Delta BB^\top\right)$, which is (3.61). □

Balance therefore closes the dynamics: since R_Δ is a function of M (Section 3.4), (3.60) is an equation for M alone, and the cores are no longer needed to specify the evolution of the function. For independent whitened blocks, $R_\Delta = M - M^\star$ and (3.60) is the end-to-end equation of a balanced linear network of depth three (Arora et al., 2018); when z and w are independent, $R_\Delta = \Sigma_{12}(M - M^\star)\Sigma_{34}$. The fractional powers are not Lipschitz at rank-deficient M, and M has rank at most $\min(D_L, D_R)$, so (3.60) should be read along the balanced trajectory of the cores, which the polynomial gradient flow determines uniquely. By (3.61), small singular values of M slow down the decrease of the loss.

Spectral form. Let $(u_i)_{i\in[d_{12}]}$ and $(v_j)_{j\in[d_{34}]}$ be orthonormal bases of left and right singular vectors of M, with $M = \sum_{i\le k} \sigma_i u_i v_i^\top$, and set $\sigma_i := 0$ for $i > k$ in both bases. By (3.59), $\mathcal{K}_\vartheta$ is diagonal in the basis $(u_i v_j^\top)$ of $\mathbb{R}^{d_{12}\times d_{34}}$:

$$\mathcal{K}_\vartheta[u_i v_j^\top] = \kappa_{ij}\, u_i v_j^\top, \qquad \kappa_{ij} := \sigma_i^{4/3} + \sigma_i^{2/3}\sigma_j^{2/3} + \sigma_j^{4/3}, \tag{3.62}$$

so that $u_i^\top \dot{M} v_j = -\kappa_{ij}\, u_i^\top R_\Delta v_j$. On the balanced manifold, the tangent operator acts on the mode $u_i v_i^\top$ with eigenvalue $3\sigma_i^{4/3}$, and it vanishes on the directions $u_i v_j^\top$ orthogonal to both singular subspaces of M. For a simple nonzero singular value, Proposition 2.50 gives

$$\dot\sigma_i = u_i^\top \dot{M} v_i = -3\sigma_i^{4/3}\, u_i^\top R_\Delta v_i, \tag{3.63}$$

that is, $\dot\sigma_i = 3\sigma_i^{4/3}(u_i^\top M^\star v_i - \sigma_i)$ for independent whitened blocks. This equation is exact on the balanced manifold, but it is not closed in the singular values: the off-diagonal entries $u_i^\top R_\Delta v_j$, $i \neq j$, rotate the singular vectors and change their overlaps $u_i^\top M^\star v_i$ with the teacher.

Aligned modes. The rotation stops when the student and the teacher share their singular vectors, and these vectors also diagonalize the Gram matrices of the data.

Proposition 3.10 (Aligned modes)**.** Let z and w be independent, and let $U = (u_1, \dots, u_k)$ and $V = (v_1, \dots, v_k)$ have orthonormal columns with $\Sigma_{12} u_i = \lambda_i^L u_i$ and $\Sigma_{34} v_i = \lambda_i^R v_i$. Assume that the teacher is $M^\star = \sum_{i=1}^k \sigma_i^\star u_i v_i^\top$, with $\sigma_i^\star \ge 0$, and that the initialization is balanced of the form (3.57), with these U and V, frames W_L and W_R, and $S(0) = \mathrm{diag}(s_1(0), \dots, s_k(0))$, $s_i(0) \ge 0$. Then the cores keep the form (3.57), with fixed U, V, W_L and W_R, and with $S(t) = \mathrm{diag}(s_1(t), \dots, s_k(t))$ solving

$$\dot s_i = \lambda_i^L \lambda_i^R\, s_i^2(\sigma_i^\star - s_i^3), \qquad \text{equivalently} \qquad \dot\sigma_i = 3\lambda_i^L\lambda_i^R\, \sigma_i^{4/3}(\sigma_i^\star - \sigma_i), \quad \sigma_i = s_i^3. \tag{3.64}$$

For independent whitened blocks, $\lambda_i^L = \lambda_i^R = 1$.

Proof. For cores of the form (3.57), $M = US^3V^\top$ and, by (3.42), $R_\Delta = \Sigma_{12}(M - M^\star)\Sigma_{34} = U\Lambda(S^3 - S^\star)V^\top$, with $\Lambda := \mathrm{diag}(\lambda_i^L\lambda_i^R)$ and $S^\star := \mathrm{diag}(\sigma_i^\star)$. By (3.41),

$$\dot A = -R_\Delta B C^\top = U\dot S W_L^\top, \qquad \dot C = -A^\top R_\Delta B = W_L \dot S W_R^\top, \qquad \dot B = -R_\Delta^\top A C = V \dot S W_R^\top,$$

with $\dot S = \Lambda S^2(S^\star - S^3)$, which is (3.64). The cores of this form therefore solve the gradient flow, which, being polynomial, has a unique solution. □

Each aligned mode thus follows the dynamics of a balanced linear network of depth three (Saxe et al., 2014), in which the three cores of the mode share the singular value s_i. Four regimes follow from (3.64).

(i) *Escape from a small initialization.* While $s_i^3 \ll \sigma_i^\star$, $\dot s_i \approx \lambda_i^L\lambda_i^R\sigma_i^\star s_i^2$, so that $s_i(t) \approx s_i(0)/(1 - \lambda_i^L\lambda_i^R\sigma_i^\star s_i(0)\, t)$, and the mode leaves the neighbourhood of zero after a time of order $1/(\lambda_i^L\lambda_i^R\sigma_i^\star s_i(0))$. This is the case $|V| = 3$ of the scaling $\varsigma^{-(|V|-2)}$ of Section 7.2.2, and the time is inversely proportional to the moment singular value $\lambda_i^L\sigma_i^\star\lambda_i^R$, as in the two-leaf computation of Experiment E1.3. From a common small initial scale, the modes are therefore acquired in decreasing order of $\lambda_i^L\lambda_i^R\sigma_i^\star$.

(ii) *Convergence.* Near $\sigma_i = \sigma_i^\star > 0$, the error $\sigma_i^\star - \sigma_i$ decays exponentially, at the rate $3\lambda_i^L\lambda_i^R\sigma_i^{\star 4/3}$.

(iii) *Extra modes.* A student mode without a teacher counterpart, $\sigma_i^\star = 0$, decays algebraically: $\sigma_i(t) = \left(\sigma_i(0)^{-4/3} + 4\lambda_i^L \lambda_i^R t\right)^{-3/4}$. Outside the aligned manifold, such a mode may instead rotate towards a teacher mode.

(iv) *Absent modes.* A mode with $s_i(0) = 0$ remains at zero.

Without common eigenvectors, Σ_{12} and Σ_{34} mix the modes, and the decoupling is lost; Section 4 studies the dynamics without alignment.

Remark 3.11 (Scope of the balanced reduction)**.**

(i) *Balance is a property of the initialization.* Small initializations are only approximately balanced: their invariants (3.55) are of the order of the square of the initial scale, which becomes negligible relative to the cores once these have grown.

(ii) *Exact balance freezes the bond rank.* Under (3.56), the row space of A and the column space of C coincide. If Π is the orthogonal projector onto this space at $t = 0$, then $X := A(I - \Pi)$ and $Y := (I - \Pi)C$ satisfy the linear equations $\dot{X} = -R_\Delta B Y^\top$ and $\dot{Y} = -X^\top R_\Delta B$, and vanish at $t = 0$, hence at all times. Therefore $\operatorname{rank} M(t) \le \operatorname{rank} C(0)$: on the balanced manifold, modes absent at initialization are never acquired, in accordance with the vanishing of $\mathcal{K}_\vartheta$ on the directions orthogonal to both singular subspaces of M. A balanced initialization must therefore have rank at least that of the teacher, whereas from generic small initializations, which have full bond rank, new modes appear in the course of training.

(iii) *Tree-specific dynamics.* For this tree, the balanced dynamics is that of a balanced linear network of depth three (Remark 3.2). Balanced dynamics specific to trees arises when the lower cores are themselves factorized (Remark 3.1), so that several non-equivalent cuts, each with its own balance conditions, are coupled.

4 Balanced dynamics without alignment

Proposition 3.10 solves the balanced dynamics of the four-leaf model of Section 3 when the student starts aligned with the teacher. From a generic initialization, the singular vectors of the student rotate while its singular values grow, and each process changes the signal that drives the other. This section studies the balanced flow (3.60) without the alignment assumption, for independent whitened blocks, so that $R_\Delta = M - M^\star$, unless stated otherwise. We do not know a closed-form solution in general, and Section 4.1 explains what stands in the way. The flow is solved exactly for a student of bond rank one (Section 4.2), which describes the escape from the origin, and the critical points are characterized by their linear rates (Section 4.3). Approximations then describe the later plateaus (Section 4.4) and the end of training (Section 4.5). Sections 4.6 and 4.7 treat non-whitened inputs and other trees, and Section 4.8 collects the time scales.

Throughout, $M^\star = \sum_{q=1}^{r^\star} \sigma_q^\star u_q^\star v_q^{\star\top}$ with $\sigma_1^\star > \cdots > \sigma_{r^\star}^\star > 0$, and we use the notation of Section 3.6: on the balanced manifold, $M = US^3V^\top = \sum_{i \le k} \sigma_i u_i v_i^\top$ with $\sigma_i = s_i^3$, and $\kappa_{ij} = \sigma_i^{4/3} + \sigma_i^{2/3}\sigma_j^{2/3} + \sigma_j^{4/3}$. We distinguish exact statements, which are proved, from leading-order asymptotics and from approximations, which are stated as such. Numerical integrations of the gradient flow of the cores, and symbolic computations, serve as consistency checks of the statements below; they are not part of the proofs.

4.1 Spectral form of the balanced flow

Let $\Theta := U^\top M^\star V \in \mathbb{R}^{k \times k}$ be the teacher written in the singular bases of the student. Its diagonal entries $\Theta_{ii} = u_i^\top M^\star v_i$ are the signals seen by the modes of the student, and its off-diagonal entries

measure how far the pairing of the left and right singular vectors of the student is from that of the teacher.

Proposition 4.1 (Spectral equations)**.**

(a) On a time interval on which a balanced trajectory has constant rank k and simple nonzero singular values, let $\Omega_U := U^\top \dot{U}$ and $\Omega_V := V^\top \dot{V}$. Then

$$\dot{\sigma}_i = 3\sigma_i^{4/3}(\Theta_{ii} - \sigma_i), \tag{4.1}$$

$$(\Omega_U)_{ij} = \kappa_{ij}\,\frac{\sigma_j\Theta_{ij} + \sigma_i\Theta_{ji}}{\sigma_j^2 - \sigma_i^2}, \qquad (\Omega_V)_{ij} = \kappa_{ij}\,\frac{\sigma_i\Theta_{ij} + \sigma_j\Theta_{ji}}{\sigma_j^2 - \sigma_i^2} \qquad (i \neq j), \tag{4.2}$$

$$(I - UU^\top)\dot{U} = (I - UU^\top)M^\star VS, \qquad (I - VV^\top)\dot{V} = (I - VV^\top)M^{\star\top}US. \tag{4.3}$$

For any input law, with $G := U^\top R_\Delta V$, the same computation gives $\dot{\sigma}_i = -3\sigma_i^{4/3}G_{ii}$, $(\Omega_U)_{ij} = -\kappa_{ij}(\sigma_j G_{ij} + \sigma_i G_{ji})/(\sigma_j^2 - \sigma_i^2)$, $(\Omega_V)_{ij} = -\kappa_{ij}(\sigma_i G_{ij} + \sigma_j G_{ji})/(\sigma_j^2 - \sigma_i^2)$, $(I - UU^\top)\dot{U} = -(I - UU^\top)R_\Delta VS$ and $(I - VV^\top)\dot{V} = -(I - VV^\top)R_\Delta^\top US$.

(b) For whitened blocks and any balanced initialization, the column and row spaces of $M(t)$ stay in range $M(0)$ + range $M^\star$ and in range $M(0)^\top$ + range $M^{\star\top}$, respectively. The dynamics therefore takes place in dimension at most $(k_0 + r^\star) \times (k_0 + r^\star)$, with $k_0 := \operatorname{rank} M(0)$, whatever d_{12} and d_{34}.

Proof. (a) By (3.62), $U^\top \dot{M} V = -\kappa \circ G$, with $\circ$ the entrywise product, and $G = S^3 - \Theta$ for whitened blocks. Differentiating $M = US^3V^\top$ gives $U^\top \dot{M} V = \Omega_U S^3 + \frac{d}{dt}S^3 - S^3\Omega_V$. Its diagonal gives (4.1), and its entries (i,j) and (j,i) form a 2×2 linear system for $(\Omega_U)_{ij}$ and $(\Omega_V)_{ij}$, whose solution is (4.2). Similarly, $(I - UU^\top)\dot{U} = (I - UU^\top)\dot{M}VS^{-3}$, and of the three terms of (3.60) only $-R_\Delta(M^\top M)^{2/3}$ survives the projection; since $(M^\top M)^{2/3}V = VS^4$ and, for whitened blocks, $(I - UU^\top)R_\Delta = -(I - UU^\top)M^\star$, this gives (4.3). The right singular vectors are treated in the same way. (b) Let P be the orthogonal projector onto the orthogonal complement of range $M(0)$+range $M^\star$. Then $PM^\star = 0$, and the flow of the cores gives $\frac{d}{dt}(PA) = -P(M - M^\star)BC^\top = -(PA)CB^\top BC^\top$, a linear equation for PA. On the balanced manifold range A = range M (Lemma 3.8), so that $PA(0) = 0$; hence $PA = 0$ and $PM = 0$ at all times. The row spaces are treated in the same way. □

Equation (4.1) has the form of the aligned equation (3.64), with the teacher singular value replaced by the time-dependent signal Θ_{ii}: amplification is driven by alignment. Conversely, (4.2) and (4.3) show that alignment is driven by amplification: the off-diagonal entries of Θ rotate the modes within the current singular subspaces, and the part of the teacher outside these subspaces rotates them out of them, at the rate $s_i = \sigma_i^{1/3}$ of the mode. We do not know a closed-form solution for $k \geq 2$, and three features stand in the way. First, every mode has its own clock: the relative growth $\dot{\sigma}_i/\sigma_i$ and the rotation of mode i both scale as s_i, so that no single change of time normalizes all modes, unlike the rank-one case below. Second, the in-span rotations (4.2) couple the clocks through the ratios $\kappa_{ij}/(\sigma_j^2 - \sigma_i^2)$, which are singular at crossings of singular values, where individual singular vectors are not defined and only singular subspaces can be followed (Remark 5.13). Third, because of the fractional powers in (3.60), the flow does not inherit the matrix Riccati structure that makes balanced linear networks with two factors solvable from arbitrary balanced initializations with whitened inputs (Braun et al., 2022).

4.2 Rank-one channels: an exact solution

Let the student have bond rank one, $k = 1$. By Remark 3.11(ii) it keeps this rank: $M(t) = s(t)^3 u(t) v(t)^\top$ with unit vectors u and v, and the cores are $A = suw_L^\top$, $C = sw_L w_R^\top$ and $B = svw_R^\top$,

whose frames w_L and w_R do not move, since $\dot{A} = -R_\Delta B C^\top$ is proportional to $w_L^\top$, and similarly for B and C. Let $\rho := u^\top M^\star v$ be the signal seen by the channel, $\sigma := s^3$ its amplitude, and $p_q^L := u(0)^\top u_q^\star$ and $p_q^R := v(0)^\top v_q^\star$ its initial overlaps with the teacher modes.

Proposition 4.2 (Exact rank-one dynamics)**.** Let $\tau(t) := \int_0^t s$, let $L := \begin{pmatrix} 0 & M^\star \\ M^{\star\top} & 0 \end{pmatrix}$ be the symmetric dilation of $M^\star$, and let $y := (u, v)/\sqrt{2}$, a unit vector. Then

$$\frac{du}{d\tau} = (I - uu^\top)M^\star v, \qquad \frac{dv}{d\tau} = (I - vv^\top)M^{\star\top} u, \qquad \frac{d\sigma}{d\tau} = 3\sigma(\rho - \sigma). \tag{4.4}$$

The orientation does not depend on the amplitude, and it is given by

$$y(\tau) = \frac{e^{\tau L} y(0)}{\|e^{\tau L} y(0)\|}, \qquad \rho(\tau) = \frac{1}{2}\frac{d}{d\tau}\log Z(\tau), \qquad Z(\tau) := \|e^{\tau L} y(0)\|^2. \tag{4.5}$$

The amplitude and the physical time then follow by quadrature:

$$\sigma(\tau) = \frac{Z(\tau)^{3/2}}{\sigma(0)^{-1} + 3\int_0^\tau Z^{3/2}}, \qquad t(\tau) = \int_0^\tau Z(q)^{-1/2}\Big(\sigma(0)^{-1} + 3\int_0^q Z^{3/2}\Big)^{1/3} dq. \tag{4.6}$$

In the singular bases of $M^\star$,

$$Z(\tau) = \sum_{q=1}^{r^\star} (\alpha_q^2 e^{2\sigma_q^\star \tau} + \beta_q^2 e^{-2\sigma_q^\star \tau}) + \gamma^2, \qquad \alpha_q := \tfrac{1}{2}(p_q^L + p_q^R), \quad \beta_q := \tfrac{1}{2}(p_q^L - p_q^R), \tag{4.7}$$

where $\gamma^2 := 1 - \sum_q (\alpha_q^2 + \beta_q^2)$ is the weight of $y(0)$ on the kernel of L, and the overlaps of the channel with the teacher modes at time τ are

$$\tfrac{1}{2}(u^\top u_q^\star + v^\top v_q^\star) = \alpha_q\, e^{\sigma_q^\star \tau} Z^{-1/2}, \qquad \tfrac{1}{2}(u^\top u_q^\star - v^\top v_q^\star) = \beta_q\, e^{-\sigma_q^\star \tau} Z^{-1/2}. \tag{4.8}$$

Proof. For $k = 1$, (4.1) and (4.3) read $\dot\sigma = 3\sigma^{4/3}(\rho - \sigma)$, $\dot u = s(I - uu^\top)M^\star v$ and $\dot v = s(I - vv^\top)M^{\star\top} u$; dividing by $\dot\tau = s = \sigma^{1/3}$ gives (4.4). Since $y^\top L y = u^\top M^\star v = \rho$, the orientation equations read $y' = Ly - (y^\top L y)y$, which is Oja's flow (Oja, 1982). If $z := e^{\tau L} y(0)$, then $(z/\|z\|)' = Lz/\|z\| - (z^\top L z) z/\|z\|^3$, so that $z/\|z\|$ solves the same equation with the same initial value (cf. Yan et al., 1994), and $Z' = 2z^\top L z = 2Z\rho$. The amplitude equation is a Bernoulli equation: $1/\sigma$ solves $(1/\sigma)' = 3 - 3\rho/\sigma$, whose solution is the first identity of (4.6), since $\exp(3\int_0^\tau \rho) = Z(\tau)^{3/2}$; and $dt = \sigma^{-1/3} d\tau$ gives the second. Finally, L has the eigenvalues $\pm\sigma_q^\star$ with eigenvectors $(u_q^\star, \pm v_q^\star)/\sqrt{2}$, and the eigenvalue 0 on the remaining space; expanding $y(0)$ in this basis gives (4.7) and (4.8). □

The non-aligned rank-one problem is therefore solved: in the rescaled time τ, the direction of the channel performs a normalized power iteration on the dilation of the teacher, and the amplitude follows by two quadratures. The change of time $\tau = \int s$ is what decouples the two processes, which in physical time run on the same clock s. Proposition 3.10 is the case $y(0) = (u_1^\star, v_1^\star)/\sqrt{2}$, for which $Z = e^{2\sigma_1^\star \tau}$. The left-right asymmetries β_q lie in the negative eigenspaces of L and decay, while the symmetric overlaps α_q grow at the rates $\sigma_q^\star$, with $\alpha_q(\tau)/\alpha_{q'}(\tau) = e^{(\sigma_q^\star - \sigma_{q'}^\star)\tau}\alpha_q(0)/\alpha_{q'}(0)$. The channel therefore converges to the teacher mode with the largest $\sigma_q^\star$ among those with $\alpha_q \neq 0$. If all α_q vanish, $\rho \leq 0$ at all times and the channel does not escape.

Corollary 4.3 (Escape from the origin)**.** Assume $\alpha_1 \neq 0$, fix $\eta \in (0,1)$, and let t_η be the first time at which $\sigma(t) = \eta\sigma_1^\star$. As $s(0) \to 0$ with $y(0)$ fixed,

$$t_\eta = \frac{I}{s(0)} + O(1), \qquad I := \int_0^\infty Z(\tau)^{-1/2}\, d\tau, \tag{4.9}$$

where the $O(1)$ term depends on η, on the spectrum and on $y(0)$, and is not uniform as $\alpha_1 \to 0$ or $\eta \to 1$. Moreover,

$$\frac{1}{\sigma_1^\star} \leq I \leq \frac{\operatorname{arsinh}(\gamma/|\alpha_1|)}{\sigma_1^\star \gamma} \leq \frac{1}{\sigma_1^\star |\alpha_1|}, \tag{4.10}$$

where the middle bound is read as its limit $1/(\sigma_1^\star|\alpha_1|)$ when $\gamma = 0$, and equality holds on the left if and only if $y(0) = \pm(u_1^\star, v_1^\star)/\sqrt{2}$.

Proof. Let $\epsilon := s(0)^3$ and $K(\tau) := 3\int_0^\tau Z^{3/2}$, so that

$$\sigma = \frac{\epsilon Z^{3/2}}{1+\epsilon K}, \qquad t(\tau) = \epsilon^{-1/3} \int_0^\tau Z^{-1/2}(1+\epsilon K)^{1/3}.$$

Since $Z \sim \alpha_1^2 e^{2\sigma_1^\star \tau}$ and $K \sim Z^{3/2}/\sigma_1^\star$ as $\tau \to \infty$, the level $\eta\sigma_1^\star$ is reached at a rescaled time τ_η with $\epsilon Z(\tau_\eta)^{3/2} \to \eta\sigma_1^\star/(1-\eta)$, so that $e^{-\sigma_1^\star \tau_\eta} = O(\epsilon^{1/3})$. Then $t_\eta = \epsilon^{-1/3}(I - \int_{\tau_\eta}^\infty Z^{-1/2}) + \epsilon^{-1/3}\int_0^{\tau_\eta} Z^{-1/2}((1+\epsilon K)^{1/3} - 1)$. The tail is at most $\epsilon^{-1/3} e^{-\sigma_1^\star\tau_\eta}/(\sigma_1^\star|\alpha_1|) = O(1)$. In the last integral, $(1+\epsilon K)^{1/3} - 1 \leq \min\{\epsilon K/3, (\epsilon K)^{1/3}\}$, and $Z^{-1/2}K \lesssim Z$ and $Z^{-1/2}K^{1/3} = O(1)$ for large τ; the integral is therefore $O(\epsilon^{1/3})$, which contributes $O(1)$ to t_η. The bounds follow from $\alpha_1^2 e^{2\sigma_1^\star\tau} + \gamma^2 \leq Z(\tau) \leq e^{2\sigma_1^\star\tau}$ and $\int_0^\infty (\gamma^2 + a^2 e^{2\sigma\tau})^{-1/2} d\tau = \operatorname{arsinh}(\gamma/a)/(\sigma\gamma)$, and the upper bound on Z is an equality if and only if $y(0)$ lies in the eigenspace of L for $\sigma_1^\star$. □

Escape time and initial geometry. To leading order, the escape time is thus the aligned escape time $1/(\sigma_1^\star s(0))$ of Proposition 3.10, multiplied by the geometric factor $\sigma_1^\star I \geq 1$, which depends only on the initial direction of the channel and on the spectrum of the teacher: misalignment always slows the escape. Consider a random initialization, with $u(0)$ and $v(0)$ uniform on the unit spheres, in the regime of a fixed teacher rank and spectral gap and of dimensions d_{12} and d_{34} of comparable sizes that tend to infinity. Then α_q^2 and β_q^2 are of order $1/d_{12} + 1/d_{34}$ in probability, and $\gamma^2 = 1 - O(r^\star/\min(d_{12}, d_{34}))$. The integrand of I stays close to one until the growing terms of (4.7) reach order one, which typically happens first for $q = 1$, at $\tau \approx \log(1/\alpha_1^2)/(2\sigma_1^\star)$, and the bound (4.10) gives $\sigma_1^\star I \leq \frac{1}{2}\log(1/\alpha_1^2) + \log 2 + o(1)$. Hence, in probability,

$$t_\eta = \frac{\log(1/\alpha_1^2) + O(1)}{2\sigma_1^\star s(0)}, \qquad \log\frac{1}{\alpha_1^2} = \log\frac{d_{12}d_{34}}{d_{12}+d_{34}} + O(1). \tag{4.11}$$

For $\sigma^\star = (2, 1.4, 0.8)$ and $d_{12} = d_{34} \in \{10^2, 10^3, 10^4\}$, over several hundred random starts for each dimension, the median of $\sigma_1^\star I - \frac{1}{2}\log(1/\alpha_1^2)$ lies between 0.6 and 0.7, its 5% and 95% quantiles lie in $[-0.6, 0.75]$, and it never exceeds $\log 2 + 0.05$. The lower tail comes from starts in which a weaker mode has a much larger overlap and drives the escape. A random start therefore lengthens the plateau by a factor of order $\log d$, which is the time spent searching for the direction of the teacher. This search proceeds at the rate $\sigma_1^\star$ of the leading mode, not at the spectral gap, because the part of $y(0)$ on the kernel of L, which dominates in high dimension, does not grow.

Alignment and amplification. In τ, alignment and amplification are exactly decoupled; in physical time, both run on the clock s. Let $w_1(\tau) := \alpha_1^2 e^{2\sigma_1^\star \tau}/Z(\tau)$ be the weight of the leading mode in the direction of the channel; it is nondecreasing, since $(\log w_1)' = 2(\sigma_1^\star - \rho) \geq 0$. Let τ_ε be the rescaled time at which $w_1 = 1-\varepsilon$, $t_{\text{align}} := t(\tau_\varepsilon)$ and $s_{\text{align}} := s(t_{\text{align}})$. For fixed ε, η and $y(0)$ with $\alpha_1 \neq 0$, as $s(0) \to 0$,

$$t_\eta - t_{\text{align}} = \frac{c_\varepsilon}{\sigma_1^\star s_{\text{align}}} + O(1), \qquad 1 \leq c_\varepsilon \leq (1-\varepsilon)^{-1/2} : \tag{4.12}$$

after alignment, the channel escapes as an aligned mode would from its current amplitude. Indeed, $s_{\text{align}} = s(0) Z(\tau_\varepsilon)^{1/2}(1+o(1))$, and for $\tau \geq \tau_\varepsilon$ the monotonicity of w_1 gives $\alpha_1^2 e^{2\sigma_1^\star \tau} \leq Z \leq \alpha_1^2 e^{2\sigma_1^\star \tau}/(1-\varepsilon)$, which bounds $\int_{\tau_\varepsilon}^\infty Z^{-1/2}$ between $Z(\tau_\varepsilon)^{-1/2}/\sigma_1^\star$ and $(1-\varepsilon)^{-1/2}$ times this value; the rest of the argument is that of Corollary 4.3. For a random start in high dimension, Z stays close to γ^2 during the search, so that $s_{\text{align}} \approx s(0)\gamma/\sqrt{\varepsilon}$: the direction is acquired during the plateau, at small amplitude, and the amplification becomes visible only at its end.

The condition $s(0) \ll \sigma_1^{\star 1/3} Z(\tau_\varepsilon)^{-1/2}$, which states that the channel is still small when it has aligned, is sufficient for alignment to precede amplification; in general, the order of the two events is decided by comparing $t(\tau_\varepsilon)$ with t_η in (4.6). When the channel escapes before it aligns, it amplifies a mixture of teacher modes, with σ following $\rho(\tau)$, and completes its alignment after the escape, at the faster rate $s \approx \rho^{1/3}$. If moreover $|\alpha_2| \gg |\alpha_1|$, the channel first approaches the strict saddle $\sigma_2^\star u_2^\star v_2^{\star\top}$ (Section 4.3), and the leading mode overtakes at $\tau_\times = \log|\alpha_2/\alpha_1|/(\sigma_1^\star - \sigma_2^\star)$. The time spent near this saddle, of order $(\tau_\times - \tau_{\text{esc}})_+/\sigma_2^{\star 1/3}$ with τ_{esc} the rescaled time of the escape, is logarithmic in the ratio of the overlaps. For fixed $y(0)$ it vanishes as $s(0) \to 0$, since τ_{esc} grows like $\log(1/s(0))$: from a small enough initialization, the channel aligns before it escapes. There is therefore no universal ordering of alignment and amplification; it depends on the initial scale through τ_{esc}, and on the gaps and the overlaps through τ_ε.

4.3 Local geometry of the critical points

On the balanced manifold, by (3.62), $\dot{M} = 0$ requires $u_i^\top (M^\star - M) v_j = 0$ whenever $\sigma_i > 0$ or $\sigma_j > 0$, i.e. $(M^\star - M)V = 0$ and $U^\top(M^\star - M) = 0$. Hence $M^\star v_i = \sigma_i u_i$ and $M^{\star\top} u_i = \sigma_i v_i$: every active triplet (u_i, v_i, σ_i) is a singular triplet of $M^\star$. For a simple teacher spectrum, the critical points of the balanced flow are therefore the partial sums $M_S := \sum_{q\in S} \sigma_q^\star u_q^\star v_q^{\star\top}$, $S \subseteq [r^\star]$, $|S| \leq \min(D_L, D_R)$; repeated teacher singular values give continuous families, parameterized by subspaces of the corresponding singular spaces.

Proposition 4.4 (Linear rates at the critical points)**.** Let $\kappa_{ij}^\star := \sigma_i^{\star 4/3} + (\sigma_i^\star \sigma_j^\star)^{2/3} + \sigma_j^{\star 4/3}$. Linearizing the spectral equations of Proposition 4.1 at M_S, along balanced perturbations that keep the rank $|S|$, gives the following rates:

(i) $-3\sigma_i^{\star 4/3}$ for $i \in S$: the amplitude of a learned mode;
(ii) $-\kappa_{ij}^\star$, twice, for $i < j$ in S: rotations within the learned singular subspaces;
(iii) $\sigma_i^{\star 1/3}(\sigma_j^\star - \sigma_i^\star)$ and $-\sigma_i^{\star 1/3}(\sigma_i^\star + \sigma_j^\star)$ for $i \in S$ and $j \in [r^\star] \setminus S$: rotations of a learned mode towards an unlearned one, by the same or by opposite angles on the left and on the right;
(iv) $-\sigma_i^{\star 4/3}$ for $i \in S$, with multiplicity $(d_{12} - r^\star) + (d_{34} - r^\star)$: rotations towards the kernels of $M^{\star\top}$ and $M^\star$.

The rates (i), (iii) and (iv) are also eigenvalues of the linearized gradient flow of the cores, i.e. of minus the Hessian of the loss at balanced cores of bond rank $|S|$ representing M_S.

Proof. (i) is the derivative of $s^2(\sigma_i^\star - s^3)$ at $s^3 = \sigma_i^\star$. For a rotation $u_i \mapsto u_i + \varepsilon_L u_j^\star$, $v_i \mapsto v_i + \varepsilon_R v_j^\star$ with $j \notin S$, (4.3) gives $\dot{\varepsilon}_L = s_i(\sigma_j^\star \varepsilon_R - \sigma_i^\star \varepsilon_L)$ and $\dot{\varepsilon}_R = s_i(\sigma_j^\star \varepsilon_L - \sigma_i^\star \varepsilon_R)$ with $s_i = \sigma_i^{\star 1/3}$, whose

eigenvalues are (iii); (iv) is the case $\sigma_j^\star = 0$, in which the left and right rotations decouple. (ii) follows from (4.14) below at $\sigma = \sigma^\star$, where both combinations decay at $\kappa_{ij}^\star$. In the cores, a common rescaling of the three cores of mode i, and the rotations of the leg-side singular vectors u_i of A and v_i of B that leave C fixed, span subspaces that are invariant under the linearized flow, because $R_\Delta = M_S - M^\star$ satisfies $R_\Delta v_i = 0$ and $u_i^\top R_\Delta = 0$; on them, the linearized flow has the rates (i), (iii) and (iv). □

In the examples that we have computed numerically, a rank-three teacher and $S = \{2\}$, $\{1,3\}$ and $\{2,3\}$, the rates (i)–(iv) are exactly the nonzero eigenvalues of the Hessian in the cores, and the remaining eigenvalues vanish, corresponding to gauge transformations, to changes of the conserved quantities (3.55), and to the creation of new modes, which enters $\dot{M}$ only quadratically by (3.62). We have not proved that this holds in general.

The proposition separates two kinds of instability. A learned mode i that is weaker than an unlearned teacher mode j gives a hyperbolic instability, with rate $\lambda_{ij} := \sigma_i^{\star 1/3}(\sigma_j^\star - \sigma_i^\star)$: a perturbation of size δ along it grows to order one in a time of order $\lambda_{ij}^{-1}\log(1/\delta)$, logarithmic in the perturbation, and for a rank-one channel the escape replaces the learned mode by the stronger one (Proposition 4.2). The local geometry of the saddle, through the gap $\sigma_j^\star - \sigma_i^\star$ and the clock $\sigma_i^{\star 1/3}$ of the learned mode, sets this escape time. Along the creation of a new mode, by contrast, the flow is degenerate and the escape is algebraic: a dormant channel of amplitude s needs a time of order I/s (Corollary 4.3), set by its amplitude and by its geometry I, not by a curvature. The origin, $S = \emptyset$, is degenerate in every direction. When every mode aligns before it escapes, as from a small enough initialization (Section 4.4), the visited critical points are $M_{[r]}$, $r = 0, 1, \dots$, which have no hyperbolic unstable direction: the plateaus are algebraic, of order $1/s(0)$, as in the saddle-to-saddle dynamics of deep linear networks (Saxe et al., 2014; Jacot et al., 2021). Hyperbolic saddles are visited only transiently, after a channel has escaped along a weaker mode, and the detour is logarithmic in the overlaps.

4.4 Plateaus of students of higher rank

Scale invariance of the early phase. While M is small, $R_\Delta = M - M^\star \approx -M^\star$, and the balanced flow reduces to $\dot{M} = \mathcal{K}_\vartheta[M^\star]$, i.e. to gradient ascent of the trilinear form $\langle M^\star, ACB^\top\rangle_F$ in the cores. This truncated equation is invariant under $(A, B, C, t) \mapsto (\lambda A, \lambda B, \lambda C, t/\lambda)$. For initializations $\vartheta(0) = s(0)\hat{\vartheta}$ with a fixed shape $\hat{\vartheta}$ from which the truncated flow escapes, one therefore expects the first mode to escape at $t_1 = s(0)^{-1}(\hat{I} + o(1))$, with $\hat{I}$ depending only on $\hat{\vartheta}$ and $M^\star$. For $k = 1$, this is Corollary 4.3, with $\hat{I} = I$. For $k \geq 2$, the modes interact through (4.2), $\hat{I}$ has no closed form, and we have not proved the expansion; for a rank-two student, $s(0)\, t_1$ changes by less than 0.4% between $s(0) = 10^{-2}$ and 10^{-3}.

Deflated normal form. After the first r modes have been acquired, they converge in a time of order one, while the next mode, still dormant, evolves on its slow clock s_{r+1}. For $i \leq r < j$, (4.2) gives $(\Omega_U)_{ij} = -s_i\Theta_{ji}$ and $(\Omega_V)_{ij} = -s_i\Theta_{ij}$, up to relative corrections of order $(\sigma_j/\sigma_i)^{2/3}$. The dormant singular vectors are thus dragged by the active ones and remain orthogonal to them, while their own motion is (4.3), restricted to the complement of the active subspaces. This suggests that the dormant channel behaves as a rank-one channel in the sense of Proposition 4.2, with $M^\star$ replaced by the deflated teacher

$$Y_r := (I - U_r U_r^\top) M^\star (I - V_r V_r^\top) \;\longrightarrow\; \sum_{q>r} \sigma_q^\star u_q^\star v_q^{\star\top}, \tag{4.13}$$

where U_r and V_r hold the active singular vectors, and the limit holds once these have aligned. The plateau that follows the acquisition of the r-th mode would then last about $I(Y_r)/s_{r+1}$, where $I(Y_r)$ is computed by (4.7) from Y_r and from the direction of the dormant channel at the beginning of the plateau, and s_{r+1} is its amplitude then. This is an approximation, which we have not proved. It requires (a) separated amplitudes, $\sigma_{r+1} \ll \sigma_r$; (b) active modes that have relaxed to teacher modes up to small angles; (c) a target Y_r that varies slowly compared with the alignment of the dormant channel, whose rate is s_{r+1} times the spectral gap of Y_r; (d) a simple leading singular value of Y_r; and (e) other dormant channels that are negligible or spectrally separated, since dormant channels started at comparable scales interact through (4.2). In the example that we have checked numerically ($d_{12} = 6$, $d_{34} = 4$, $D_L = 2$, $D_R = 3$, $\sigma^\star = (2, 1.4, 0.8)$, a rank-two student with random balanced initial cores of amplitudes $3 \cdot 10^{-3} \times (1, 0.8)$), it predicts the length of the second plateau to within 0.1%. When the modes are aligned, it reduces to the sequential acquisition of Proposition 3.10: the $(r+1)$-th mode escapes after a time $1/(\sigma^\star_{r+1} s_{r+1})$.

4.5 Near the aligned manifold and near the optimum

Assume that $k = r^\star$, that the singular values of the student are simple, with gaps $|\sigma_i - \sigma_j|$ bounded below, that its modes are matched with those of the teacher, and that, within the singular spaces of the teacher, $U = U^\star e^X$ and $V = V^\star e^Y$ with small skew-symmetric X and Y. Rotations out of the singular spaces of the teacher are governed by (4.3), whose rates at the optimum are Proposition 4.4(iv). Then $\Theta = S^\star - XS^\star + S^\star Y + O(\|X\|^2 + \|Y\|^2)$, with $S^\star := \mathrm{diag}(\sigma^\star_q)$, so that $\Theta_{ii} = \sigma^\star_i + O(\|X\|^2 + \|Y\|^2)$, and by (4.1) small alignment errors do not affect the amplitudes at first order. Linearizing (4.2) for $\xi^\pm_{ij} := X_{ij} \pm Y_{ij}$ gives

$$\dot\xi^+_{ij} = -\kappa_{ij} \frac{\sigma^\star_i - \sigma^\star_j}{\sigma_i - \sigma_j} \xi^+_{ij}, \qquad \dot\xi^-_{ij} = -\kappa_{ij} \frac{\sigma^\star_i + \sigma^\star_j}{\sigma_i + \sigma_j} \xi^-_{ij}, \tag{4.14}$$

up to terms of second order. The coupling is one-way: the amplitudes, given by (3.64), set the rates at which the angles decay. The rate of the symmetric error ξ^+ depends on the gaps, and that of the left-right mispairing ξ^- on the sums. If the amplitudes of the student are ordered as those of the teacher, both errors decay. If they are ordered oppositely, ξ^+_{ij} grows at the rate $\kappa_{ij}(\sigma^\star_j - \sigma^\star_i)/(\sigma_i - \sigma_j)$: the current pairing is linearly unstable, and a swap of the modes is expected. The linearization fails near crossings of the singular values of the student, where blocks of singular vectors must be followed instead.

Near the optimum, $M = M^\star + \Delta$, and the right-hand side of (3.60) is $-\mathcal{K}_{M^\star+\Delta}[\Delta] = -\mathcal{K}_{M^\star}[\Delta] + o(\|\Delta\|)$, by continuity of the fractional powers in M; the remainder is in general of fractional order, not quadratic. Complete $U^\star$ and $V^\star$ into orthonormal bases and set $\sigma^\star_q := 0$ for $q > r^\star$. In these bases, $\frac{d}{dt}(U^{\star\top}\Delta V^\star)_{ij} = -\kappa^\star_{ij}(U^{\star\top}\Delta V^\star)_{ij} + o(\|\Delta\|)$. Amplitude errors decay at the rates $3\sigma_i^{\star 4/3}$ and pairing errors at the rates $\kappa^\star_{ij}$, so that reducing an error by a factor ε takes $\log(1/\varepsilon)/(3\sigma_i^{\star 4/3})$ and $\log(1/\varepsilon)/\kappa^\star_{ij}$, respectively. In the block orthogonal to both singular spaces of the teacher, which a student of rank larger than $r^\star$ occupies, $\kappa^\star$ vanishes; there the linearization is degenerate, and extra modes decay algebraically, as in the regimes listed after Proposition 3.10.

4.6 Non-whitened inputs

For independent blocks with Gram matrices Σ_{12} and Σ_{34}, $R_\Delta = \Sigma_{12}(M - M^\star)\Sigma_{34}$, and the rank-one reduction is no longer exact: in the rescaled time, the orientation equation acquires the term $-\sigma(v^\top \Sigma_{34} v)(I - uu^\top)\Sigma_{12} u$, which depends on the amplitude unless u is an eigenvector of Σ_{12},

and similarly for v. During the escape of a rank-one channel, however, $\sigma \ll 1$ and this term is negligible. To leading order in σ, Proposition 4.2 then holds with $M^\star$ replaced by the moment matrix $H := \Sigma_{12} M^\star \Sigma_{34}$. If H has a simple leading singular value and the channel has a nonzero symmetric overlap with its leading singular pair, the channel aligns with this pair, and the time to reach any fixed amplitude below that of its limit is $I(H)/s(0) + o(1/s(0))$, where $I(H)$ is computed by (4.7) with H in place of $M^\star$; numerically, $s(0)$ times this time converges to $I(H)$. Only after the escape does the channel relax towards the data-weighted rank-one optimum, which, for invertible Gram matrices, is $\Sigma_{12}^{-1/2} \hat{N} \Sigma_{34}^{-1/2}$, with $\hat{N}$ the best rank-one approximation of $\Sigma_{12}^{1/2} M^\star \Sigma_{34}^{1/2}$. For a single channel, this supports the conjecture of Experiment E1.3: the moment matrix sets the time of acquisition, and the data-weighted spectrum sets the decrease of the loss. For several interacting channels, H is the matrix that drives the early phase, but we have not shown that it sets the order of acquisition.

4.7 Rank-one channels of other trees, and the role of the cuts

The rank-one reduction extends to any TTN. Consider a tree with N cores, in which the cores that carry open legs group the legs into m groups $G_1, \dots, G_m$. In the four-leaf model, $N = 3$, $m = 2$, $G_1 = \{1, 2\}$ and $G_2 = \{3, 4\}$; in its version with leaf cores (Figure 1(c)), $N = 7$ and $m = 4$. Consider cores that are exactly balanced at every bond, with bond rank one and a common nonzero scale s. By the argument of Remark 3.11(ii) applied to each bond, the gradient flow keeps them of bond rank one, and the represented tensor is $T = s^N w_1 \otimes \cdots \otimes w_m$, with unit vectors w_g over the legs of G_g. For whitened blocks, with $\rho := \langle T^\star, w_1 \otimes \cdots \otimes w_m \rangle$ and $\mu := s^N$,

$$\frac{dw_g}{d\tau} = (I - w_g w_g^\top)\, T^\star(w_1, \dots, w_{g-1}, \cdot, w_{g+1}, \dots, w_m), \qquad \frac{d\mu}{d\tau} = N\mu(\rho - \mu), \qquad \frac{d\tau}{dt} = s^{N-2}. \tag{4.15}$$

Indeed, each core of a rank-one channel is s times a tensor of unit norm, which is fixed except for the directions w_g of the cores that carry legs, and the gradient of the loss with respect to a core is the contraction of the residual with the other cores, whose total scale is s^{N-1}. The triangular structure persists: in τ, the orientation does not depend on the amplitude, and $\mu(\tau) = e^{NA}/(\mu(0)^{-1} + N \int_0^\tau e^{NA})$ with $A := \int_0^\tau \rho$. If the orientation flow converges to a component with $\rho > 0$, so that $\int_0^\infty e^{-(N-2)A}$ is finite, the escape takes $t \approx s(0)^{-(N-2)} \int_0^\infty e^{-(N-2)A(\tau)} d\tau$. This is the scaling $\varsigma^{-(|V|-2)}$ of Section 7.2.2, with a geometric prefactor; for $N = 3$, $e^{-A} = Z^{-1/2}$ and Corollary 4.3 is recovered.

What changes with the tree is the orientation flow. For $m = 2$, it is Oja's flow on the dilation of the matricization $T^{\star\langle G_1 \rangle}$, solved by Proposition 4.2: alignment is a power iteration at a single cut, exponentially fast in τ. For $m \geq 3$, it is a continuous-time version of the higher-order power method (De Lathauwer et al., 2000b; Anandkumar et al., 2014), which we do not know how to solve in closed form. For an orthogonally decomposable teacher $T^\star = \sum_q \sigma_q^\star\, a_{1,q} \otimes \cdots \otimes a_{m,q}$, the overlaps $p_{g,q} := w_g^\top a_{g,q}$ satisfy $p'_{g,q} = \sigma_q^\star \prod_{h \neq g} p_{h,q} - \rho\, p_{g,q}$. For a rank-one teacher and equal positive initial overlaps $p(0)$ in all groups, $p' = \sigma^\star p^{m-1}(1 - p^2)$, and alignment takes $\tau_{\text{align}} = p(0)^{-(m-2)}/((m-2)\sigma^\star) + O(\log(1/p(0)))$, i.e. of order $d_g^{(m-2)/2}/((m-2)\sigma^\star)$ for overlaps of order $d_g^{-1/2}$, with d_g the dimension of each group. For $m = 2$, the same computation gives $p' = \sigma^\star p(1 - p^2)$ and $\tau_{\text{align}} = \log d_g/(2\sigma^\star) + O(1)$; when each group gathers two legs of dimension d, as in the three-core model, $d_g = d^2$ and $\tau_{\text{align}} = \log d/\sigma^\star + O(1)$. From a random initialization the overlaps are unequal and of random signs, which changes the constants and the winning component; the scaling $d_g^{(m-2)/2}$ for $m \geq 3$ is the robust conclusion, and it is analogous to the role of the information exponent in the escape from a random start in single-index models (Ben Arous et al., 2021). Finer leg groups thus make the search for the direction of the teacher polynomially slow.

Time	Setting	Expression	Status
Escape from the origin	rank-one channel, $\alpha_1 \neq 0$	$I/s(0) + O(1)$, $I = \int_0^\infty Z^{-1/2} \in \left[1/\sigma_1^\star,\ 1/(\sigma_1^\star\lvert\alpha_1\rvert)\right]$	leading asymptotic
Escape from a random start	rank-one channel; fixed rank and gap, $d \to \infty$	$(\log(1/\alpha_1^2) + O(1))/(2\sigma_1^\star s(0))$	in probability
Alignment up to ε	rank-one channel	$t(\tau_\varepsilon)$, with $w_1(\tau_\varepsilon) = 1 - \varepsilon$	exact
Amplification after alignment	rank-one channel, $s(0) \to 0$	$c_\varepsilon/(\sigma_1^\star s_{\text{align}})$, $1 \leq c_\varepsilon \leq (1-\varepsilon)^{-1/2}$	leading asymptotic
Plateau after r modes	separated amplitudes, relaxed active modes, slowly varying Y_r	$I(Y_r)/s_{r+1}$	approximation
Escape from a strict saddle	learned mode i weaker than an unlearned mode j	$\lambda_{ij}^{-1}\log(1/\delta)$, $\lambda_{ij} = \sigma_i^{\star 1/3}(\sigma_j^\star - \sigma_i^\star)$	linearization
Swap of mispaired modes	amplitudes ordered oppositely to the teacher	$\log(1/\delta)\,(\sigma_i - \sigma_j)/\big(\kappa_{ij}(\sigma_j^\star - \sigma_i^\star)\big)$	linearization
Convergence	near the optimum, transverse blocks	$\log(1/\varepsilon)/(3\sigma_i^{\star 4/3})$, $\log(1/\varepsilon)/\kappa_{ij}^\star$; algebraic in the null block	linearization
Alignment, deeper trees	rank-one channel and teacher, equal overlaps, $m \geq 3$ groups	$\tau_{\text{align}} \approx d_g^{(m-2)/2}/((m-2)\sigma^\star)$	leading asymptotic

Table 1: Time scales of the balanced dynamics without alignment, for whitened blocks; δ is the size of the initial perturbation along the unstable direction. "Exact" and "leading asymptotic" refer to Proposition 4.2, Corollary 4.3 and Section 4.7; "linearization" to Proposition 4.4 and Section 4.5; "approximation" to the deflated normal form of Section 4.4, which is not proved.

This also clarifies how the dynamics varies across cuts. A rank-one channel has the single singular value s^N at every cut, so that its amplitude is acquired simultaneously at all cuts. Alignment, by contrast, is cut-dependent. For an orthogonally decomposable teacher, at a cut $A \mid A^c$ of the tree, the overlaps of the left and right singular vectors of the channel with those of a teacher component are $\prod_{g\subseteq A} p_{g,q}$ and $\prod_{g\subseteq A^c} p_{g,q}$, while their product, the overlap of the whole channel with the component, is the same at every cut. The singular vector on one side of a cut is therefore aligned no earlier than the leg groups on that side, and an interface of the tree is identified only once the groups below it are. In the three-core model, which groups the legs into two cores, the non-aligned dynamics is that of a matrix (Remark 3.2); factorizing the lower cores into leaf maps changes the search for the direction of the teacher, not only the parameterization.

4.8 Summary: which times to distinguish

There is no single learning time in the non-aligned setting. Table 1 collects the time scales derived above, for whitened blocks. They call for separate observables: amplitudes σ_i, or the diagonal of the coupling matrix (Definition 5.15), for acquisition; overlaps of singular subspaces for alignment, followed as blocks near degeneracies (Remark 5.13); the signals Θ_{ii}; and the off-diagonal of the coupling matrix for the pairing. The decomposition of the error in Proposition 5.16 separates these contributions.

5 How to compare TTNs and DNNs

We want to test experimentally whether the training dynamics of TTNs is similar to that of DNNs with standard activation functions, such as ReLU, GELU or SiLU. Hypothesis 1.2 phrases this

question in terms of observables, i.e. quantities that can be evaluated along the trajectory of any architecture. This section constructs such observables in the setting in which they are easiest to interpret: a teacher–student experiment in which the teacher is a TTN. The teacher's tree then provides a canonical hierarchy of interfaces, its bonds, and the singular value decomposition across each bond, taken in the geometry of the data, provides a gauge-invariant list of candidate features, which we call *atoms*. Comparing a TTN with a DNN then amounts to asking, for each atom and each student, when the student acquires the atom, where it computes it, and whether it relies on it.

These questions call for different kinds of evidence, which should not be conflated (Figure 3):

(i) *output*: the predictor of the student contains the atom, with the teacher's strength;
(ii) *representation*: the atom, or the two factors of which it is the product, can be read off a hidden layer of the student;
(iii) *use*: the computation downstream of that layer is sensitive to this internal code, and causally relies on it.

The answers also depend on how the input is distributed over the leaves of the teacher. We distinguish three regimes:

(a) *product law*: the leaves receive distinct blocks of the input, and the blocks are independent;
(b) *dependent blocks*: the leaves receive distinct blocks of the input, with an arbitrary joint law;
(c) *repeated inputs*: all the leaves receive copies of the same input, as in the tied TTN of a deep polynomial network (Section 2.6).

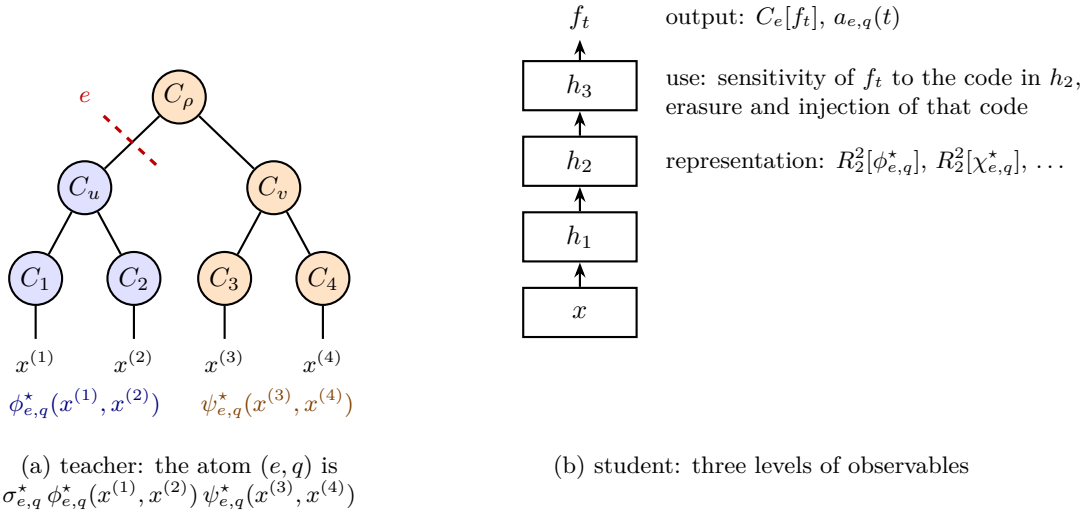
Independence of the blocks is not needed to define any of the observables below. It is a source of orthogonality and of exact computability: it makes functions of disjoint blocks uncorrelated, it makes the population loss of a TTN depend on the data only through the second moments of the leaf embeddings, and it turns the decomposition of the teacher across each bond into a Schmidt decomposition in the geometry of the loss. We therefore define every object for an arbitrary joint law, and state separately what independence adds. Regime (c) is not a limit of the other two: its leaves are copies of one variable rather than distinct variables, and its decompositions across bonds describe a computation rather than a function.

Section 5.1 fixes the setting. Section 5.2 defines the data-weighted HSVD of a TTN, whose singular functions are the hierarchical features of the teacher, and of any TTN student, in the geometry of the data; Section 5.3 defines the atoms. Section 5.4 defines output observables in the coordinates of the teacher, and Section 5.5 a teacher-free decomposition of any student, its functional HSVD, which coincides with the data-weighted HSVD for TTNs. Section 5.6 treats teachers with repeated inputs. Sections 5.7 and 5.8 define the representation and use observables, Section 5.9 explains which observables can be computed exactly for TTN students, and Section 5.10 turns the observables into acquisition times, comparisons between students and an estimation protocol. Section 5.11 summarizes them in Tables 2 and 3.

5.1 The teacher–student setting

Teacher. Fix a binary tree whose N leaves carry one open leg each, indexed by $a \in [N]$; a balanced tree, with N a power of two, is the simplest choice. We root the tree at an internal vertex ρ and, as in Proposition 2.14, let A_e be the set of legs below the bond e, so that A_e and A_e^c are the two sides of the cut of Lemma 2.8. A TTN on this tree (Definition 2.7) represents a tensor $T^\star$. Leg a receives the block $x^{(a)} \in \mathcal{X}_a$ of the input through a fixed embedding $\xi_a : \mathcal{X}_a \to \mathbb{R}^{d_a}$, for example $\xi_a(z) = z$, $\xi_a(z) = (1, z)$ or a one-hot encoding, and, as in (2.3), the teacher is

$$f^\star(x) = \Big\langle T^\star, \bigotimes_{a=1}^{N} \xi_a(x^{(a)}) \Big\rangle. \tag{5.1}$$



(a) teacher: the atom (e, q) is $\sigma^\star_{e,q}\, \phi^\star_{e,q}(x^{(1)}, x^{(2)})\, \psi^\star_{e,q}(x^{(3)}, x^{(4)})$

(b) student: three levels of observables

Figure 3: Comparing a student with a TTN teacher. (a) Cutting the bond e splits the teacher into the subtree below e (blue), which computes the left factors $\phi^\star_{e,q}$ of the inputs $x^{(1)}, x^{(2)}$, and its environment (orange), which computes the right factors $\psi^\star_{e,q}$ of $x^{(3)}, x^{(4)}$. An atom is the product of two such factors, weighted by a singular value of the data-weighted HSVD (Sections 5.2 and 5.3). (b) For a student, one asks whether its output contains the atom (Section 5.4), whether the factors and the atom can be read off a hidden layer, here h_2 (Section 5.7), and whether the layers above use and rely on that code (Section 5.8).

The labels are $y = f^\star(x)$, possibly plus independent zero-mean noise. This is a *physical* TTN in the sense of Section 2.2: distinct legs receive distinct blocks of the input. For example, the balanced bilinear teacher of Surrogate note (2026, Section 6.2), which applies linear maps at the leaves, planted bilinear maps at the internal vertices and a linear readout at the root, is of this form with $\xi_a(z) = z$. Formula (5.1) defines $f^\star$ at every point of $\prod_a \mathcal{X}_a$, not only on the support of the data; this will matter when the blocks are dependent. Teachers with repeated inputs are introduced in Section 5.6.

Data. The blocks $X = (X^{(1)}, \dots, X^{(N)})$ of a random input have an arbitrary joint law ν_x. For $A \subseteq [N]$, we write $x_A = (x^{(a)})_{a\in A}$, ν_A for the law of X_A, $\mu_a := \nu_{\{a\}}$ for the law of the block a, $\Xi_A(x_A) = \bigotimes_{a\in A} \xi_a(x^{(a)})$, and

$$\Sigma_A := \mathbb{E}[\Xi_A(X_A)\, \Xi_A(X_A)^\top], \qquad \Sigma_a := \Sigma_{\{a\}}, \tag{5.2}$$

for the Gram matrix of the embedding of the legs in A. We assume that $\mathbb{E}\|\Xi_A(X_A)\|^2 < \infty$ for every A. Norms and inner products of functions are those of $L^2(\nu_x)$, or of $L^2(\nu_A)$ for functions of x_A. The *product law* is the special case

$$\nu_x = \mu_1 \otimes \cdots \otimes \mu_N, \tag{5.3}$$

in which $\Sigma_A = \bigotimes_{a\in A} \Sigma_a$ for every A. In neither case do the blocks need to be Gaussian, centred, whitened or identically distributed.

Two further notions will be used. At each bond e, the *decoupled law* $\nu_e^\otimes := \nu_{A_e} \otimes \nu_{A_e^c}$ gives the two sides of the cut their correct marginal laws but draws them independently. It coincides with ν_x if and only if X_{A_e} and $X_{A_e^c}$ are independent, and a sample from it, which we call a *recombined input*,

is obtained by joining the A_e-part of one data point to the A_e^c-part of another, independent one. The law ν_x is *nondegenerate* if $\Sigma_{[N]}$ is nonsingular, i.e. if no function $\langle D, \Xi_{[N]} \rangle$ with $D \neq 0$ vanishes almost surely. This holds under the product law when every Σ_a is nonsingular; for polynomial embeddings with linearly independent coordinates and a law with a density that is positive on a nonempty open set, since a nonzero polynomial cannot vanish almost everywhere on an open set; and for one-hot embeddings of discrete blocks when every combination of values has positive probability. It fails when the support of ν_x is degenerate, for instance when a block is a deterministic function of other blocks. Synthetic experiments realize the product law by construction. Dependence can then be introduced in a controlled way, for instance with correlated Gaussian blocks, which give a nondegenerate law, or with a discrete latent variable, which makes ν_x a finite mixture of product laws and keeps exact computations available (Remark 5.7).

Students and time. A DNN f_θ of the form (2.6), with a standard activation, and a TTN g_ϑ of the form (2.3), normally on the teacher's tree, are trained on the same examples with the same loss. We write f_t for the predictor of a student at time t, whichever its architecture: $f_t = f_{\theta(t)}$ or $f_t = g_{\vartheta(t)}$. Time is the integrated learning rate $t = \sum_n \eta_n$ of Hypothesis 1.2. Students that share their parameters, such as a DNN and its polynomial proxy or its TTN surrogate (Section 2.7), are compared under identical updates from the same optimizer state. A DNN and a TTN, by contrast, have different parameter metrics, so the same raw learning rate can move them at very different speeds in function space. The optimizer family, the tuning budget and the time convention must then be fixed before any feature curve is examined. A single global calibration of the time scale, fixed in advance, for instance by matching the initial speeds $\|\partial_t f_t\|$ at $t = 0$, is compatible with Hypothesis 1.2: for gradient descent, it amounts to a constant multiplier in the parameterization of the TTN, i.e. to a choice of architecture. Feature-specific or post hoc time warps are not, since they would make the hypothesis unfalsifiable. Both the number of updates and the integrated learning rate should be reported. Three further precautions apply (Surrogate note, 2026, Sections 6.6 and 6.7). Every student should be able to fit the teacher under a generous training budget, since an underfitting student says nothing about dynamics. There is no single notion of equal capacity for a DNN and a TTN, so widths, ranks and initialization scales should be bracketed rather than matched by one number. Finally, a TTN on a wrong tree is a useful structural control.

5.2 The data-weighted HSVD of the teacher

Fix a bond e of the teacher. It splits the legs into the legs A_e below e and the others, A_e^c, and contracting (5.1) gives, up to the ordering of the legs,

$$f^\star(x) = \Xi_{A_e}(x_{A_e})^\top M_e^\star \, \Xi_{A_e^c}(x_{A_e^c}), \qquad M_e^\star := T^{\star\langle A_e \rangle}, \tag{5.4}$$

at every point x, whatever the law of the data. The HSVD of Definition 2.13 decomposes $M_e^\star$ in the Euclidean geometry of coefficients. This is the right geometry for comparing tensors, but not for comparing functions: learning is driven by the population loss, hence by the geometry of $L^2(\nu_x)$. The two geometries agree when the embeddings are orthonormal under the data, $\Sigma_A = I$ for every A, for instance for independent whitened blocks and $\xi_a(z) = z$. Otherwise they differ, and the Euclidean singular values can be misleading about which features matter (Example 5.4). We therefore express each side of the cut in coordinates that are orthonormal under its own law, and take the SVD there. No preprocessing is applied to the inputs seen by the students: the change of coordinates only concerns the description of the teacher. Since the same construction will be applied to TTN students, we state it for a general tensor.

Definition 5.1 (Data-weighted HSVD)**.** Let T be a tensor over the legs of the tree, for instance the tensor represented by a TTN on it, and let $g_T := \langle T, \Xi_{[N]} \rangle$ be the function it defines. Write $(\cdot)^{1/2}$ for the positive semidefinite square root and $(\cdot)^+$ for the Moore–Penrose pseudoinverse, and let $\tilde{\Xi}_A := (\Sigma_A^{1/2})^+ \, \Xi_A$ be the *whitened embedding* of the legs in A. For each bond e, let

$$\tilde{M}_e(T) := \Sigma_{A_e}^{1/2} \, T^{\langle A_e \rangle} \, \Sigma_{A_e^c}^{1/2} = U_e S_e V_e^\top, \qquad S_e = \operatorname{diag}\left(\sigma_{e,1}, \dots, \sigma_{e,\tilde{r}_e}\right), \quad \sigma_{e,1} \geq \cdots \geq \sigma_{e,\tilde{r}_e} > 0,$$

be a compact SVD, with columns $u_{e,q}$ of U_e and $v_{e,q}$ of V_e. The *data-weighted singular functions* of T at e are

$$\phi_{e,q}(x_{A_e}) := u_{e,q}^\top \, \tilde{\Xi}_{A_e}(x_{A_e}), \qquad \psi_{e,q}(x_{A_e^c}) := v_{e,q}^\top \, \tilde{\Xi}_{A_e^c}(x_{A_e^c}), \qquad q \leq \tilde{r}_e. \tag{5.5}$$

The family $(\sigma_{e,q}, \phi_{e,q}, \psi_{e,q})_{e \in E_{\mathrm{int}}, \, q \leq \tilde{r}_e}$ is the *data-weighted HSVD* of T, and the left singular functions $\phi_{e,q}$ are its *data-weighted hierarchical features*. For the teacher, $T = T^\star$, we write $\tilde{M}_e^\star$, $U_e^\star$, $S_e^\star$, $V_e^\star$, $\sigma_{e,q}^\star$, $\phi_{e,q}^\star$, $\psi_{e,q}^\star$ and $r_e^\star$.

In the rest of this section, singular values and singular vectors always refer to data-weighted SVDs. The data enter only through the Gram matrices Σ_{A_e} and $\Sigma_{A_e^c}$ of the embeddings of the two sides of each cut. These involve the law of each side, but never the dependence between the two sides. Under the product law they are Kronecker products of the leaf Gram matrices Σ_a, which are computed analytically or estimated once, from a large sample independent of all other data; for $\xi_a(z) = (1, z)$, Σ_a is the matrix of the first and second moments of $x^{(a)}$. Under dependence, Σ_{A_e} also contains mixed moments of the embeddings of different leaves on the same side; Proposition 5.6 below shows how to compute the decomposition without forming these exponentially large matrices. The data-weighted rank $\tilde{r}_e = \operatorname{rank} \tilde{M}_e(T)$ is at most $\operatorname{rank} T^{\langle A_e \rangle}$, with equality when Σ_{A_e} and $\Sigma_{A_e^c}$ are nonsingular: directions of coefficients that vanish on the data do not count.

Proposition 5.2 (Properties of the data-weighted HSVD)**.** For every joint law ν_x, every tensor T and every bond e:

(i) $(\phi_{e,q})_q$ is orthonormal in $L^2(\nu_{A_e})$, and $(\psi_{e,q})_q$ is orthonormal in $L^2(\nu_{A_e^c})$;

(ii) the expansion

$$g_T(x) = \sum_{q=1}^{\tilde{r}_e} \sigma_{e,q} \, \phi_{e,q}(x_{A_e}) \, \psi_{e,q}(x_{A_e^c}) \tag{5.6}$$

holds for ν_x-almost every x and for $\nu_e^\otimes$-almost every x;

(iii) the products $\phi_{e,i}\psi_{e,j}$ are orthonormal in $L^2(\nu_e^\otimes)$, and (5.6) is the Schmidt decomposition of g_T in $L^2(\nu_e^\otimes)$ across the cut $(A_e \mid A_e^c)$. Its singular values, and the spans of its singular functions that belong to equal singular values, depend only on g_T as an element of $L^2(\nu_e^\otimes)$. In particular, they are gauge invariant, and they do not change if the embeddings and the tensor are changed in a way that leaves g_T unchanged on recombined inputs;

(iv) if X_{A_e} and $X_{A_e^c}$ are independent, in particular under the product law, then $\nu_e^\otimes = \nu_x$: the products $\phi_{e,i}\psi_{e,j}$ are orthonormal in $L^2(\nu_x)$, (5.6) is the Schmidt decomposition of g_T across the physical cut $(A_e \mid A_e^c)$ of Remark 2.39, and $\tilde{r}_e$ is the rank of g_T across this cut.

Proof. Let $\Pi_A := \Sigma_A^{1/2}(\Sigma_A^{1/2})^+$ be the orthogonal projector onto the range of Σ_A. If $\Sigma_A w = 0$, then $\mathbb{E}[(w^\top \Xi_A)^2] = w^\top \Sigma_A w = 0$; hence $\Xi_A(X_A)$ lies in the range of Σ_A almost surely, i.e. $\Xi_A = \Pi_A \Xi_A = \Sigma_A^{1/2} \tilde{\Xi}_A$ almost surely. (i) $\mathbb{E}[\tilde{\Xi}_A \tilde{\Xi}_A^\top] = (\Sigma_A^{1/2})^+ \Sigma_A (\Sigma_A^{1/2})^+ = \Pi_A$, and the columns of U_e lie in $\operatorname{col}(\tilde{M}_e)$, which is contained in the range of $\Sigma_{A_e}^{1/2}$, so $\mathbb{E}[\phi_{e,q}\phi_{e,q'}] = u_{e,q}^\top \Pi_{A_e} u_{e,q'} = \delta_{qq'}$; the same argument applies to the $\psi_{e,q}$. (ii) The events $\{\Xi_{A_e} = \Sigma_{A_e}^{1/2} \tilde{\Xi}_{A_e}\}$ and $\{\Xi_{A_e^c} = \Sigma_{A_e^c}^{1/2} \tilde{\Xi}_{A_e^c}\}$

have probability one under ν_x and under $\nu_e^{\otimes}$, since these two laws have the same marginals. On their intersection, $g_T = \Xi_{A_e}^{\top} T^{\langle A_e \rangle} \Xi_{A_e^c} = \tilde{\Xi}_{A_e}^{\top} \tilde{M}_e \tilde{\Xi}_{A_e^c}$, and inserting the SVD gives (5.6). (iii) Under $\nu_e^{\otimes}$, $\mathbb{E}[\phi_{e,i}\psi_{e,j}\phi_{e,k}\psi_{e,l}] = \mathbb{E}[\phi_{e,i}\phi_{e,k}]\,\mathbb{E}[\psi_{e,j}\psi_{e,l}] = \delta_{ik}\delta_{jl}$ by (i). The function g_T is the kernel of the Hilbert–Schmidt operator $K : L^2(\nu_{A_e^c}) \to L^2(\nu_{A_e})$, $(K\psi)(x_{A_e}) = \mathbb{E}[g_T(x_{A_e}, X'_{A_e^c})\,\psi(X'_{A_e^c})]$ with $X'_{A_e^c} \sim \nu_{A_e^c}$, and by (i) and (ii), $KK^* = \sum_q \sigma_{e,q}^2\, \phi_{e,q} \langle \phi_{e,q}, \cdot \rangle$. The $\sigma_{e,q}^2$ are therefore the nonzero eigenvalues of KK^*, with eigenspaces spanned by the $\phi_{e,q}$ of equal $\sigma_{e,q}$, and KK^* depends only on g_T as an element of $L^2(\nu_e^{\otimes})$; the same argument with K^*K applies to the $\psi_{e,q}$. Gauge invariance holds because $\tilde{M}_e(T)$ depends only on T (Proposition 2.12). (iv) If X_{A_e} and $X_{A_e^c}$ are independent, their joint law is the product of their marginal laws; apply (iii). □

Proposition 5.2 says what the data-weighted HSVD is in every regime: at each bond, it is the Schmidt decomposition of the function in the *decoupled geometry* $L^2(\nu_e^{\otimes})$, in which the subtree below e and its environment receive independent inputs with the correct marginal laws. Under the product law, this is the geometry of the loss, and what the HSVD of Section 2.4 says about the Frobenius norm holds for the population loss: the squared singular values are the energies carried by the modes, and truncations are optimal (Proposition 5.12). Under dependence, the squared singular values are energies on recombined inputs, which can differ from the energies on the data; Sections 5.3 and 5.4 account for the difference. The decoupled geometry is nevertheless the natural one for a physical TTN: its formula (5.1) is defined on recombined inputs, and the subtree below e computes the same functions whatever its environment receives.

Under the product law, the data-weighted HSVD is moreover the HSVD of a single TTN.

Proposition 5.3 (Product law: one data-weighted TTN)**.** Under the product law (5.3), for every tensor T:

(i) $\tilde{\Xi}_A = \bigotimes_{a \in A} \tilde{\xi}_a$, with $\tilde{\xi}_a := (\Sigma_a^{1/2})^+ \xi_a$, and $\tilde{M}_e(T)$ is the A_e-matricization of the *data-weighted tensor* $\tilde{T}$ obtained from T by multiplying each leg a by $\Sigma_a^{1/2}$;

(ii) if T is represented by a TTN, then $\tilde{T}$ is represented by the same TTN with the open leg of each leaf core multiplied by $\Sigma_a^{1/2}$, with the same tree and bond dimensions. Hence the data-weighted HSVD of T is the HSVD of $\tilde{T}$: Propositions 2.14 and 2.16 and Remark 2.17 apply to it;

(iii) $g_T(x) = \langle \tilde{T}, \bigotimes_a \tilde{\xi}_a(x^{(a)}) \rangle$ for ν_x-almost every x.

Proof. (i) Under (5.3), $\Sigma_A = \bigotimes_{a \in A} \Sigma_a$, and the square root and the pseudoinverse of a Kronecker product of positive semidefinite matrices are the Kronecker products of the square roots and of the pseudoinverses. Matricizing $\tilde{T}$ across $(A_e \mid A_e^c)$ gives $\Sigma_{A_e}^{1/2} T^{\langle A_e \rangle} \Sigma_{A_e^c}^{1/2}$. (ii) Multiplying an open leg of a core by a matrix multiplies the corresponding mode of the contraction (2.2) by that matrix. (iii) $\langle \tilde{T}, \bigotimes_a \tilde{\xi}_a \rangle = \langle T, \bigotimes_a \Pi_a \xi_a \rangle$ with $\Pi_a := \Sigma_a^{1/2} (\Sigma_a^{1/2})^+$, and $\Pi_a \xi_a(X^{(a)}) = \xi_a(X^{(a)})$ almost surely (proof of Proposition 5.2). □

Example 5.4 (Why the data weighting matters)**.** Take $N = 2$ blocks $x^{(1)}, x^{(2)} \in \mathbb{R}^2$, $\xi_a(z) = z$, and the teacher $f^\star(x) = \langle x^{(1)}, x^{(2)} \rangle = x_1^{(1)} x_1^{(2)} + x_2^{(1)} x_2^{(2)}$, a TTN with two leaves and one bond. Its coefficient matrix is $M^\star = I_2$, so the Euclidean HSVD has a degenerate pair $\sigma_1 = \sigma_2 = 1$ and suggests two equally important atoms, $x_1^{(1)} x_1^{(2)}$ and $x_2^{(1)} x_2^{(2)}$. Now let $x^{(1)}$ and $x^{(2)}$ be independent, with $x^{(2)} \sim \mathcal{N}(0, I_2)$ and $x^{(1)} \sim \mathcal{N}(0, \Sigma_1)$, where $\Sigma_1 = \left(\begin{smallmatrix} 1 & \rho \\ \rho & 1 \end{smallmatrix}\right)$ and $0 < \rho < 1$. Then $\tilde{M}^\star = \Sigma_1^{1/2}$ has the singular values $\sqrt{1+\rho}$ and $\sqrt{1-\rho}$, and

$$f^\star = \sqrt{1+\rho}\, \phi_+^\star \psi_+^\star + \sqrt{1-\rho}\, \phi_-^\star \psi_-^\star, \qquad \phi_\pm^\star = \frac{x_1^{(1)} \pm x_2^{(1)}}{\sqrt{2(1 \pm \rho)}}, \qquad \psi_\pm^\star = \frac{x_1^{(2)} \pm x_2^{(2)}}{\sqrt{2}}.$$

Under the data, there is one strong atom, which carries the fraction $(1+\rho)/2$ of $\|f^\star\|^2$, and one weak atom. For ρ close to 1, the best predictor of rank one across the cut already has the small population loss $(1-\rho)/2$ (Proposition 5.12). Which atoms the loss favours, and plausibly which atoms a loss-driven learner acquires first, is decided by the data-weighted singular values, not by the Euclidean ones. The blocks are independent here: the weighting matters even under the product law, as soon as the coordinates of an embedding are correlated or not normalized.

Remark 5.5 (Which moments of the data enter)**.** For every joint law and all tensors T and T',

$$\langle g_T, g_{T'} \rangle_{L^2(\nu_x)} = \mathrm{vec}(T)^\top \, \Sigma_{[N]} \, \mathrm{vec}(T'), \tag{5.7}$$

so the population loss of a TTN student depends on the data only through the Gram matrix $\Sigma_{[N]}$ of the full product embedding. The data-weighted SVD at e uses less: it replaces $\Sigma_{[N]}$ by $\Sigma_{A_e} \otimes \Sigma_{A_e^c}$ (up to the ordering of the legs), which is the Gram matrix of $\Xi_{[N]}$ under the decoupled law $\nu_e^{\otimes}$. Under the product law, $\Sigma_{[N]} = \bigotimes_a \Sigma_a$ and the two coincide: the data enter only through the second moments of the leaf embeddings. This is not a Gaussian assumption; it holds because a physical TTN is linear in each leaf embedding separately. Moreover, Σ_a is a second moment of the *embedding*, not of the raw input: for a scalar block and $\xi_a(z) = (1, z, z^2, z^3)$, it contains the moments of z up to order six. Under dependence, Σ_{A_e} contains mixed moments of the embeddings of different leaves on the same side of the cut, and $\Sigma_{[N]}$ contains mixed moments across the cut; the difference between $\Sigma_{[N]}$ and $\Sigma_{A_e} \otimes \Sigma_{A_e^c}$ is what the joint Gram matrices of Section 5.4 correct for. When the same input feeds several legs, $\Sigma_{[N]}$ contains the moments of the input up to twice the number of copies (Section 5.6; Local TN note, 2026, Section 8.1).

The data-weighted SVD at a bond can be computed from the two messages that the network sends through it.

Proposition 5.6 (Data-weighted SVD from the bond messages)**.** Let e be a bond, T a tensor, and $T^{\langle A_e \rangle} = F_e H_e^\top$ a factorization through $\mathbb{R}^{D_e}$, for instance the one given by the cores of a TTN (Lemma 2.8). Let

$$h_e(x_{A_e}) := F_e^\top \, \Xi_{A_e}(x_{A_e}), \qquad n_e(x_{A_e^c}) := H_e^\top \, \Xi_{A_e^c}(x_{A_e^c})$$

be the messages that the two sides of the cut send through the bond, so that $g_T(x) = h_e(x_{A_e})^\top n_e(x_{A_e^c})$, and let $\Sigma_e^h := \mathbb{E}[h_e h_e^\top]$ and $\Sigma_e^n := \mathbb{E}[n_e n_e^\top]$. Then the data-weighted singular values of T at e are the nonzero singular values of the $D_e \times D_e$ matrix $(\Sigma_e^h)^{1/2}(\Sigma_e^n)^{1/2}$. If $(\Sigma_e^h)^{1/2}(\Sigma_e^n)^{1/2} = WSY^\top$ is a compact SVD, the singular vectors can be chosen so that, almost surely,

$$\Phi_e = W^\top ((\Sigma_e^h)^{1/2})^+ h_e, \qquad \Psi_e = Y^\top ((\Sigma_e^n)^{1/2})^+ n_e,$$

where $\Phi_e := (\phi_{e,1}, \dots, \phi_{e,\tilde{r}_e})^\top$ and $\Psi_e := (\psi_{e,1}, \dots, \psi_{e,\tilde{r}_e})^\top$. In particular, the data-weighted hierarchical features at e are linear readouts of the message h_e.

Proof. Let $\tilde{F}_e := \Sigma_{A_e}^{1/2} F_e$ and $\tilde{H}_e := \Sigma_{A_e^c}^{1/2} H_e$, so that $\tilde{M}_e(T) = \tilde{F}_e \tilde{H}_e^\top$, $\Sigma_e^h = F_e^\top \Sigma_{A_e} F_e = \tilde{F}_e^\top \tilde{F}_e$ and $\Sigma_e^n = \tilde{H}_e^\top \tilde{H}_e$. Write $\Sigma^h := \Sigma_e^h$ and $\Sigma^n := \Sigma_e^n$, and let $U := \tilde{F}_e((\Sigma^h)^{1/2})^+ W$ and $V := \tilde{H}_e((\Sigma^n)^{1/2})^+ Y$. Since $WS = (\Sigma^h)^{1/2}(\Sigma^n)^{1/2} Y$, the columns of W lie in the range of Σ^h, so $U^\top U = W^\top((\Sigma^h)^{1/2})^+ \Sigma^h ((\Sigma^h)^{1/2})^+ W = W^\top W = I$; likewise $V^\top V = I$. Moreover $USV^\top = \tilde{F}_e \Pi^h \Pi^n \tilde{H}_e^\top$, where Π^h and Π^n are the orthogonal projectors onto the ranges of Σ^h and Σ^n. Since $\ker \Sigma^h = \ker \tilde{F}_e$, $\tilde{F}_e \Pi^h = \tilde{F}_e$, and likewise $\Pi^n \tilde{H}_e^\top = \tilde{H}_e^\top$. Hence $\tilde{M}_e(T) = USV^\top$ is a compact SVD. Finally, $h_e = F_e^\top \Sigma_{A_e}^{1/2} \tilde{\Xi}_{A_e} = \tilde{F}_e^\top \tilde{\Xi}_{A_e}$ almost surely (proof of Proposition 5.2), so $U^\top \tilde{\Xi}_{A_e} = W^\top((\Sigma^h)^{1/2})^+ h_e$; the same computation applies to Ψ_e. □

Remark 5.7 (Computing the data-weighted HSVD)**.** The matrices $\tilde{M}_e$ are exponentially large and are never formed. By Proposition 5.6, the data-weighted SVD at e only requires the two $D_e \times D_e$ second-moment matrices of the bond messages, and the values of the singular functions at a sample are obtained by contracting the sample through the two halves of the network. Under the product law, these matrices are computed exactly by propagating second moments through the tree. For a leaf bond e with leaf core $C_a \in \mathbb{R}^{d_a \times D_e}$, $\Sigma_e^h = C_a^\top \Sigma_a C_a$; at an internal vertex u with core C_u, child bonds e_1, e_2 and parent bond e, $(h_e)_k = \sum_{i,j} (C_u)_{ijk} (h_{e_1})_i (h_{e_2})_j$, so $(\Sigma_e^h)_{kk'} = \sum_{i,i',j,j'} (C_u)_{ijk} (C_u)_{i'j'k'} (\Sigma_{e_1}^h)_{ii'} (\Sigma_{e_2}^h)_{jj'}$, because $X_{A_{e_1}}$ and $X_{A_{e_2}}$ are independent. The environment moments Σ_e^n are propagated in the same way, from the root downwards. The cost is polynomial in the bond and leg dimensions and linear in N. Equivalently, one multiplies the leaf cores by $\Sigma_a^{1/2}$ (Proposition 5.3(ii)), orthonormalizes the network towards e by QR decompositions, and takes the SVD of the resulting small centre matrix (Grasedyck, 2010), (Diagonal TTN note, 2026, Section 9.2). Under dependence, the recursion fails, since in general $\mathbb{E}[h_{e_1} h_{e_1}^\top \otimes h_{e_2} h_{e_2}^\top] \neq \Sigma_{e_1}^h \otimes \Sigma_{e_2}^h$, and Σ_e^h and Σ_e^n are estimated by Monte Carlo, on a large sample independent of all other data; the estimates are unbiased, and only two $D_e \times D_e$ matrices are needed per bond. If ν_x is a finite mixture of product laws, $\nu_x = \sum_k \pi_k \bigotimes_a \mu_a^{(k)}$, the recursion is run for each component and the results are averaged with the weights π_k, which keeps the computation exact.

The data-weighted features of a TTN are hierarchical: the features at a bond are built from the features at the bonds below it.

Proposition 5.8 (Hierarchical composition of the features)**.** Let $u \neq \rho$ be an internal vertex with parent bond e and child bonds e_1 and e_2, so that $A_e = A_{e_1} \sqcup A_{e_2}$, and let T be a tensor.

(i) Under the product law,

$$\phi_{e,q}(x_{A_e}) = \sum_{i \le \tilde{r}_{e_1}} \sum_{j \le \tilde{r}_{e_2}} (B_u)_{(i,j),q}\, \phi_{e_1,i}(x_{A_{e_1}})\, \phi_{e_2,j}(x_{A_{e_2}}) \tag{5.8}$$

with $B_u := (U_{e_1} \otimes U_{e_2})^\top U_e$, which has orthonormal columns, and the products $\phi_{e_1,i}\phi_{e_2,j}$ are orthonormal in $L^2(\nu_{A_e})$.

(ii) For an arbitrary joint law, if $\Sigma_{A_{e'}}$ and $\Sigma_{A_{e'}^c}$ are nonsingular for $e' \in \{e, e_1, e_2\}$, in particular if ν_x is nondegenerate, then (5.8) holds with

$$B_u := (U_{e_1}^\top \Sigma_{A_{e_1}}^{1/2} \otimes U_{e_2}^\top \Sigma_{A_{e_2}}^{1/2})\, \Sigma_{A_e}^{-1/2} U_e.$$

The products $\phi_{e_1,i}\phi_{e_2,j}$ are in general not orthonormal, and $B_u^\top \Gamma_u B_u = I$, where Γ_u is their Gram matrix in $L^2(\nu_{A_e})$.

Proof. Write $\Sigma_{e'} := \Sigma_{A_{e'}}$ and $M_{e'} := T^{\langle A_{e'} \rangle}$, and order the legs so that $\Xi_{A_e} = \Xi_{A_{e_1}} \otimes \Xi_{A_{e_2}}$. (i) By Proposition 5.3, the $U_{e'}$ are the left singular vectors of the matricizations of $\tilde{T}$, so Proposition 2.14, applied to $\tilde{T}$, gives $U_e = (U_{e_1} \otimes U_{e_2}) B_u$. Since $\tilde{\Xi}_{A_e} = \tilde{\Xi}_{A_{e_1}} \otimes \tilde{\Xi}_{A_{e_2}}$, $\Phi_e = U_e^\top \tilde{\Xi}_{A_e} = B_u^\top (\Phi_{e_1} \otimes \Phi_{e_2})$, which is (5.8). The columns of B_u are orthonormal because those of U_e are orthonormal and lie in the range of the isometry $U_{e_1} \otimes U_{e_2}$, and the products are orthonormal by the independence of $X_{A_{e_1}}$ and $X_{A_{e_2}}$ and Proposition 5.2(i). (ii) For $i = 1, 2$, since $\Sigma_{A_{e_i}^c}$ is nonsingular, $\mathrm{col}(U_{e_i}) = \mathrm{col}(\tilde{M}_{e_i}) = \Sigma_{e_i}^{1/2} \mathrm{col}(M_{e_i})$, i.e. $\mathrm{col}(M_{e_i}) = \mathrm{col}(\Sigma_{e_i}^{-1/2} U_{e_i})$. Since Σ_e is nonsingular, $\phi_{e,q} = (\Sigma_e^{-1/2} u_{e,q})^\top \Xi_{A_e}$, and $\Sigma_e^{-1/2} u_{e,q} \in \mathrm{col}(M_e \Sigma_{A_e^c}^{1/2}) \subseteq \mathrm{col}(M_e) \subseteq \mathrm{col}(M_{e_1}) \otimes \mathrm{col}(M_{e_2})$ by (2.5). Hence $\Sigma_e^{-1/2} u_{e,q} = (\Sigma_{e_1}^{-1/2} U_{e_1} \otimes \Sigma_{e_2}^{-1/2} U_{e_2})\, b_q$ for a unique vector b_q, which is the q-th column of B_u because $U_{e_i}^\top \Sigma_{e_i}^{1/2} \Sigma_{e_i}^{-1/2} U_{e_i} = I$. Since $\Phi_{e_i} = U_{e_i}^\top \Sigma_{e_i}^{-1/2} \Xi_{A_{e_i}}$, this gives $\phi_{e,q} = b_q^\top (\Phi_{e_1} \otimes \Phi_{e_2})$, which is (5.8), and $B_u^\top \Gamma_u B_u = I$

expresses the orthonormality of the $\phi_{e,q}$. Finally, if ν_x is nondegenerate and $c^\top \Xi_A = 0$ almost surely for some A, then $(c \otimes c')^\top \Xi_{[N]} = (c^\top \Xi_A)(c'^\top \Xi_{A^c}) = 0$ almost surely for every c', hence $c = 0$: all the matrices Σ_A are nonsingular. □

Remark 5.9 (Features of the teacher are built from features of the teacher)**.** Proposition 5.8 is the precise sense in which the data-weighted features are hierarchical. Each left singular function at a bond is a *bilinear* function of the left singular functions at the two bonds below it, and at a *leaf bond*, i.e. a bond e that joins a leaf a to its parent, $A_e = \{a\}$ and $\phi_{e,q} = u_{e,q}^\top \tilde{\xi}_a$ is linear in the whitened embedding. This motivates looking for the features of the teacher in successive layers of a DNN (Hypothesis 5.27). Under the product law the composition is orthonormal, and the data-weighted SVDs at all bonds form the HSVD of a single tensor. Under dependence, each bond has its own whitening, so the data-weighted SVDs at different bonds are not the HSVD of a single tensor; the composition (5.8) survives under the assumption of Proposition 5.8(ii), but without orthonormality. Without that assumption it can fail. Take three scalar blocks on the tree $((1,2),3)$, with $\xi_a(z) = (1, z)$, $X^{(1)}$ and $X^{(2)}$ independent and non-constant, $X^{(3)} = X^{(2)}$, and the teacher $f^\star = \ell_a(x^{(1)})\, x^{(2)} + \ell_b(x^{(1)})\, x^{(3)}$, where $\ell_a = a^\top \xi_1$ and $\ell_b = b^\top \xi_1$ for linearly independent $a, b \in \mathbb{R}^2$. The environment of the leaf bond of leaf 1 always receives $x^{(2)} = x^{(3)}$, so the only left singular function there is proportional to $\ell_a + \ell_b$; at the leaf bond of leaf 2, the left singular functions span 1 and $x^{(2)}$. At the bond e above leaves 1 and 2, whose environment receives $x^{(3)}$ independently of $(x^{(1)}, x^{(2)})$ on recombined inputs, they span $\ell_a(x^{(1)})\, x^{(2)}$ and $\ell_b(x^{(1)})$, and no nonzero combination of these two functions is a combination of the products $(\ell_a + \ell_b)(x^{(1)})$ and $(\ell_a + \ell_b)(x^{(1)})\, x^{(2)}$. Finally, the two child bonds of the root define the same bipartition, with left and right exchanged, and therefore have the same singular values and atoms.

Remark 5.10 (When are the data-weighted features properties of the data?)**.** The data-weighted SVD is gauge invariant, but it is computed from the tensor $T^\star$, whereas the data only determine $f^\star$ as an element of $L^2(\nu_x)$. Let T and T' define the same function ν_x-almost surely. Since $\mathbb{E}_{\nu_e^\otimes}[g_{T-T'}^2] = \|\tilde{M}_e(T) - \tilde{M}_e(T')\|_F^2$, they have the same data-weighted SVD at e if and only if $g_{T-T'}$ also vanishes $\nu_e^\otimes$-almost surely. Hence the data-weighted SVD at e is determined by the values of the function on the data, for every tensor with the given embeddings, if and only if

$$\text{every function } g_D = \langle D, \Xi_{[N]} \rangle \text{ that vanishes } \nu_x\text{-almost surely also vanishes } \nu_e^\otimes\text{-almost surely.} \tag{5.9}$$

This holds under the product law, whenever $\nu_e^\otimes$ is absolutely continuous with respect to ν_x, and whenever ν_x is nondegenerate, since then no nonzero g_D vanishes almost surely. If ν_x and $\nu_e^\otimes$ are mutually absolutely continuous, as for jointly Gaussian blocks with a nonsingular covariance, almost sure identities are the same under both laws, and $r_e^\star$ is also the rank of $f^\star$ across the cut on the data, i.e. the smallest r such that $f^\star = \sum_{k \le r} a_k(x_{A_e})\, b_k(x_{A_e^c})$ ν_x-almost surely, with $a_k \in L^2(\nu_{A_e})$ and $b_k \in L^2(\nu_{A_e^c})$. Condition (5.9) fails when the support of ν_x is degenerate: in the example of Remark 5.9, the teachers $f^\star$ and $(\ell_a + \ell_b)(x^{(1)})\, x^{(2)}$ coincide on the data but have data-weighted ranks 2 and 1 at the bond e, and for teachers with repeated inputs the condition fails at every bond (Section 5.6). The data-weighted features are then properties of the teacher's computation, planted by the choice of $T^\star$, rather than of the function that it computes on the data.

5.3 Atoms: identity and strength

Definition 5.11 (Atoms)**.** For a bond e and $i, j \le r_e^\star$, let $\chi_{e,ij}^\star(x) := \phi_{e,i}^\star(x_{A_e})\, \psi_{e,j}^\star(x_{A_e^c})$, and $\chi_{e,q}^\star := \chi_{e,qq}^\star$. The *atom* (e, q) of the teacher is the function $\sigma_{e,q}^\star \chi_{e,q}^\star$. We call $\chi_{e,q}^\star$ the *normalized*

atom, $\phi^\star_{e,q}$ its *left factor* and $\psi^\star_{e,q}$ its *right factor*. For a set B of modes at e, the *block atom* is $m^\star_{e,B} := \sum_{q\in B} \sigma^\star_{e,q}\chi^\star_{e,q}$.

An atom is one mode of the data-weighted SVD at one bond: one "atom of computation" of the teacher. Its left factor is a function computed by the subtree below e, its right factor is a function computed by the rest of the tree, and the atom is their product, weighted by the strength $\sigma^\star_{e,q}$ with which the two sides are coupled.

Proposition 5.12 (Atoms)**.** Let e be a bond.

(i) For every joint law, $f^\star = \sum_{q\le r^\star_e} \sigma^\star_{e,q}\chi^\star_{e,q}$ almost surely, and in $L^2(\nu^\otimes_e)$: the $\chi^\star_{e,ij}$, $i,j \le r^\star_e$, are orthonormal; $\|\sigma^\star_{e,q}\chi^\star_{e,q}\| = \sigma^\star_{e,q}$ and $\|f^\star\|^2 = \sum_q \sigma^{\star 2}_{e,q}$; and, for $r < r^\star_e$, the truncation $f^\star_{e,r} := \sum_{q\le r} \sigma^\star_{e,q}\chi^\star_{e,q}$ minimizes $\|f^\star - g\|$ over the functions g of rank at most r across $(A_e \mid A^c_e)$, i.e. of the form $g = \sum_{k\le r} a_k(x_{A_e})\, b_k(x_{A^c_e})$, with $\|f^\star - f^\star_{e,r}\|^2 = \sum_{q>r} \sigma^{\star 2}_{e,q}$.

(ii) If X_{A_e} and $X_{A^c_e}$ are independent, in particular under the product law, all the statements of (i) hold in $L^2(\nu_x)$.

Proof. (i) The expansion is (5.6), and orthonormality is Proposition 5.2(iii). The truncation property is the Eckart–Young theorem (Eckart and Young, 1936), which holds verbatim for Hilbert–Schmidt operators, applied to the operator K of the proof of Proposition 5.2: its approximations of rank at most r are exactly the kernels of rank at most r across the cut. (ii) In this case $\nu^\otimes_e = \nu_x$. □

Dependent blocks. Under dependence, the energies of Proposition 5.12(i) are energies on recombined inputs. On the data, the inner products $\langle \chi^\star_{e,ij}, \chi^\star_{e,kl}\rangle = \mathbb{E}[\phi^\star_{e,i}\phi^\star_{e,k}\psi^\star_{e,j}\psi^\star_{e,l}]$ involve mixed moments across the cut, so the atoms are correlated, $\|f^\star\|^2 \neq \sum_q \sigma^{\star 2}_{e,q}$ in general, and $\sigma^{\star 2}_{e,q}$ is not the part of $\mathbb{E}[f^\star(X)^2]$ carried by the atom. The truncation need not be optimal either. For instance, if $X_{A^c_e}$ is a deterministic function of X_{A_e}, every function of x coincides on the data with a function of x_{A_e} alone, hence has rank one across the cut; the best approximation of rank one in $L^2(\nu_x)$ is then $f^\star$ itself, whereas the truncation $f^\star_{e,1}$ generally has a positive error. Section 5.4 defines output observables that take these correlations into account.

Identity and strength. The normalized factors $\phi^\star_{e,q}$ and $\psi^\star_{e,q}$ specify *which* computation the atom performs on each side of the interface; the singular value specifies its *strength*. Under the product law, $\sigma^{\star 2}_{e,q}$ is the part of the teacher's second moment $\mathbb{E}[f^\star(X)^2] = \|f^\star\|^2$ carried by the atom, and replacing the truncation $f^\star_{e,q-1}$ by $f^\star_{e,q}$, i.e. adding the atom to the stronger atoms at e, lowers the population loss $\mathcal{L}(f) = \frac{1}{2}\mathbb{E}[(f(x)-y)^2]$ by $\sigma^{\star 2}_{e,q}/2$. Under dependence, these statements hold on recombined inputs, while on the data the decrease of the loss due to an atom also depends on its correlations with the other atoms. In all cases, a student can have acquired the identity of an atom, for instance by computing its factors internally, without having amplified it in its output; conversely, its output can contain a weak version of the right atom. Direction acquisition and amplification must therefore be tracked separately (Diagonal TTN note, 2026, Sections 3.2 and 9.4).

Remark 5.13 (Degenerate atoms)**.** If $\sigma^\star_{e,q}$ is a simple singular value, then $(\phi^\star_{e,q}, \psi^\star_{e,q})$ is unique up to a simultaneous change of sign, which leaves $\chi^\star_{e,q}$ unchanged: the atom is canonical. If instead a set B of $k \ge 2$ singular values equals a common value $\sigma^\star_B$, then $(U^\star_{e,B}R,\, V^\star_{e,B}R)$ is another valid choice of the corresponding singular vectors for any $R \in \mathrm{O}(k)$ (Remark 2.15). The canonical objects are then the left and right subspaces $\mathrm{span}\{\phi^\star_{e,q} : q \in B\}$ and $\mathrm{span}\{\psi^\star_{e,q} : q \in B\}$, and the block atom $m^\star_{e,B} = \sigma^\star_B \sum_{q\in B} \chi^\star_{e,q}$, which is invariant because $\sum_q (UR)_{:,q} \otimes (VR)_{:,q} = \sum_q u_q \otimes v_q$ when $RR^\top = I$. Individual atoms inside a degenerate block have no separate meaning, and Example 2.42 shows that such blocks occur in natural tasks. When singular values are distinct but their gaps

are comparable to the estimation error, the cluster should also be treated as a block, since singular functions are stable only relative to the gaps (Remark 2.17). The block atom is then still stable, because it corresponds to the part of $\tilde{M}_e^\star$ whose singular values lie in a window separated from the rest of the spectrum. The observables below are defined for a partition of the modes at e into such blocks, and reduce to single atoms for simple singular values.

Remark 5.14 (Atoms are attached to bonds)**.** For each bond e, Proposition 5.12 decomposes the same function $f^\star$ into features with strengths. Under the product law, the features are orthonormal, exactly as in (1.1), with $\chi_m = \chi^\star_{e,q}$ and $\lambda_m = \sigma^{\star 2}_{e,q}$; under dependence, they are orthonormal only in the decoupled geometry, and the coefficients of (1.1) must be replaced by coordinates of projections (Section 5.4). Different bonds give different decompositions. The teacher therefore supplies not one flat dictionary of features but one decomposition per bond, and these decompositions are linked by the composition (5.8). Accordingly, observables are indexed by a bond and a mode, or a block.

5.4 Output observables: has the predictor acquired an atom?

The most direct question is whether the predictor f_t of a student contains the atom $\sigma^\star_{e,q}\chi^\star_{e,q}$. The distance $\|f_t - \sigma^\star_{e,q}\chi^\star_{e,q}\|$ is not the right observable: a student that has learned this atom *and* the other ones is far from it. The right operation is a projection, which measures the component of f_t along the atom. Since a student can also couple the teacher's factors in ways that the teacher does not, pairing $\phi^\star_{e,i}$ with $\psi^\star_{e,j}$ for $i \neq j$, we measure all pairings at once. Under dependence, the products $\chi^\star_{e,ij}$ are not orthonormal in $L^2(\nu_x)$, so inner products with them are not coordinates. All output observables are therefore defined through orthogonal projections in $L^2(\nu_x)$, the geometry of the loss; they reduce to inner products when the two sides of the cut are independent.

Definition 5.15 (Coupling matrix and acquisition coefficients)**.** Let $\Phi^\star_e := (\phi^\star_{e,1}, \dots, \phi^\star_{e,r^\star_e})^\top$, $\Psi^\star_e := (\psi^\star_{e,1}, \dots, \psi^\star_{e,r^\star_e})^\top$, and let $\chi^\star_e := \mathrm{vec}(\Phi^\star_e \Psi^{\star\top}_e)$ be the vector of the products $\chi^\star_{e,ij}$. The *joint Gram matrix* at e is $\Gamma_e := \mathbb{E}[\chi^\star_e(X)\,\chi^\star_e(X)^\top]$, with entries $\langle \chi^\star_{e,ij}, \chi^\star_{e,kl}\rangle$, and we let $\mathcal{D}_e := \mathrm{span}\{\chi^\star_{e,ij} : i,j \le r^\star_e\}$. The *coupling matrix* of $f \in L^2(\nu_x)$ at e is the matrix $C_e[f] \in \mathbb{R}^{r^\star_e \times r^\star_e}$ defined by

$$\mathrm{vec}\, C_e[f] := \Gamma_e^+\, \mathbb{E}[f(X)\,\chi^\star_e(X)]. \tag{5.10}$$

Fix a partition of the modes $\{1, \dots, r^\star_e\}$ into blocks B (Remark 5.13), let $\mathcal{A}_e := \mathrm{span}\{m^\star_{e,B}\}$ and $\Lambda_e := \big(\langle m^\star_{e,B}, m^\star_{e,B'}\rangle\big)_{B,B'}$, and let

$$a_e[f] := \Lambda_e^+ \big(\langle f, m^\star_{e,B}\rangle\big)_B,$$

with components $a_{e,B}[f]$. Along a trajectory f_t, the *output acquisition coefficient* of the block B is $a_{e,B}(t) := a_{e,B}[f_t]$, and that of the atom (e,q) is $a_{e,q}(t) := a_{e,\{q\}}(t)$ when $\{q\}$ is a block.

When X_{A_e} and $X_{A_e^c}$ are independent, Proposition 5.16(iv) below shows that $\Gamma_e = I$, $C_e[f]_{ij} = \langle f, \chi^\star_{e,ij}\rangle$, and

$$a_{e,q}(t) = \frac{C_e[f_t]_{qq}}{\sigma^\star_{e,q}} = \frac{\langle f_t, \chi^\star_{e,q}\rangle}{\sigma^\star_{e,q}}, \qquad a_{e,B}(t) = \frac{\langle S^\star_{e,BB}, C_e[f_t]_{BB}\rangle_F}{\|S^\star_{e,BB}\|_F^2},$$

where $C_{BB'}$ denotes the submatrix of a matrix C with rows in B and columns in B'; for a block of equal singular values, $a_{e,B}(t) = \mathrm{tr}\, C_e[f_t]_{BB} / \mathrm{tr}\, S^\star_{e,BB}$. By Remark 5.14, $a_{e,q}$ is then the coefficient (1.1) of the decomposition of $f^\star$ at e. In general, $a_e[f]$ collects the coefficients of the best approximation of f by a combination of the teacher's block atoms at e: $a_{e,B}(t) = 0$ means that the block is absent from the predictor, and $a_{e,B}(t) = 1$ that it is present with the teacher's strength. The coupling matrix contains more information than these coefficients.

Proposition 5.16 (Exact error decomposition at a bond)**.** Let Π_e, $P_{\mathcal{W}_e}$ and $P_{\mathcal{A}_e}$ be the orthogonal projectors of $L^2(\nu_x)$ onto $\mathcal{D}_e$, onto $\mathcal{W}_e := \operatorname{span}\{\chi^\star_{e,ij} : i,j \in B \text{ for some block } B\}$ and onto $\mathcal{A}_e$, so that $\mathcal{A}_e \subseteq \mathcal{W}_e \subseteq \mathcal{D}_e$. For every joint law and every $f \in L^2(\nu_x)$:

(i) $\Pi_e f = \Phi_e^{\star\top} C_e[f]\, \Psi_e^\star$, $\|\Pi_e f\|^2 = \operatorname{vec}(C_e[f])^\top \Gamma_e \operatorname{vec}(C_e[f])$, and $P_{\mathcal{A}_e} f = \sum_B a_{e,B}[f]\, m^\star_{e,B}$;

(ii) if Γ_e is nonsingular, $C_e[f^\star] = S^\star_e$, and if Λ_e is nonsingular, $a_e[f^\star] = \mathbf{1}$. If Γ_e (resp. Λ_e) is singular, a nonzero combination of the products $\chi^\star_{e,ij}$ (resp. of the block atoms) vanishes almost surely, and the coordinates of $C_e[f]$ (resp. of $a_e[f]$) along such combinations are not determined by f;

(iii) the excess error splits into four nonnegative terms,

$$\|f - f^\star\|^2 = \underbrace{\|P_{\mathcal{A}_e}(f - f^\star)\|^2}_{\text{amplitudes}} + \underbrace{\|(P_{\mathcal{W}_e} - P_{\mathcal{A}_e})f\|^2}_{\text{within blocks}} + \underbrace{\|(\Pi_e - P_{\mathcal{W}_e})f\|^2}_{\text{across blocks}} + \underbrace{\|(I - \Pi_e)f\|^2}_{\text{outside}}, \tag{5.11}$$

and, if Λ_e is nonsingular, the first term is $(a_e[f] - \mathbf{1})^\top \Lambda_e\, (a_e[f] - \mathbf{1})$;

(iv) if X_{A_e} and $X_{A_e^c}$ are independent, in particular under the product law, then $\Gamma_e = I$, $C_e[f]_{ij} = \langle f, \chi^\star_{e,ij}\rangle$ and $\Lambda_e = \operatorname{diag}(\|S^\star_{e,BB}\|_F^2)_B$, and, writing $C = C_e[f]$, $S = S^\star_e$ and $a_B = a_{e,B}[f] = \langle S_{BB}, C_{BB}\rangle_F / \|S_{BB}\|_F^2$, the four terms of (5.11) are

$$\sum_B \|S_{BB}\|_F^2 (a_B - 1)^2, \qquad \sum_B \left(\|C_{BB}\|_F^2 - a_B^2 \|S_{BB}\|_F^2\right), \qquad \sum_{B \neq B'} \|C_{BB'}\|_F^2, \qquad \|f\|^2 - \|C\|_F^2;$$

the first two add up to $\sum_B \|C_{BB} - S_{BB}\|_F^2$, and when all blocks are single modes they are $\sum_q \sigma^{\star 2}_{e,q}(a_{e,q} - 1)^2$ and 0;

(v) if the blocks are unions of sets of equal singular values, then $\mathcal{A}_e$, $\mathcal{W}_e$, $\mathcal{D}_e$, the block atoms, the coefficients $a_{e,B}[f]$, the four terms of (5.11) and the singular values of $C_e[f]$ do not depend on the choices of singular functions allowed by Remark 5.13.

Proof. (i) If $\Gamma_e c = 0$, then $\mathbb{E}[(c^\top \chi^\star_e)^2] = 0$, so $c^\top \chi^\star_e = 0$ almost surely and $c^\top \mathbb{E}[f \chi^\star_e] = 0$: the vector $\mathbb{E}[f\chi^\star_e]$ lies in the range of Γ_e. The normal equations $\Gamma_e \beta = \mathbb{E}[f\chi^\star_e]$ of the least-squares problem $\min_\beta \|f - \beta^\top \chi^\star_e\|$ are therefore solved by $\beta = \operatorname{vec} C_e[f]$, so $\Pi_e f = \operatorname{vec}(C_e[f])^\top \chi^\star_e = \Phi_e^{\star\top} C_e[f] \Psi^\star_e$ and $\|\Pi_e f\|^2 = \beta^\top \Gamma_e \beta$. The same argument applies to the block atoms and Λ_e. (ii) By (5.6), $f^\star = \operatorname{vec}(S^\star_e)^\top \chi^\star_e = \sum_B m^\star_{e,B}$ almost surely, so $\mathbb{E}[f^\star \chi^\star_e] = \Gamma_e \operatorname{vec} S^\star_e$ and $(\langle f^\star, m^\star_{e,B}\rangle)_B = \Lambda_e \mathbf{1}$, which gives the first statement. If $\Gamma_e c = 0$ with $c \neq 0$, then $c^\top \chi^\star_e = 0$ almost surely, so the coefficient vectors β and $\beta + c$ define the same function; the same holds for Λ_e. (iii) Since the subspaces are nested, $P_{\mathcal{A}_e}$, $P_{\mathcal{W}_e} - P_{\mathcal{A}_e}$, $\Pi_e - P_{\mathcal{W}_e}$ and $I - \Pi_e$ are the orthogonal projectors onto mutually orthogonal subspaces whose sum is $L^2(\nu_x)$. Applying them to $f - f^\star$ gives an orthogonal decomposition, and the last three annihilate $f^\star \in \mathcal{A}_e$. The expression of the first term follows from (i) and (ii). (iv) By Proposition 5.12(ii), the $\chi^\star_{e,ij}$ are orthonormal, so $\Gamma_e = I$ and $C_e[f]_{ij} = \langle f, \chi^\star_{e,ij}\rangle$. The block atoms $m^\star_{e,B} = \sum_{q \in B} \sigma^\star_{e,q} \chi^\star_{e,qq}$ are orthogonal, with $\|m^\star_{e,B}\|^2 = \|S_{BB}\|_F^2$ and $\langle f, m^\star_{e,B}\rangle = \langle S_{BB}, C_{BB}\rangle_F$, which gives Λ_e, a_B, $\|P_{\mathcal{A}_e}(f - f^\star)\|^2 = \sum_B (a_B - 1)^2 \|S_{BB}\|_F^2$ and $\|P_{\mathcal{A}_e} f\|^2 = \sum_B a_B^2 \|S_{BB}\|_F^2$. Since $\mathcal{W}_e$ and $\mathcal{D}_e$ have orthonormal bases of products, $\|P_{\mathcal{W}_e} f\|^2 = \sum_B \|C_{BB}\|_F^2$ and $\|\Pi_e f\|^2 = \|C\|_F^2$, and the last three terms follow by taking differences. Adding the first two and using $a_B \|S_{BB}\|_F^2 = \langle S_{BB}, C_{BB}\rangle_F$ gives $\sum_B \|C_{BB} - S_{BB}\|_F^2$; for single modes, $a_q = C_{qq}/\sigma^\star_{e,q}$ and the second term vanishes. (v) An allowed change of singular functions replaces $\Phi^\star_e$ and $\Psi^\star_e$ by $R^\top \Phi^\star_e$ and $R^\top \Psi^\star_e$, where R is block diagonal and orthogonal: a sign for each simple singular value and an orthogonal matrix for each set of equal singular values. It leaves the spans of $\{\phi^\star_{e,q} : q \in B\}$ and $\{\psi^\star_{e,q} : q \in B\}$

unchanged, hence $\mathcal{W}_e$ and $\mathcal{D}_e$, and it leaves each block atom $m^\star_{e,B} = \Phi^{\star\top}_{e,B} S^\star_{e,BB} \Psi^\star_{e,B}$ unchanged, because $R_B^\top S^\star_{e,BB} R_B = S^\star_{e,BB}$; hence $\mathcal{A}_e$, Λ_e, $a_e[f]$ and the four terms are unchanged. Finally, $C_e[f] \mapsto R^\top C_e[f] R$, which preserves singular values. □

Interpretation. For noiseless labels, $\|f - f^\star\|^2 = 2\mathcal{L}(f)$; with independent zero-mean label noise, it is twice the excess loss. For every bond e, (5.11) therefore splits the excess loss of a student into four nonnegative parts:

(a) an *amplitude* error: the student contains the teacher's block atoms with the wrong amplitudes, typically because an atom has not yet been amplified ($a_{e,B} < 1$) or has overshot;
(b) an error *within blocks*: inside the factor spaces of a block, the student departs from the teacher's block atom, for instance by pairing the left and right subspaces of a degenerate block differently from the teacher (a coupling block $C_{BB} = \sigma^\star_B R$ with an orthogonal $R \neq I$ has the right strengths and subspaces but the wrong pairing), or by giving the modes of a cluster the wrong relative strengths; this term vanishes when all blocks are single modes;
(c) an error *across blocks*: the student couples a left factor of the teacher with a right factor that belongs to a different block;
(d) an *outside* error: the energy of f outside the teacher's factor spaces at e, i.e. the student uses functions of x_{A_e} or of $x_{A_e^c}$ that the teacher does not use across this cut.

Each bond gives a different split of the same number. Matching the coefficients $a_{e,q}$ while carrying a large outside error is not a success (Surrogate note, 2026, Section 7.1), so the four terms should be reported together, for instance divided by $\|f^\star\|^2$, so that they add up to the relative error $\|f_t - f^\star\|^2 / \|f^\star\|^2$. Under dependence, the decomposition is still exact, but the subspaces spanned by different pairings are no longer orthogonal: the nesting $\mathcal{A}_e \subseteq \mathcal{W}_e \subseteq \mathcal{D}_e$ attributes the variance that a mispaired product shares with a teacher atom to the atom. The terms must then be computed with the Gram matrices Γ_e and Λ_e: inserting the raw inner products $\mathbb{E}[f \chi^\star_{e,ij}]$ into the formulas of (iv) gives terms that no longer add up to $\|f - f^\star\|^2$, and that can be negative. If only the leading $r < r^\star_e$ modes are used, the decomposition holds with the projectors onto the corresponding smaller subspaces, applied to $f - f^\star$; the terms then also contain the omitted part of the teacher.

Identifiability. The matrix Λ_e is small, of the size of the number of blocks, and it is nonsingular as soon as the block atoms are linearly independent in $L^2(\nu_x)$; the joint Gram matrix Γ_e has size $r^{\star 2}_e$. Both are nonsingular when ν_x is nondegenerate, because the products $\chi^\star_{e,ij}$, and hence the block atoms, are functions g_D with linearly independent coefficient tensors D. Both are computed once from the teacher, analytically or on a large independent sample, and their condition numbers should be reported. When one of them is singular or badly conditioned, the corresponding coordinates are not identifiable (Proposition 5.16(ii)), but the projections are: blocks should then be merged, and the subspace terms of (5.11), which are always identifiable, reported. Under dependence, the coefficients of non-orthogonal features also depend on the family of features against which they are computed: $a_{e,q}$ controls for the other teacher atoms, whereas the diagonal entry $C_e[f]_{qq}/\sigma^\star_{e,q}$ of the coupling matrix controls in addition for the mispaired products. The two coincide when the two sides of the cut are independent.

Remark 5.17 (Removing an atom from the output is not informative)**.** One may try to assess whether an atom is useful to the student by removing the student's own component along it and measuring the resulting change of the loss. When the two sides of the cut are independent, this component is $C_e[f]_{qq}\, \chi^\star_{e,q}$, and for $f^{(-q)} := f - C_e[f]_{qq}\, \chi^\star_{e,q}$, expanding the square, with or without

independent label noise, and using $\langle f - f^\star, \chi^\star_{e,q}\rangle = C_e[f]_{qq} - \sigma^\star_{e,q}$ gives

$$\mathcal{L}(f^{(-q)}) - \mathcal{L}(f) = \frac{\sigma^{\star 2}_{e,q}}{2}\Big(1 - (1 - a_{e,q})^2\Big), \qquad a_{e,q} = \frac{C_e[f]_{qq}}{\sigma^\star_{e,q}}.$$

This is a function of $a_{e,q}$ alone: it is positive for $0 < a_{e,q} < 2$, maximal at $a_{e,q} = 1$, and negative for $a_{e,q} < 0$ or $a_{e,q} > 2$. It carries no information beyond $a_{e,q}$. Under dependence, it also involves the correlations between the components of the output, but still not how the student computes the atom. Causal questions have to be asked inside the network (Section 5.8).

5.5 The functional HSVD of a student

The coupling matrix describes a student in the coordinates of the teacher. A complementary, teacher-free description uses the student's own decomposition across the same cuts, in the geometry of Proposition 5.2.

Definition 5.18 (Functional HSVD)**.** Let e be a bond of the teacher's tree and $f \in L^2(\nu_e^\otimes)$. The operator $K_e[f] : L^2(\nu_{A_e^c}) \to L^2(\nu_{A_e})$, $(K_e[f]\psi)(x_{A_e}) := \mathbb{E}[f(x_{A_e}, X'_{A_e^c})\,\psi(X'_{A_e^c})]$ with $X'_{A_e^c} \sim \nu_{A_e^c}$, is Hilbert–Schmidt, with Hilbert–Schmidt norm $\|f\|_{L^2(\nu_e^\otimes)}$, and its SVD gives the Schmidt decomposition

$$f(x) = \sum_{q \geq 1} s_{e,q}(f)\,\phi^f_{e,q}(x_{A_e})\,\psi^f_{e,q}(x_{A_e^c}) \qquad \text{in } L^2(\nu_e^\otimes), \tag{5.12}$$

with $s_{e,1}(f) \geq s_{e,2}(f) \geq \cdots \geq 0$ and orthonormal families $(\phi^f_{e,q})_q$ in $L^2(\nu_{A_e})$ and $(\psi^f_{e,q})_q$ in $L^2(\nu_{A_e^c})$. The family of these decompositions over the bonds of the teacher's tree is the *functional HSVD* of f (Surrogate note, 2026, Section 7.4), (Diagonal TTN note, 2026, Section 9.2).

The functional HSVD only uses the values of f, so it is defined in the same way for a DNN, for a TTN on any tree and for a polynomial surrogate, and its ranks are not bounded by any width. By Proposition 5.2(iii), the functional HSVD of $f^\star$, and that of any TTN student on the teacher's tree with the same leaf embeddings, is its data-weighted HSVD. For TTNs, the data-weighted hierarchical features of Definition 5.1 and the functional HSVD are therefore the same object, computed in two ways: from the cores and the moments of the data, or from the values of the function. This is what makes the comparison between a TTN and a DNN meaningful at the level of decompositions. The computational HSVD of Definition 2.34, which describes how a polynomial network computes, is a different object (Section 5.6).

Proposition 5.19 (What the functional HSVD measures)**.** Let e be a bond and $f, g \in L^2(\nu_e^\otimes)$.

(i) $|s_{e,q}(f) - s_{e,q}(g)| \leq \|f - g\|_{L^2(\nu_e^\otimes)}$ for every q.

(ii) If X_{A_e} and $X_{A_e^c}$ are independent, in particular under the product law, then (5.12) is the Schmidt decomposition of f in $L^2(\nu_x)$; it depends only on f as an element of $L^2(\nu_x)$, and $\sum_q s_{e,q}(f)^2 = \|f\|^2$.

(iii) In general, the functional HSVD depends on the values of f on the support of $\nu_e^\otimes$, which can be larger than the support of ν_x. If ν_x has a density w with respect to $\nu_e^\otimes$ and $w \geq c > 0$ $\nu_e^\otimes$-almost everywhere, then $\|h\|_{L^2(\nu_e^\otimes)} \leq c^{-1/2}\|h\|_{L^2(\nu_x)}$ for every h, so that $|s_{e,q}(f) - s_{e,q}(g)| \leq c^{-1/2}\|f - g\|_{L^2(\nu_x)}$: functions that are close on the data have close functional HSVDs.

Proof. (i) The cut operator is linear in f, so $K_e[f] - K_e[g] = K_e[f - g]$, whose operator norm is at most its Hilbert–Schmidt norm $\|f - g\|_{L^2(\nu_e^\otimes)}$; apply Weyl's inequality for the singular values of compact operators. (ii) In this case $\nu_e^\otimes = \nu_x$. (iii) $\|h\|^2_{L^2(\nu_x)} = \int h^2 w\, d\nu_e^\otimes \geq c\,\|h\|^2_{L^2(\nu_e^\otimes)}$; combine with (i). □

Dependent blocks. Proposition 5.19 makes precise what the functional HSVD of a student measures when the blocks are dependent. It is a decomposition on recombined inputs, in which the student is evaluated on combinations $(x_{A_e}, x'_{A_e^c})$ of the two sides of different data points. If these combinations lie in the support of the data, as for jointly Gaussian blocks, they are merely reweighted; otherwise the student is evaluated where it has never been trained. In the extreme case where $X_{A_e^c}$ is a deterministic function of X_{A_e}, every function has rank one across the cut in $L^2(\nu_x)$, so no decomposition of $L^2(\nu_x)$ across the cut can separate the teacher's modes, while the functional HSVD still does, using values of the student that the training data do not constrain. Under dependence, the functional HSVD therefore asks whether a student has learned the teacher's *computation*, in which the two sides of each bond interact only through the bond, rather than only its values on the data. This is a meaningful question for the surrogate hypothesis, and the teacher, TTN students and DNN students can all be compared in this geometry, but the answer is not a decomposition of the training objective. It should be labelled as such, and complemented by the on-distribution output observables of Section 5.4; Proposition 5.19(iii) indicates when the two agree.

Comparison with the teacher. The strengths $s_{e,q}(f_t)$ are compared with the $\sigma^\star_{e,q}$, and the directions through the alignment of the leading singular subspaces,

$$\mathrm{Al}_{e,r}(t) := \frac{1}{r}\,\mathrm{tr}\,\big(P^{f_t}_{e,r} P^\star_{e,r}\big) \in [0,1],$$

where $P^{f_t}_{e,r}$ and $P^\star_{e,r}$ are the orthogonal projectors of $L^2(\nu_{A_e})$ onto the spans of the r leading left singular functions of f_t and of $f^\star$, with r chosen at a gap of both spectra (Diagonal TTN note, 2026, Section 9.4); right subspaces are compared in the same way. Alignment measures the acquisition of directions, and singular values measure amplification. Both are stable: by Proposition 5.19(i), $|s_{e,q}(f_t) - \sigma^\star_{e,q}| \le \|f_t - f^\star\|_{L^2(\nu_e^\otimes)}$, and the deviation of the subspaces is controlled by the Davis–Kahan theorem relative to the gap (Local TN note, 2026, Section 8.3). When the two sides of the cut are independent, the coupling matrix sees the part of this decomposition that lies in the teacher's factor spaces: the singular values of $C_e[f]$ are those of $\Pi_e f$, and $s_{e,q}(\Pi_e f) \le s_{e,q}(f)$ for every q, because the cut operator of $\Pi_e f$ is that of f multiplied on both sides by orthogonal projectors. Under dependence, Π_e is a projection in $L^2(\nu_x)$ while the functional HSVD lives in $L^2(\nu_e^\otimes)$, and no such inequality holds in general.

Estimation. Draw $x^{\mathrm{L}}_1, \dots, x^{\mathrm{L}}_{n_L} \sim \nu_{A_e}$ and $x^{\mathrm{R}}_1, \dots, x^{\mathrm{R}}_{n_R} \sim \nu_{A_e^c}$ independently, for instance as the A_e-parts and the A_e^c-parts of two disjoint sets of data points, and form the crossed-grid matrix

$$\hat K_e[f] := \left[\frac{f(x^{\mathrm{L}}_k, x^{\mathrm{R}}_l)}{\sqrt{n_L n_R}}\right]_{k \le n_L,\ l \le n_R}. \tag{5.13}$$

Every entry is evaluated at a recombined input, which is exactly what the decoupled geometry requires, so no conditional sampler is needed, even under dependence. If $\hat K_e[f] = \hat U \hat S \hat V^\top$, the singular values estimate the $s_{e,q}(f)$, and $\sqrt{n_L}\,\hat U_{:,q}$ and $\sqrt{n_R}\,\hat V_{:,q}$ estimate the values of $\phi^f_{e,q}$ at the x^{L}_k and of $\psi^f_{e,q}$ at the x^{R}_l (Local TN note, 2026, Section 8.4). Indeed, substituting (5.12) gives $\hat K_e[f] = \sum_q s_{e,q}(f)\, p_q w_q^\top$ with $p_q = (\phi^f_{e,q}(x^{\mathrm{L}}_k)/\sqrt{n_L})_k$ and $w_q = (\psi^f_{e,q}(x^{\mathrm{R}}_l)/\sqrt{n_R})_l$, and these vectors are orthonormal up to sampling error by the law of large numbers. Letting first n_R and then n_L tend to infinity reduces the consistency of the leading singular values to the convergence of the spectra of kernel matrices to those of integral operators (Koltchinskii and Giné, 2000). The same grid should be used for the teacher and all students. Since the $n_L n_R$ entries share rows and columns, uncertainty should be estimated with independent grids or by resampling rows and columns, not by

an i.i.d. bootstrap of the entries; a 64×64 grid with a stability check at 128×128 is a reasonable start (Surrogate note, 2026, Sections 9 and 10.2).

Remark 5.20 (Cost, and proxies for large inputs)**.** The size of $\hat{K}_e$ is chosen by the analyst and does not grow with the input dimension; the matrix costs $n_L n_R$ evaluations of the student. The number of samples needed is set by the number of modes to resolve, by their gaps and by the tails of f, rather than by $\dim x$. When even this is too expensive, a Galerkin version needs fewer evaluations. Let Φ^D and Ψ^D be finite families of functions of x_{A_e} and of $x_{A_e^c}$, orthonormal in $L^2(\nu_{A_e})$ and $L^2(\nu_{A_e^c})$ and containing the teacher's singular functions. The SVD of $C_e^D[f] := \mathbb{E}_{\nu_e^\otimes}[f(X)\,\Phi^D(X_{A_e})\Psi^D(X_{A_e^c})^\top]$ is the functional SVD of the projection of f onto $\operatorname{span}\Phi^D \otimes \operatorname{span}\Psi^D$ in $L^2(\nu_e^\otimes)$, and the captured fraction $\|C_e^D[f]\|_F^2/\|f\|^2_{L^2(\nu_e^\otimes)}$ certifies how much of f the dictionaries resolve. Natural dictionaries add products of whitened leaf embeddings, or orthonormal polynomials of the blocks, to the teacher's singular functions (Local TN note, 2026, Section 8.2). As the dictionaries become complete, the projections converge to f, and their singular values and well-separated singular subspaces converge to those of (5.12). The expectation defining $C_e^D[f]$ is over recombined inputs, and its Monte Carlo estimate from M of them has error $O(M^{-1/2})$ when the integrands have finite variance; under the product law, recombined inputs are ordinary samples. Under dependence, the SVD of the matrix $\mathbb{E}_{\nu_x}[f\,\Phi^D\Psi^{D\top}]$ computed on ordinary samples is *not* a decomposition of f: when ν_x has a density w with respect to $\nu_e^\otimes$, it is the matrix $C_e^D[fw]$, which mixes the student with the dependence of the data.

Remark 5.21 (Scope: real data)**.** For real data, the blocks are dependent and their joint law is unknown, which is part of the reason for using DNNs in the first place. Recombined inputs $(x_k^{\mathrm{L}}, x_l^{\mathrm{R}})$ can then be far from the data, and the functional HSVD probes how a student extrapolates rather than what it has learned on the data. Conditional samplers of $\nu(dx_{A_e^c} \mid x_{A_e})$, when they are trustworthy, give access to on-distribution operators such as $\psi \mapsto \mathbb{E}[f(X)\,\psi(X_{A_e^c}) \mid X_{A_e} = \cdot\,]$, but their SVDs describe f weighted by the dependence of the data, as in Remark 5.20, not an expansion of f. Comparisons in representation space (Section 5.7) need no recombination at all (Surrogate note, 2026, Section 6.5). In the teacher–student benchmark, the joint law is chosen, and the functional HSVD under the product law serves as a ground truth against which the more scalable observables can be validated.

5.6 Teachers with repeated inputs

Consider now a teacher whose legs all receive the same input, $f^\star(x) = \langle T^\star, \bar{x}^{\otimes N}\rangle$ with $\bar{x} = (1, x)$, for instance the tied TTN of a deep polynomial network (Theorem 2.30); for a deep monomial network, $\bar{x}$ is replaced by x (Proposition 2.24). This is a *computational* TTN. By Remark 2.25, its legs are formal copies of one variable $X \sim \mu$, with $\bar{X} := (1, X)$, and the joint law of the legs is the image of μ under $x \mapsto (x, \dots, x)$, which is supported on the diagonal. This is not a small perturbation of the physical setting:

(i) *The geometry of the loss.* By (5.7), $\langle g_T, g_{T'}\rangle_{L^2(\mu)} = \operatorname{vec}(T)^\top \Sigma_{[N]} \operatorname{vec}(T')$, now with $\Sigma_{[N]} = \mathbb{E}[\bar{X}^{\otimes N}(\bar{X}^{\otimes N})^\top]$, which contains the moments of X up to order $2N$. For Gaussian X, its entries are computed exactly by Wick contractions, but they do not reduce to products of leaf covariances. Since $\bar{X}^{\otimes N}$ is a symmetric tensor, $\Sigma_{[N]}$ is singular, and the population loss depends on T only through $\operatorname{Sym} T$ (Remark 2.31(v)).

(ii) *No physical cut.* The two sides of a bond are functions of the same variable, so every function of x has rank one across any cut in $L^2(\mu)$. Condition (5.9) fails: if D is antisymmetric in two legs that lie on opposite sides of the bond, $\langle D, \bar{x}^{\otimes N}\rangle$ vanishes identically, but in general not

on recombined inputs. Different tensors computing the same function therefore have different decompositions across the bonds, as in Example 2.40.

Two kinds of observables are then needed: function-level observables, which depend only on the predictor and on μ, and computation-level observables, which describe the tied TTN itself.

Function track. For a law μ with enough moments, let $\mathcal{P}_k \subseteq L^2(\mu)$ be the space of polynomials of degree at most k, and $\mathcal{H}_k := \mathcal{P}_k \ominus \mathcal{P}_{k-1}$ the orthogonal complement of $\mathcal{P}_{k-1}$ in $\mathcal{P}_k$. Every $f \in L^2(\mu)$ has the orthogonal *degree decomposition* $f = \sum_{k \le n} P_{\mathcal{H}_k} f + (I - P_{\mathcal{P}_n})f$, for every n, and a polynomial teacher of degree at most N is the sum of its *degree components* $f_k^\star := P_{\mathcal{H}_k} f^\star$, $k \le N$, with $\|f^\star\|^2 = \sum_k \|f_k^\star\|^2$. For Gaussian μ, $\mathcal{H}_k$ is the k-th Wiener chaos, spanned by the Hermite polynomials of total degree k (Janson, 1997); for a general law, an empirically orthonormalized polynomial dictionary is used. The degree components, their energies and finer decompositions inside each component are function-level objects: they depend only on $f^\star$ and μ, and they are defined in the same way for every student. For instance, for standard Gaussian μ, the quadratic component is $f_2^\star(x) = x^\top Q x - \operatorname{tr} Q$ for a symmetric matrix Q, and it decomposes into the orthonormal features $((w_i^\top x)^2 - 1)/\sqrt{2}$, along the eigenvectors w_i of Q, with coefficients $\sqrt{2}\,\lambda_i(Q)$. The output observables of Section 5.4 apply with these features in place of the atoms: acquisition coefficients $a_k(t) := \langle f_t, f_k^\star \rangle / \|f_k^\star\|^2$, and the split of $\|f_t - f^\star\|^2$ into the errors $\|P_{\mathcal{H}_k}(f_t - f^\star)\|^2$ of each degree and the residual $\|(I - P_{\mathcal{P}_N})f_t\|^2$, all estimated on ordinary samples. A crossed grid would instead evaluate the multilinear extension on pairs (x, x') of different inputs, which depends on the off-diagonal part of T and not on the function; it should not be used as a function-level observable.

Computation track. The decompositions across the bonds of the tied TTN describe how the teacher computes. The Euclidean HSVD of Definition 2.34 is one such description. Definition 5.1, applied to the law of the legs on the diagonal, gives a data-weighted one: by Proposition 5.2(iii), it is the Schmidt decomposition of the function $(x, x') \mapsto \langle T^\star, \bar{x}^{\otimes K^\ell} \otimes \bar{x}'^{\otimes (N - K^\ell)} \rangle$ in $L^2(\mu \otimes \mu)$ at a bond of height ℓ, in which the subtree below the bond receives copies of x and its environment receives copies of an independent input x'. For a polynomial network, it has a simple expression in terms of the network.

Proposition 5.22 (Data-weighted computational HSVD of a polynomial network)**.** In the setting of Theorem 2.30, assume that $\mathbb{E}\|X\|^{2N} < \infty$, let $1 \le \ell \le L - 1$, let $f_{>\ell}$ be the map computed by the layers above ℓ, so that $f_\theta = f_{>\ell} \circ h_\ell$, and let $m_\ell := K^{L-\ell}$. Define

$$\bar{g}_\ell(x) := \Big(m_\ell f_\theta(x) - \big(h_\ell(x) - \mathbb{E}h_\ell\big)^\top \nabla f_{>\ell}(h_\ell(x)), \;\; \operatorname{Cov}(h_\ell)^{1/2}\, \nabla f_{>\ell}(h_\ell(x)) \Big) \in \mathbb{R}^{1+d_\ell},$$

where $\mathbb{E}h_\ell$ and $\operatorname{Cov}(h_\ell)$ are taken under $X \sim \mu$. Then, at every bond of height ℓ, the squared data-weighted singular values of $T(\theta)$ are the nonzero eigenvalues of $m_\ell^{-2}\, \mathbb{E}[\bar{g}_\ell(X)\, \bar{g}_\ell(X)^\top]$, and, if w_q are corresponding orthonormal eigenvectors, the left singular functions are

$$\phi_{\ell,q}(x) = w_q^\top \Big(1, \;\; \big(\operatorname{Cov}(h_\ell)^{1/2}\big)^+ \big(h_\ell(x) - \mathbb{E}h_\ell\big) \Big).$$

Proof. Let e be a bond of height ℓ. By the proof of Theorem 2.30, the message of the subtree below e is $h_e(x) = \Phi_\ell(\bar{x}, \dots, \bar{x}) = \bar{h}_\ell(x)$. Let $\mathcal{F}_\ell$ be the m_ℓ-linear form computed by the vertices above height ℓ, whose arguments are the bonds of height ℓ, and let $F_\ell(\bar{h}) := \mathcal{F}_\ell(\bar{h}, \dots, \bar{h})$; it is a homogeneous polynomial of degree m_ℓ on H_ℓ, and $F_\ell(1, h) = f_{>\ell}(h)$ by (2.10). The environment message of e at an input x' is the linear form obtained from $\mathcal{F}_\ell$ by leaving the argument of e open and setting all other arguments to $\bar{h}_\ell(x')$. As in the proof of Proposition 2.33, $\mathcal{F}_\ell$ is invariant under the

automorphisms of the tree above height ℓ, which act transitively on its arguments, so the m_ℓ partial maps of this form coincide, and $n_e(x') = m_\ell^{-1}\nabla F_\ell(\bar{h}_\ell(x'))$. By Euler's identity for homogeneous polynomials, the derivative of F_ℓ along the constant coordinate is $m_\ell f_{>\ell}(h) - h^\top \nabla f_{>\ell}(h)$ at $(1, h)$. Now let E be the invertible matrix with $E(a, h) = (a, h - a\,\mathbb{E}h_\ell)$, and apply Proposition 5.6 to the factorization $T^{\langle A_e\rangle} = (F_e E^\top)(H_e E^{-1})^\top$, whose messages are $h'_e = Eh_e = (1, h_\ell - \mathbb{E}h_\ell)$ and $n'_e = E^{-\top} n_e$. The second moment of h'_e is $\Sigma_e^{h'} = \mathrm{diag}(1, \mathrm{Cov}(h_\ell))$, and a direct computation gives $(\Sigma_e^{h'})^{1/2} n'_e(x') = m_\ell^{-1}\bar{g}_\ell(x')$. Under the decoupled law, the environment receives an input $X' \sim \mu$, so $(\Sigma_e^{h'})^{1/2}\Sigma_e^{n'}(\Sigma_e^{h'})^{1/2} = m_\ell^{-2}\mathbb{E}[\bar{g}_\ell \bar{g}_\ell^\top]$. Its nonzero eigenvalues are the squared singular values of $(\Sigma_e^{h'})^{1/2}(\Sigma_e^{n'})^{1/2}$, whose left singular vectors are the w_q, and Proposition 5.6 gives the singular functions. □

The data-weighted computational HSVD at height ℓ thus ranks the whitened directions of the representation of layer ℓ, augmented by the constant direction, by the mean squared sensitivity of the rest of the network to them. The lower-right block of $\mathbb{E}[\bar{g}_\ell \bar{g}_\ell^\top]$ is the average gradient outer product AGOP_ℓ of Definition 5.28 below. The decomposition is invariant under the hidden-unit symmetries of Remark 2.26, which act on the bond by gauge transformations, and it is estimated from samples of X, with one forward and one backward pass per sample. For a DNN with a standard activation, AGOP_ℓ is defined in the same way, although it is no longer the matrix of a cut SVD; for its polynomial proxy and its TTN surrogate (Section 2.7), Proposition 5.22 applies exactly. The spectra and leading eigenspaces of AGOP_ℓ at each layer are therefore computation-level observables shared by a DNN and its polynomial counterparts. Like the Euclidean HSVD, however, the data-weighted computational HSVD is an invariant of the computation, not of the function: in Example 2.40, the two networks compute the same function but have data-weighted ranks 1 and 2 at height one, for every law of x with a density that is positive on a nonempty open set.

Comparing students with a teacher with repeated inputs. Function-level observables are computed for every student from its predictor, in the same way. Computation-level observables are compared through representations: the teacher's hidden states $\bar{h}^\star_\ell$ and computational singular functions $\phi^\star_{\ell,q}$, which are functions of x, are probed in the layers of each student (Section 5.7), and the AGOP spectra of the students are compared with each other. Agreement in computation-level observables is evidence about mechanisms, but it does not identify copy-cut singular values with canonical features of the input–output function.

5.7 Representation observables: where is an atom represented?

Output observables say what a student computes, not where it computes it (Local TN note, 2026, Section 8.5). Let $h_\ell(x) \in \mathbb{R}^{d_\ell}$ be the representation at layer ℓ of a DNN student, as in (2.6), with $h_0 = x$. For a physical teacher, we ask whether the left factor $\phi^\star_{e,q}$, the right factor $\psi^\star_{e,q}$ and the normalized atom $\chi^\star_{e,q}$ can be read off h_ℓ. These are functions of the whole input, by composition with $x \mapsto x_{A_e}$ or $x \mapsto x_{A_e^c}$, so decoding them from $h_\ell(x)$ is well defined. The first two targets test whether the factors are available, and the third whether their product has been formed. For a teacher with repeated inputs, the targets are the teacher's computational singular functions $\phi^\star_{\ell,q}$, its augmented hidden states $\bar{h}^\star_\ell$, as a vector target, and its degree components $f^\star_k$ (Section 5.6). As decoders we use affine maps, because the next layer acts on h_ℓ through an affine map.

Definition 5.23 (Linear acquisition scores)**.** Let $h(x) \in \mathbb{R}^d$ be a representation with $\mathbb{E}\|h(X)\|^2 < \infty$. For a target $\zeta \in L^2(\nu_x)$ with $\mathrm{Var}\,\zeta > 0$, let

$$R^2_h[\zeta] := 1 - \frac{\min_{w\in\mathbb{R}^d,\, b\in\mathbb{R}} \mathbb{E}\big[(\zeta(X) - w^\top h(X) - b)^2\big]}{\mathrm{Var}\,\zeta} \in [0, 1]$$

be the population coefficient of determination of the best affine decoder. For a vector target $Z(x) \in \mathbb{R}^p$, let

$$R^2_h[Z] := \frac{1}{p_Z} \operatorname{tr}\Big(\operatorname{Cov}(Z)^+ \operatorname{Cov}(Z,h) \operatorname{Cov}(h)^+ \operatorname{Cov}(h,Z)\Big), \qquad p_Z := \operatorname{rank} \operatorname{Cov}(Z) \geq 1.$$

For layer ℓ of a student at time t, we write $R^2_\ell[\zeta](t)$. The three scalar scores of the atom (e,q) are $R^2_\ell[\phi^\star_{e,q}]$, $R^2_\ell[\psi^\star_{e,q}]$ and $R^2_\ell[\chi^\star_{e,q}]$; for a block B, the vector targets are $(\phi^\star_{e,q})_{q\in B}$, $(\psi^\star_{e,q})_{q\in B}$ and the products $(\chi^\star_{e,ij})_{i,j\in B}$.

Proposition 5.24 (Closed forms and invariances)**.** Let $h^{\mathrm{w}} := (\operatorname{Cov}(h)^{1/2})^+(h - \mathbb{E}h)$ be the whitened representation.

(i) $R^2_h[\zeta] = \operatorname{Cov}(h,\zeta)^\top \operatorname{Cov}(h)^+ \operatorname{Cov}(h,\zeta)/\operatorname{Var}\zeta = \|\mathbb{E}[h^{\mathrm{w}}\zeta]\|^2/\operatorname{Var}\zeta$, and $w = \operatorname{Cov}(h)^+ \operatorname{Cov}(h,\zeta)$ is an optimal decoder.

(ii) $R^2_h[Z]$ is the sum of the squared canonical correlations between h and Z (Hotelling, 1936), divided by p_Z. Equivalently, let $Z^{\mathrm{w}} := (\operatorname{Cov}(Z)^{1/2})^+(Z - \mathbb{E}Z)$ be the whitened target; then $p_Z\, R^2_h[Z]$ is the total variance of Z^{w} explained by the best affine function of h. For a scalar target, $R^2_h[Z] = R^2_h[\zeta]$ with $\zeta = Z$.

(iii) $R^2_h[\zeta]$ and $R^2_h[Z]$ do not change when h is replaced by $Qh + c$ and Z by $RZ + c'$, for invertible matrices Q and R. In particular, they are invariant under permutations and rescalings of hidden units, and the block scores $R^2_h[Z]$ are invariant under the rotations within degenerate blocks of Remark 5.13; the scalar scores of individual atoms inside a degenerate block are not.

Proof. (i) The optimal b centres the decoder, so the minimum equals $\min_w \mathbb{E}[(\zeta - \mathbb{E}\zeta - w^\top(h - \mathbb{E}h))^2]$. If $\operatorname{Cov}(h)w = 0$, then $w^\top(h - \mathbb{E}h) = 0$ almost surely, hence $w^\top \operatorname{Cov}(h,\zeta) = 0$. Thus $\operatorname{Cov}(h,\zeta)$ lies in the range of $\operatorname{Cov}(h)$, the normal equations $\operatorname{Cov}(h)w = \operatorname{Cov}(h,\zeta)$ are solved by $w = \operatorname{Cov}(h)^+ \operatorname{Cov}(h,\zeta)$, and the minimum is $\operatorname{Var}\zeta - \operatorname{Cov}(h,\zeta)^\top \operatorname{Cov}(h)^+ \operatorname{Cov}(h,\zeta)$. The second expression follows from $\mathbb{E}[h^{\mathrm{w}}\zeta] = (\operatorname{Cov}(h)^{1/2})^+ \operatorname{Cov}(h,\zeta)$, which holds because h^{w} is centred. (ii) The canonical correlations are the singular values of $(\operatorname{Cov}(h)^{1/2})^+ \operatorname{Cov}(h,Z)(\operatorname{Cov}(Z)^{1/2})^+$, and by cyclicity of the trace the sum of their squares is the trace in the definition of $R^2_h[Z]$. By (i) applied to each coordinate, the same trace is the total variance of the whitened target explained by h. (iii) Canonical correlations and best affine predictions depend only on the spaces of affine functions of h and of Z, which invertible affine maps leave unchanged. □

None of these definitions involves the law of the blocks: they apply verbatim under dependence and for repeated inputs.

Reading the three scores. High scores $R^2_\ell[\phi^\star_{e,q}]$ and $R^2_\ell[\psi^\star_{e,q}]$ with a low $R^2_\ell[\chi^\star_{e,q}]$ mean that both factors are available at layer ℓ but that their product has not been formed as an explicit feature; a later rise of $R^2_\ell[\chi^\star_{e,q}]$ means that the composition has become explicit. At the last hidden layer, $f_\theta = c^\top h_L + b_{\mathrm{out}}$ is itself an affine readout, so a student with $f_\theta \approx f^\star$ represents the sum of the atoms there, but not necessarily each atom separately.

Remark 5.25 (What linear decodability does and does not mean)**.** Linear decodability is deliberately restrictive: it says that a feature has become explicitly and cheaply accessible to the next layer. A low score does not show that the information is absent, since it can be encoded nonlinearly. Conversely, unrestricted nonlinear decoders are too permissive: if h_ℓ is an injective function of x, every function of x can be decoded from it, whether or not the network has computed it. A controlled hierarchy of decoders interpolates between the two, for instance polynomial decoders of a low degree k, which are linear decoders on $(h, h^{\otimes 2}, \dots, h^{\otimes k})$ so that Proposition 5.24 applies, or

small multilayer perceptrons of fixed width; the score is then reported as a function of the complexity of the decoder. In every case, the same decoder should be applied to controls (Alain and Bengio, 2016): the input $h_0 = x$, the network at initialization, and *control targets* of the same complexity that play no role in the task, for instance the singular functions of an independently drawn teacher with the same tree and ranks. A representation that decodes the control targets as well as the teacher's factors reflects the capacity of the probe and the richness of the representation rather than the learning of the task; this is the selectivity criterion of Hewitt and Liang (2019). The input baseline $R_0^2[\zeta]$ is essential. With linear leaf embeddings, the left factors at leaf bonds are linear functions of x and are perfectly decodable from the input itself, so only targets with a small input baseline say something about what the network has computed. Finally, decodability shows that a feature is present, not that it is used (Surrogate note, 2026, Section 7.3).

Remark 5.26 (Dependent blocks create a predictability confound)**.** When the blocks are dependent, a representation can decode a function of some blocks without receiving them, through their dependence on the blocks it does receive. If $h = h(x_A)$ depends only on x_A and $\zeta = \zeta(x_{A^c})$, then

$$R_h^2[\zeta] \le \frac{\operatorname{Var}\left(\mathbb{E}[\zeta \mid X_A]\right)}{\operatorname{Var}\zeta},$$

the fraction of the variance of ζ that is predictable from X_A, because no function of X_A predicts ζ better than $\mathbb{E}[\zeta \mid X_A]$; under the product law this bound is zero. For a TTN student, whose bond states only see the blocks below them (Section 5.9), this bound is the baseline against which decoding a right factor $\psi^\star_{e,q}$ from a bond state must be compared. A DNN layer sees the whole input, and the analogous control is the *residual target* $\zeta - \mathbb{E}[\zeta \mid X_{A_e}]$, which is uncorrelated with every function of x_{A_e}: a positive score for the residual of a right factor shows that the layer carries information about the other side of the cut beyond what x_{A_e} predicts. The conditional expectations are known analytically for Gaussian blocks, and are otherwise estimated by nonparametric regression on held-out data.

For a DNN, one may expect the order of the layers to reflect the order of the bonds in the tree. To state this precisely, let the *height* $\mathrm{ht}(e)$ of a bond be the height of the subtree below it, with $\mathrm{ht}(e) = 1$ for a leaf bond, and let

$$\ell_{e,q}(t;\delta) := \min\left\{\ell \ge 1 :\ R_\ell^2[\phi^\star_{e,q}](t) \ge \delta\right\} \qquad (\min\emptyset = +\infty)$$

be the first layer in which the left factor of the atom (e, q) is explicitly represented at time t.

Hypothesis 5.27 (Layer–bond correspondence)**.** In a DNN student trained on a TTN teacher, once the factors have been acquired, $\ell_{e,q}(t;\delta)$ is nondecreasing in $\mathrm{ht}(e)$: the factors of bonds closer to the leaves are first represented in earlier layers, and the factors of bonds closer to the root in later layers.

The first layer above the threshold, rather than the layer with the highest score, is the right statistic, because a feature computed early can remain decodable in all later layers, for instance through residual connections. The hypothesis is natural in view of (5.8): once the factors at the two bonds below e are linearly available, the factors at e are bilinear functions of them, which one layer with quadratic activations makes exactly available, and one layer with a standard activation approximately. It can be tested with the rank correlation between $\mathrm{ht}(e)$ and $\ell_{e,q}$ across atoms with a small input baseline. In a TTN student on the teacher's tree, the correspondence is built into the architecture (Section 5.9); in a DNN it has to be discovered, and whether it is discovered is part of the comparison.

5.8 Use and causal reliance

A feature can be represented without being used: residual connections and wide layers can carry many decodable features that the downstream computation ignores. Conversely, a student can contain an atom in its output without any single layer encoding it linearly. We therefore add observables of *use*, defined by derivatives of the downstream computation, and of *causal reliance*, defined by interventions. Raw weights are not suitable for this purpose. They are not invariant under the symmetries of hidden units (Remark 2.26), and unsupervised spectra of weight or activation Gram matrices can change considerably without revealing which computation has been learned; they should be paired with task-conditioned statistics (Literature review, 2026). All the expectations below are taken over ordinary samples of the data, whatever their law.

Write $f_\theta = f_{>\ell} \circ h_\ell$, where $f_{>\ell} : \mathbb{R}^{d_\ell} \to \mathbb{R}$ is the map computed by the layers above ℓ, and use the whitened representation $h_\ell^{\mathrm{w}} := (\mathrm{Cov}(h_\ell)^{1/2})^{+}(h_\ell - \mathbb{E}h_\ell)$, with the moments of the current checkpoint. In whitened coordinates, every unit direction in the range of $\mathrm{Cov}(h_\ell)$ has unit variance and orthogonal directions are uncorrelated, so that a perturbation of a given size means the same thing in every direction. For a target ζ with $R^2_\ell[\zeta] > 0$, the *code direction*

$$u_\ell[\zeta] := \frac{\mathbb{E}[h_\ell^{\mathrm{w}}\zeta]}{\|\mathbb{E}[h_\ell^{\mathrm{w}}\zeta]\|}$$

is the direction of the optimal linear decoder (Proposition 5.24(i)), and the *code* $u_\ell[\zeta]^\top h_\ell^{\mathrm{w}}(x)$ has unit variance and squared correlation $R^2_\ell[\zeta]$ with ζ. For a vector target Z, the code directions form an orthonormal basis $U_\ell[Z]$ of the span of the columns of $\mathbb{E}[h_\ell^{\mathrm{w}} Z^\top]$.

Definition 5.28 (Downstream sensitivity)**.** Let $\partial_\ell f(x) := \mathrm{Cov}(h_\ell)^{1/2}\,\nabla f_{>\ell}(h_\ell(x))$ be the gradient of the output with respect to the whitened representation, and let $\mathrm{AGOP}_\ell := \mathbb{E}[\partial_\ell f(X)\,\partial_\ell f(X)^\top]$ be the corresponding average gradient outer product (cf. Radhakrishnan et al., 2024). The *sensitivity* of the output to the code of ζ, and its normalized version, are

$$\mathrm{sens}_\ell[\zeta] := u_\ell[\zeta]^\top \mathrm{AGOP}_\ell\, u_\ell[\zeta] = \mathbb{E}\Big[\big(\partial_\ell f(X)^\top u_\ell[\zeta]\big)^2\Big], \qquad \overline{\mathrm{sens}}_\ell[\zeta] := \frac{\mathrm{sens}_\ell[\zeta]}{\mathrm{tr}(\mathrm{AGOP}_\ell)/\operatorname{rank}\mathrm{Cov}(h_\ell)},$$

and, for a vector target, $\mathrm{sens}_\ell[Z] := \mathrm{tr}\,\big(U_\ell[Z]^\top \mathrm{AGOP}_\ell\, U_\ell[Z]\big)$.

The sensitivity is the mean squared first-order change of the output per standard deviation of the code. The denominator of $\overline{\mathrm{sens}}_\ell$ is the average of $u^\top \mathrm{AGOP}_\ell u$ over uniformly random unit directions u in the range of the whitened representation, so $\overline{\mathrm{sens}}_\ell[\zeta] > 1$ means that the downstream computation is more sensitive to the code than to a typical direction. A high $R^2_\ell[\zeta]$ together with $\overline{\mathrm{sens}}_\ell[\zeta] \lesssim 1$ indicates a feature that is present but not preferentially used. For orientation: if h_ℓ contained $\phi^\star_{e,q}$ as a coordinate uncorrelated with the other coordinates, and $f_{>\ell}$ used it as the teacher does, through the term $\sigma^\star_{e,q}\phi^\star_{e,q}\psi^\star_{e,q}$, then $\mathrm{sens}_\ell[\phi^\star_{e,q}] = \sigma^{\star 2}_{e,q}\,\mathrm{Var}\,\phi^\star_{e,q}$. For a polynomial network, the eigendecomposition of AGOP_ℓ, augmented by the constant direction, is the data-weighted computational HSVD at height ℓ (Proposition 5.22).

The sensitivity has no sign. To know whether strengthening the code would increase the atom in the output, inject the atom along its code direction, $h_\ell^{\mathrm{w}}(x) \mapsto h_\ell^{\mathrm{w}}(x) + \varepsilon\,\chi^\star_{e,q}(x)\,u$ with $u = u_\ell[\chi^\star_{e,q}]$, and let $f^{(\varepsilon)}$ be the resulting output. Since $\frac{d}{d\varepsilon} f^{(\varepsilon)}(x)\big|_{\varepsilon=0} = \chi^\star_{e,q}(x)\,\partial_\ell f(x)^\top u$ by the chain rule, and $f \mapsto a_e[f]$ is linear, the *gain*

$$\mathrm{gain}_\ell(e,q) := \frac{d}{d\varepsilon}\Big|_{\varepsilon=0} a_{e,q}[f^{(\varepsilon)}] = a_{e,q}[\chi^\star_{e,q}\,\partial_\ell f^\top u]$$

is positive when a slightly stronger internal code would increase the output acquisition coefficient. When the two sides of the cut are independent, $\mathrm{gain}_\ell(e,q) = \mathbb{E}[\chi^\star_{e,q}(X)^2\,\partial_\ell f(X)^\top u]/\sigma^\star_{e,q}$. The gain is meaningful only once the atom is represented, i.e. once $R^2_\ell[\chi^\star_{e,q}]$ is clearly above its controls (Remark 5.25), since before that the code direction is dominated by noise. For a block B, the same definition applies to the normalized block atom $m^\star_{e,B}/\|m^\star_{e,B}\|$ and to $a_{e,B}$; both are invariant under the rotations of Remark 5.13.

Remark 5.29 (Two other Jacobian observables)**.** (a) *Transmission.* Replacing $f_{>\ell}$ by the map from h_ℓ to the whitened representation of the next layer measures whether that layer transmits the code: $\mathbb{E}\|(\mathrm{Cov}(h_{\ell+1})^{1/2})^+ J_{\ell\to\ell+1}\,\mathrm{Cov}(h_\ell)^{1/2}u\|^2$, with $J_{\ell\to\ell+1} = \partial h_{\ell+1}/\partial h_\ell$, compared with the same quantity for random directions u. Unlike $\|W_{\ell+1}u\|$, it includes the current derivatives of the activations. (b) *Input side.* If ζ is a differentiable function of h_ℓ, then by the chain rule $\nabla_x\zeta$ lies in the row space of $\partial h_\ell/\partial x$, so a fraction equal to one of $\|\nabla_x\zeta\|^2$ lying in that row space, at almost every x, is necessary for ζ to be an exact differentiable function of h_ℓ. It is uninformative whenever the layer is at least as wide as the input and its Jacobian has full column rank, which is the typical case, so we do not use it as a primary observable.

Interventions. Neither projections nor derivatives show that a student relies on a feature: a derivative is local, and a code to which the output is sensitive may be bypassed. Causal tests intervene on the code and measure the response.

Definition 5.30 (Erasure and injection)**.** Fix a layer ℓ and a target ζ, which may be a factor, an atom or the vector target of a block, with code directions $U = U_\ell[\zeta]$. The *erasure* of ζ at layer ℓ replaces h^{w}_ℓ by $(I - UU^\top)h^{\mathrm{w}}_\ell$, that is,

$$h_\ell \;\mapsto\; h_\ell - \mathrm{Cov}(h_\ell)^{1/2}\,UU^\top h^{\mathrm{w}}_\ell,$$

and passes the result to the layers above ℓ. The *injection* of a scalar target ζ along a unit direction u in the range of the whitened representation replaces $h^{\mathrm{w}}_\ell(x)$ by $h^{\mathrm{w}}_\ell(x) + \varepsilon\,\zeta(x)\,u$; a vector target is injected along an orthonormal family of directions. Once ζ is represented, u can be its code direction $u_\ell[\zeta]$; before that, the code direction is undefined or dominated by noise, and u is a fixed direction chosen independently of the data, for instance at random. Either intervention can be applied once, at a checkpoint, or kept in place during training from some time on.

Proposition 5.31 (Erasure removes all linear information)**.** If h'_ℓ denotes the erased representation, then $\mathrm{Cov}(h'_\ell, \zeta) = 0$ (and $\mathrm{Cov}(h'_\ell, Z) = 0$ for a vector target). Hence $R^2_{h'_\ell}[\zeta] = 0$: no affine function of h'_ℓ predicts ζ better than a constant.

Proof. Since $\mathbb{E}[h^{\mathrm{w}}_\ell\zeta]$ lies in the span of U, $\mathrm{Cov}(h'_\ell,\zeta) = \mathrm{Cov}(h_\ell,\zeta) - \mathrm{Cov}(h_\ell)^{1/2}\mathbb{E}[h^{\mathrm{w}}_\ell\zeta] = \mathrm{Cov}(h_\ell,\zeta) - \mathrm{Cov}(h_\ell)^{1/2}(\mathrm{Cov}(h_\ell)^{1/2})^+\,\mathrm{Cov}(h_\ell,\zeta) = 0$, because $\mathrm{Cov}(h_\ell,\zeta)$ lies in the range of $\mathrm{Cov}(h_\ell)$ (proof of Proposition 5.24). The conclusion follows from Proposition 5.24(i). □

This erasure is the closed-form linear concept erasure (LEACE) of Belrose et al. (2023), which removes all linear information about ζ with the smallest possible change of the representation, in the sense made precise there. This is why the projection is done in whitened coordinates; it also replaces the code by its mean rather than by zero. Information about ζ can survive nonlinearly, so a null effect of an erasure is weaker evidence than a positive one.

What to measure. After an erasure at a checkpoint, measure the immediate response: the change of the coupling matrices $C_e[f]$ and of the acquisition coefficients, in particular the drop of $a_{e,q}$, the change of the loss, and the collateral changes of all other atoms. When the erasure is kept during training, measure the subsequent response, chiefly the acquisition times of the atoms at the bonds above e. By (5.8), the factors of these atoms are built from the left factors at e, not from the atoms at e, which also contain the right factors $\psi^\star_{e,q}$ of the environment; the target to erase for this purpose is therefore the left factor $\phi^\star_{e,q}$, or the vector $(\phi^\star_{e,q})_{q\in B}$ for a block. The moments $\mathbb{E}h_\ell$ and $\mathrm{Cov}(h_\ell)$ and the code directions can then be frozen at the checkpoint, which applies a fixed affine map to the network, or re-estimated at every checkpoint. Proposition 5.31 holds for the moments with which the erasure is computed, so only the second version keeps the feature linearly erased while the student changes and possibly re-encodes it in other directions, and only the second comes close to blocking the feature; the two versions answer different questions. An injection of $\phi^\star_{e,q}$ made before it is represented naturally asks the converse questions: does $a_{e,q}$ rise, do the atoms above e appear earlier, does the loss decrease faster, or does the student ignore the injected code? Immediate and subsequent responses should be reported separately (Surrogate note, 2026, Section 7.6).

Relation with acceleration. Kept in place during training, the erasure and the injection of a left factor $\phi^\star_{e,q}$ implement, for a DNN, the "blocked" and "learned" interventions of Definition 1.1 on the earlier feature, the later feature being an atom (e', q') at a bond e' above e, measured by $a_{e',q'}$ with target value 1. In a TTN student, the same questions are asked by clamping cores (Diagonal TTN note, 2026, Section 3), or by applying the same erasure to the representation at a bond (Section 5.9). This is what makes the comparison mechanistic: at this level, a TTN is a faithful surrogate of a DNN only if interventions on the same teacher feature have comparable effects on both (Surrogate note, 2026, Section 7.6). A complementary intervention acts on the teacher instead of the students and does not require locating a code: weaken a prerequisite atom, for instance by replacing $f^\star$ with $f^\star - \frac{1}{2}\sigma^\star_{e,q}\chi^\star_{e,q}$, train both students on the modified teacher, and compare the shifts of the acquisition times of the atoms above e. Since the modification also changes the data-weighted HSVD at the other bonds, the atoms of the modified teacher must be recomputed; what matters is that both students are compared on the same modified teacher.

Controls. Every intervention needs controls: the same intervention along random directions of the same dimension in whitened coordinates, perturbations of matched norm, and sham interventions that run the same machinery with $U = 0$ (Diagonal TTN note, 2026, Section 3.3). The data used to estimate the whitening and the code directions should be disjoint from the data on which effects are measured. Temporal order alone is not evidence of dependence: two atoms with different strengths are learned at different times even when they are independent (Diagonal TTN note, 2026, Section 3.2). Only interventions separate such staging from a genuine dependence of one acquisition on another.

5.9 Observables for TTN students

All the observables above are defined for every student. For a TTN student g_ϑ on the teacher's tree with the same leaf embeddings, many of them can be computed exactly, to an extent that depends on the law of the data.

Product law.

(i) *Functional HSVD.* By Proposition 5.2(iv), the functional HSVD (5.12) of g_ϑ is its data-weighted HSVD, which is computed exactly by the moment recursion of Remark 5.7, without crossed grids.

(ii) *Coupling matrices.* Let $\tilde{M}_e(\vartheta) := \tilde{M}_e(T(\vartheta))$ be the data-weighted matricization of the student. Then $C_e[g_\vartheta] = U_e^{\star\top} \tilde{M}_e(\vartheta) V_e^\star$, which is computed exactly by contracting the student with the two orthonormalized halves of the teacher, and by (5.7), $\|g_\vartheta - f^\star\|^2 = \operatorname{vec}(T(\vartheta) - T^\star)^\top \big(\bigotimes_a \Sigma_a\big) \operatorname{vec}(T(\vartheta) - T^\star)$. The decomposition (5.11) is therefore exact, with no sampling error.

(iii) *Representations.* The role of a hidden layer is played by the message that the subtree below a bond e sends through it, $h_e(x_{A_e}) = F_e^\top \Xi_{A_e}(x_{A_e}) \in \mathbb{R}^{D_e}$, where F_e is the contraction of the student's cores below e (Lemma 2.8). Unlike a hidden layer of a DNN, h_e only sees the blocks below e, and by independence it is uncorrelated with every function of the other blocks. No bond sees all the blocks, and no bond sees exactly the blocks of A_e^c unless e is a child bond of the root. Hence the atoms $\chi^\star_{e,q}$ cannot be fully read off any bond, and neither can the right factors $\psi^\star_{e,q}$, except at a child bond of the root, whose right factors are functions of the blocks below the other child bond. Representation observables at bonds therefore concern left factors: the probes $R^2_{h_{e'}}[\phi^\star_{e,q}]$, for the bond $e' = e$ and for the bonds e' above e, and the corresponding code directions and erasures. Since $g_\vartheta(x) = h_e(x_{A_e})^\top n_e(x_{A_e^c})$, with $n_e = H_e^\top \Xi_{A_e^c}$ the message that the rest of the network sends through e, the gradient of the output with respect to the whitened h_e, the other blocks being fixed, is $\operatorname{Cov}(h_e)^{1/2} n_e(x_{A_e^c})$; it replaces $\partial_\ell f$ in the sensitivities. All these moments are computable exactly from the first and second moments $\mathbb{E}\xi_a$ and Σ_a of the embeddings. Finally, by Proposition 5.6, the student's own data-weighted features at e are linear readouts of h_e, as in Proposition 2.35(ii).

For (ii), Proposition 5.2(ii) applied to the student gives $g_\vartheta = \tilde{\Xi}_{A_e}^\top \tilde{M}_e(\vartheta) \tilde{\Xi}_{A_e^c}$ almost surely. Multiply by $\phi^\star_{e,i}\psi^\star_{e,j} = (u^{\star\top}_{e,i}\tilde{\Xi}_{A_e})(v^{\star\top}_{e,j}\tilde{\Xi}_{A_e^c})$ and take expectations: by independence, and since $\mathbb{E}[\tilde{\Xi}_A \tilde{\Xi}_A^\top]$ is the orthogonal projector onto the range of $\Sigma_A^{1/2}$ (proof of Proposition 5.2), which contains the singular vectors of the teacher, $C_e[g_\vartheta]_{ij} = u^{\star\top}_{e,i} \tilde{M}_e(\vartheta) v^\star_{e,j}$.

Dependent blocks. The algebraic structure is unchanged: $g_\vartheta(x) = h_e(x_{A_e})^\top n_e(x_{A_e^c})$ at every bond, and by Propositions 5.2(iii) and 5.6, the functional HSVD of the student is still its data-weighted HSVD, obtained from the $D_e \times D_e$ matrices $\mathbb{E}[h_e h_e^\top]$ and $\mathbb{E}[n_e n_e^\top]$. These contain mixed moments of the embeddings within each side, and are estimated by Monte Carlo. The same computation as above shows that $U_e^{\star\top} \tilde{M}_e(\vartheta) V_e^\star$ is now the matrix of inner products $\mathbb{E}_{\nu_e^\otimes}[g_\vartheta\, \phi^\star_{e,i}\psi^\star_{e,j}]$ in the decoupled geometry, not the on-distribution coupling matrix $C_e[g_\vartheta]$, which requires moments across the cut, as do Γ_e and the loss. Exact computations survive when the law has a compatible structure, for instance for a finite mixture of product laws (Remark 5.7). A bond state h_e is no longer independent of the blocks outside A_e: it can predict functions of them, including right factors, through the dependence of the blocks, up to the bound of Remark 5.26.

Repeated inputs. For the tied TTN of a polynomial network, the subtree state at height ℓ is the augmented hidden state $\bar{h}_\ell(x)$, and the gradient with respect to a shared core is the sum of its gradients over all occurrences. The loss is governed by the moment matrix $\Sigma_{[N]}$ of Section 5.6. The Euclidean HSVD of Definition 2.34 is computed exactly from the cores, and the data-weighted computational HSVD from samples, by Proposition 5.22; neither is an invariant of the function.

The difference between the two kinds of students is now clear. In a TTN student on the teacher's tree, the correspondence between the teacher's bonds and the student's representations is fixed by the architecture, and the question is only when each left factor and each atom are acquired. In a

DNN student, this correspondence has to be discovered (Hypothesis 5.27), and the atoms need not be localized in any layer. A TTN on a wrong tree is an intermediate control: its output observables are those of any student, but its bonds do not match those of the teacher.

5.10 Acquisition times, comparison of students, and estimation

Definition 5.32 (Acquisition times)**.** Fix a threshold $\delta \in (0,1)$ and a persistence window $\Delta \geq 0$. The *output acquisition time* of a block B at the bond e and the *representation acquisition times* of the atom (e,q) are

$$\tau^{\mathrm{out}}_{e,B}(\delta) := \inf\{t \geq 0 :\ a_{e,B}(s) \geq \delta \text{ for all } s \in [t, t+\Delta]\},$$
$$\tau^{\zeta}_{e,q}(\delta) := \inf\Big\{t \geq 0 :\ \max_{\ell \geq 1} R^2_\ell[\zeta](s) \geq \delta \text{ for all } s \in [t, t+\Delta]\Big\}, \qquad \zeta \in \{\phi^\star_{e,q}, \psi^\star_{e,q}, \chi^\star_{e,q}\},$$

with $\inf \emptyset = +\infty$; we write $\tau^{\mathrm{out}}_{e,q}$ when $B = \{q\}$, and use the vector targets $R^2_\ell[Z]$ for blocks. For a TTN student, $\tau^{\phi^\star_{e,q}}_{e,q}$ is defined with a maximum over bonds instead of layers. Under the product law, $\tau^{\psi^\star_{e,q}}_{e,q}$ and $\tau^{\chi^\star_{e,q}}_{e,q}$ are then in general infinite, since a bond only sees the blocks below it; under dependence, they can be finite through the dependence of the blocks, and must be compared with the bound of Remark 5.26.

For $\Delta = 0$ and independent sides of the cut, $\tau^{\mathrm{out}}_{e,q}$ is the acquisition time $\tau_m(\delta)$ of (1.1). The persistence window prevents noisy, transient crossings from being counted as acquisitions; with discrete checkpoints, it is a number of consecutive checkpoints. A time that is not observed before the end of training is reported as right-censored, not dropped (Surrogate note, 2026, Section 7.3). When the acquisition coefficients are not identifiable (Proposition 5.16(ii)), acquisition is defined through a statistic of projections instead, such as $1 - \|P_{\mathcal{A}_e}(f_t - f^\star)\|^2/\|f^\star\|^2$, and the conditioning of Λ_e is reported. For teachers with repeated inputs, the same definitions are used with the degree coefficients $a_k(t)$ and the representation targets of Section 5.6.

Choosing the thresholds. No threshold is canonical. One should report the curves $a_{e,q}(t)$ and $R^2_\ell[\zeta](t)$ themselves, together with acquisition times for several thresholds, for example $\delta \in \{0.1, 0.25, 0.5\}$ (Surrogate note, 2026, Section 7.3), and draw only conclusions that hold over this range. In saddle-to-saddle dynamics, transitions are typically sharp and well separated, so that orders of acquisition do not depend on δ; when two curves cross, the order depends on δ and should be reported as unresolved. For representation scores, the threshold must in addition exceed the input and initialization baselines, the scores of the control targets of Remark 5.25 and, under dependence, the bounds of Remark 5.26.

Comparing a TTN and a DNN. All the observables are functions of the predictor, or of a representation and of fixed teacher targets, so they can be computed for both students and compared in the same time variable (Hypothesis 1.2). The comparison has several levels:

(i) *trajectories*: $\sup_t |a^{\mathrm{DNN}}_{e,q}(t) - a^{\mathrm{TTN}}_{e,q}(t)|$, or a root mean square over t, for every atom, and the time course of the four terms of (5.11) at every bond;

(ii) *orders*: the rank correlation, for instance Spearman's, of the acquisition times $\tau^{\mathrm{out}}_{e,q}(\delta)$ across atoms, and whether atoms at lower bonds are acquired before atoms at higher bonds in both students;

(iii) *delays*: the delay $\tau^{\mathrm{out}}_{e,q} - \tau^{\phi^\star_{e,q}}_{e,q}$ between the representation of the left factor and the appearance of the atom in the output, which is defined for both students; for a DNN, also whether the factors are represented before their product;

(iv) *interventions*: whether erasing or injecting the same left factor, or weakening the same teacher atom, shifts the same acquisition times by comparable amounts in both students.

Functional HSVDs, through singular values and alignments, complement (i) and (ii) with teacher-free decompositions; under dependence, they compare the students on recombined inputs (Section 5.5). For teachers with repeated inputs, the same comparisons are made with the degree components in place of the atoms, and with AGOP spectra as computation-level observables.

Estimation protocol.

1. *Teacher.* Under the product law, compute the Σ_a analytically, or estimate them on a large sample independent of all other data, and compute the data-weighted HSVD at every bond exactly (Remark 5.7). Under dependence, estimate the moments Σ_e^h and Σ_e^n of the teacher's bond messages on such a sample (Proposition 5.6). Record the singular values, their gaps, and the blocks of singular values whose gaps are below the estimation error. Estimate the Gram matrices Γ_e and Λ_e once, on the same independent sample ($\Gamma_e = I$ exactly under the product law); report their condition numbers, regularize small eigenvalues, and merge blocks whose atoms are nearly collinear.
2. *Output observables.* Fix a held-out audit set $x_1, \dots, x_M$ of ordinary samples, independent of the training data, and use it at every checkpoint; reusing the same inputs reduces the noise of differences over time. Estimate the inner products $\mathbb{E}[f_t \chi_e^\star]$ and $(\langle f_t, m_{e,B}^\star \rangle)_B$, and $\|f_t\|^2$, on this set, with error bars from a bootstrap over audit examples or from independent audit sets. These estimates are unbiased, with error $O(M^{-1/2})$ when the integrands have finite variance (Diagonal TTN note, 2026, Section 9.1), and so are the estimates of $C_e[f_t]$ and $a_e[f_t]$, which are fixed linear maps of them once Γ_e and Λ_e are known. The four terms of (5.11) are quadratic, and plugging in estimates biases them. Under the product law, for instance, $\mathbb{E}\|\hat{C} - S_e^\star\|_F^2$ exceeds $\|C - S_e^\star\|_F^2$ by the sum of the variances of the entries of $\hat{C}$, of order $r_e^{\star 2}/M$, so the amplitude, within-block and across-block terms are overestimated, the outside term is underestimated, and the across-block term has a positive floor even when the student pairs the factors correctly. Sample splitting removes this bias. Each term is a difference of quantities of the form $\|Pg\|^2 = c[g]^\top G^+ c[g]$, where P is the orthogonal projector onto the span of a family of functions with Gram matrix G, and $c[g]$ is the vector of inner products of g with that family. If $\hat{c}^{(1)}$ and $\hat{c}^{(2)}$ are estimates of $c[g]$ computed on two disjoint halves of the audit set, then $\hat{c}^{(1)\top} G^+ \hat{c}^{(2)}$ is an unbiased estimate of $\|Pg\|^2$; it can be negative when the true value is small. Under dependence, raw inner products must not be inserted into the formulas of Proposition 5.16(iv), and samples should be recombined only to estimate functional HSVDs, which are then labelled as such. For TTN students, use the exact formulas of Section 5.9 where they apply.
3. *Representation observables.* At each checkpoint and layer, collect $h_\ell(x_n)$ on separate sets for fitting probes, for validation and for testing. Fit ridge-regularized probes, choose the regularization on the validation set, and report the held-out R^2. Held-out estimation is essential: the in-sample R^2 of an affine probe tends to 1 as d_ℓ approaches the number of probe examples, whatever the target. For blocks, use multivariate ridge regression on whitened targets, or canonical correlation analysis after reducing h_ℓ to its leading principal components, as in SVCCA (Raghu et al., 2017). Include the controls of Remark 5.25 and, under dependence, the bounds and residual targets of Remark 5.26; for teachers with repeated inputs, use the targets of Section 5.6.
4. *Use and interventions.* Since the output is scalar, one backward pass per audit example gives the gradient $\partial_\ell f(x)$, hence sens_ℓ, gain_ℓ and $\mathrm{tr}(\mathrm{AGOP}_\ell) = \mathbb{E}\|\partial_\ell f(X)\|^2$; Jacobian–vector products give transmissions without forming Jacobians.
5. *Statistics.* Pair the students: same teacher, same data and same seeds wherever pairing is

Level	Question	Observables
Output	Does the predictor contain the atom, with the right factors, pairing and strength?	$a_{e,B}(t)$, $C_e[f_t]$ and the decomposition (5.11); functional HSVD $s_{e,q}(f_t)$ and alignment $\mathrm{Al}_{e,r}(t)$; for repeated inputs, degree components
Representation	Can the factors and the atom be read off a layer? For a TTN: can the left factor be read off a bond?	$R^2_\ell[\phi^\star_{e,q}]$, $R^2_\ell[\psi^\star_{e,q}]$, $R^2_\ell[\chi^\star_{e,q}]$; $R^2_\ell[Z]$ for blocks; first layer $\ell_{e,q}$; for a TTN, $R^2_{h_{e'}}[\phi^\star_{e,q}]$; residual targets under dependence
Use	Is the downstream computation sensitive to the code, with the right sign?	sens_ℓ, $\overline{\mathrm{sens}}_\ell$, $\mathrm{gain}_\ell(e,q)$, transmission; spectra of AGOP_ℓ
Causal reliance	Does erasing or injecting the code of the left factor change the output, the loss, or the acquisition of the atoms above it?	erasure and injection, teacher-strength interventions; random-direction, matched-norm and sham controls

Table 2: Levels of evidence that a student has acquired an atom (e, q) of the teacher, and the corresponding observables.

meaningful. Use about five seeds to screen and at least twenty for confirmatory claims about orders of acquisition or effects of interventions, declare the primary bonds and atoms in advance, and report unresolved gaps and censored times (Surrogate note, 2026, Sections 9 and 10).

5.11 Summary

Table 2 collects the observables by level of evidence. The levels should not be collapsed: an atom can be present in the output without being localized in any layer, represented in a layer without being used, and used locally without the student relying on it. Table 3 collects what each regime of input laws allows. The hierarchical features of a TTN are its data-weighted singular functions; the decomposition of any student, DNN or TTN, is its functional HSVD; the two coincide for TTNs on the teacher's tree. Under the product law both are Schmidt decompositions in the geometry of the loss; under dependence they live on recombined inputs, and the on-distribution comparison uses Gram-corrected projections; for repeated inputs, function-level and computation-level observables must be kept apart.

For an atom that a DNN acquires hierarchically, one expects the following sequence: its factors become decodable in some layer; then their product becomes decodable; the downstream computation becomes sensitive to its code; its output coefficient $a_{e,q}(t)$ grows towards 1; and erasing the code of its left factor removes the output component and delays the acquisition of the atoms above it. A TTN student on the teacher's tree has no layer in which the product of the two factors could appear, but it has the same left factors at its bonds and the same output coefficients. The surrogate hypothesis is at its strongest if the trajectories of a TTN student predict, for a DNN student, when the left factors become accessible, when the output coefficients grow, and how interventions on one feature change the acquisition of the others.

The methodological conclusion is two-tiered. Experiments should start with the product law, where the measurement apparatus can be validated against exact and additive decompositions. The same observables, with the Gram corrections and the predictability baselines, then carry over to dependent blocks, and the function and computation tracks to repeated inputs. There, success means agreement in the degree components of the output, in the hidden and computational feature

Object	Product law	Dependent blocks	Repeated inputs
Teacher features	Data-weighted HSVD: the HSVD of one weighted TTN; Schmidt decompositions in $L^2(\nu_x)$; orthonormal composition	Data-weighted SVDs: Schmidt decompositions on recombined inputs; one whitening per bond; composition if the Gram matrices are nonsingular; properties of the data under (5.9)	Data-weighted computational HSVD = eigendecomposition of the augmented AGOP; a property of the computation, not of the function
Output observables	Coupling matrix of inner products; additive per-atom errors; optimal truncations	Gram-corrected coupling matrix and projection coefficients; nested decomposition (5.11); identifiable if Γ_e, Λ_e are nonsingular	Degree components (Wiener chaos for Gaussian inputs) and their coefficients
Student decomposition	Functional HSVD in the geometry of the loss; crossed grid	Functional HSVD on recombined inputs; close to the data if $w \geq c > 0$	No physical cut; degree decomposition; crossed grids not meaningful
Representation and use	Probes, sensitivity, erasure, injection; bond states independent of the environment	Same, on joint samples; predictability bounds and residual targets	Same; teacher hidden states and computational singular functions as targets; AGOP spectra
Exact TTN computations	Second-moment recursion on the tree	Monte Carlo for the bond-message moments; exact for mixtures of product laws	Moments up to order $2N$; Wick contractions for Gaussian inputs

Table 3: What each regime of input laws allows. Independence of the blocks is a source of orthogonality and of exact computability, not a prerequisite for comparing the training dynamics of TTNs and DNNs.

subspaces, and in the responses to interventions; it does not identify copy-cut singular values with canonical features of the input–output function.

6 First experiments

A first round of experiments would explore the training dynamics of TTNs empirically. We start with a simple teacher–student setup in which the teacher and the student are TTNs on the same binary tree, of depth between 2 and 4, that is, with between 4 and 16 leaves. We write $T^\star$ for the teacher and mark all of its quantities with a star: $M_e^\star = U_e^\star S_e^\star V_e^{\star\top}$ is the SVD of the teacher's matricization at a bond e (Definition 2.13), and $M_e = U_e S_e V_e^\top$ is that of the student, which evolves during training.

Setting. We start with small bond dimensions and ranks, for example a teacher with bond dimension 3 and rank $r = 3$, and an overparametrized student with bond dimension 4 or 5. The student is trained on the quadratic loss, with inputs sampled from $\mathcal{N}(0, I_d)$, where d is the input dimension. The entries of its cores are initialized i.i.d. from $\mathcal{N}\big(0, \epsilon^2/(D_L D_R)\big)$, where D_L and D_R are the dimensions of the left and right bonds of the core and ϵ is the initialization scale.

Teacher. We specify the spectrum of the teacher through the singular values of its matricization across one bond e, $M_e^\star = U_e^\star S_e^\star V_e^{\star\top}$. The singular values $S_e^\star = \mathrm{diag}(\sigma_{e,1}^\star, \ldots, \sigma_{e,r}^\star)$ follow a power

law with exponent ν,

$$\sigma^\star_{e,k} = \frac{k^{-\nu}}{\left(\sum_{j=1}^{r} j^{-2\nu}\right)^{1/2}}, \qquad k = 1, \dots, r,$$

where the denominator normalizes the teacher to unit norm, $\|T^\star\|_F^2 = \sum_{k=1}^{r} (\sigma^\star_{e,k})^2 = 1$. We then sample two random matrices $U_e^\star$ and $V_e^\star$ with orthonormal columns, which together with $S_e^\star$ give the SVD of $T^\star$ at e. Finally, we record the spectra that this construction induces at the other bonds.

Observables. During training, we track:

- the population loss;
- the singular values $\sigma_{e,k}$ of the student at every bond e;
- the alignment between the singular vectors of the teacher and of the student at every bond, for example $\|U_e^{\star\top} U_e\|_F$;
- the matricization M_e of the student expressed in the singular bases of the teacher, $U_e^{\star\top} M_e V_e^\star$;
- the movement of the neural tangent kernel (NTK) away from its initialization, to identify the training regime.

7 A possible range of other Experiments

The experiments have two goals. The first is to understand the training dynamics of TTNs: which dynamical regimes exist, what a TTN student acquires from a teacher, in what order and on what time scales, and which properties of the teacher, the data, the initialization and the optimizer control these. The second is to measure how far the training dynamics of DNNs is from that of TTNs, that is, to test Hypothesis 1.2 and, where it fails, to locate the failure. We pursue the two goals in parallel rather than one after the other. Experiments on TTNs are cheap, and they fix the vocabulary in which the comparison is phrased: regimes, canonical configurations, and laws for orders and time scales. Conversely, an early comparison on a few configurations keeps the study of TTNs focused on the regimes that DNNs actually visit.

All experiments are teacher–student experiments with TTN teachers, in the setting of Section 5, so that every student can be described in terms of the atoms of the teacher. Section 7.1 states the principles of the design, and Section 7.2 the common setting: teachers and data, students and their initialization, training protocols and observables. Sections 7.3–7.7 describe the experiments in five phases: the validation of the measurements; the dynamics of TTNs; the comparison of TTNs and DNNs; optimizers, noise and finite samples; and deferred experiments. Each experiment states a question, its motivation, a design, a prediction when there is one, and what its outcome changes in the rest of the program. Section 7.8 sets priorities and divides the work between two people, and Table 6 lists the experiments.

7.1 Principles

Questions rather than knobs. The list of things one could vary is long: the tree, size and spectra of the teacher, the leaf embeddings and the law of the data, the architecture, width and depth of the student, its initialization, the optimizer and its hyperparameters, the batch size, weight decay, label noise and the loss. A factorial design over these factors is out of reach, and most of its cells would be uninterpretable. Each experiment therefore starts from a question. Factors are varied one at a time around a base configuration (Table 4), and interactions between factors are studied only where a question requires them.

Exact before sampled. Under the product law (5.3), the population loss of a TTN student on the teacher's tree, its gradient and all its output observables are computed exactly from the cores and the second moments of the leaf embeddings (Remark 5.7 and Section 5.9). The population gradient flow of a TTN is therefore available without sampling error, and it is the reference dynamics. Gradient noise, finite samples and DNN students, for which population quantities must be estimated, are introduced one at a time, so that their effects are not confounded.

One change at a time. When two students disagree, we want to know why. Comparisons between architectures are organized on a grid in which neighbouring students differ in a single respect (Section 7.2.2), and the comparison asks which change first breaks the agreement.

Learnability is a controlled variable. A teacher that no student learns within the budget says nothing about dynamics. As shown in Section 7.2.1, the simplest TTN teachers are of this kind for every student, TTNs included. The difficulty of the teacher is therefore an explicit parameter, set to a learnable value in the base configuration and varied in a dedicated experiment.

Validated measurements. The observables of Section 5 are delicate: plug-in estimates of the error decomposition are biased, probes overfit, and singular subspaces are unstable near degeneracies. Every observable is checked on cases with a known answer, and the sample-based estimators are checked against exact values on TTN students, before any scientific comparison (Section 7.3).

Statistics. We follow Section 5.10: paired designs, five seeds for screening and at least twenty for confirmatory claims, primary bonds and atoms declared in advance, and acquisition times reported for several thresholds, and as right-censored when they are not observed. One rule is added for sequential dynamics. There, acquisition curves are close to step functions whose times vary across seeds, and their average over seeds is a smooth curve that no seed follows. Curves are therefore shown per seed, and aggregates are computed from event times, such as the distribution of $\tau^{\mathrm{out}}_{e,q}(\delta)$, or after aligning the curves on such times.

7.2 Common setting

7.2.1 Teachers and data

Sizes. Teachers are TTNs on balanced binary trees, or on comb trees for staircases, with $N \in \{4, 8, 16\}$ leaves, leaf embeddings of dimension at most about ten, and ranks $r^\star_e \leq 8$. Three facts motivate these small sizes. First, cost: an internal core of a binary TTN with bond dimension D has D^3 entries, so that bond dimensions of 10^3 to 10^4 give 10^9 to 10^{12} entries per core, and students several times wider are out of reach. Second, the rank across a bond is bounded by the dimensions of its matricization (Definition 2.5); at a leaf bond, $r_e \leq d_a$, so that large ranks are only possible high in the tree. Third, exactness: with a suitable order of contraction, one step of exact population gradient descent for a student of bond dimension D, against a teacher of smaller bond dimension, costs $O(ND^4)$ operations, so that thousands of runs with $N \leq 16$ and $D \leq 64$ are cheap. Deep trees, such as a balanced tree of depth ten with 1024 leaves, are not considered.

Homogeneous teachers are hard for every student. Suppose that the blocks are centred Gaussian vectors and that $\xi_a(z) = z$, the simplest instance of the setting of Section 5.1. The teacher (5.1) is then a homogeneous polynomial of degree N, linear in each block, and all its Hermite components of degree less than N vanish. Equivalently, $\mathbb{E}[f^\star \mid X_A] = 0$ for every proper subset

$A \subsetneq [N]$ of the blocks: no function of fewer than N blocks is correlated with the teacher. More generally, we call a teacher *homogeneous* when $\mathbb{E}[f^\star \mid X_A] = \mathbb{E}f^\star$ for every proper subset A, i.e. when its ANOVA components (Section 7.2.3) involving between one and $N-1$ blocks vanish; the parity of N bits is an example. Homogeneity has three consequences.

(i) For DNN students, the teacher has leap N in the sense of Abbe et al. (2023). Results on single- and multi-index models (Ben Arous et al., 2021; Abbe et al., 2023) indicate that online SGD on two-layer networks then needs a number of samples that grows like d_0^{N-1}, where d_0 is the input dimension.

(ii) For TTN students, the same obstruction appears as a high-order saddle. At the constant point of Section 7.2.2, the TTN analogue of a standard initialization of a DNN, the gradient depends on the teacher only through its mean and its components involving a single block, and the latter vanish. For the parity of N bits on a balanced tree, this rank-one point is a saddle of order at least $N/2$ (Furman et al., 2026, Proposition 5.3) (Remark 2.21).

(iii) For Gaussian blocks, a single product term $\prod_a u_a^\top x^{(a)}$ with unit vectors u_a has kurtosis 3^N, which is 6561 for $N = 8$. Stochastic gradients and empirical losses are then dominated by rare samples.

With $N = 8$ leaves, such a teacher already makes every student look as if it learned nothing, for reasons unrelated to the differences between architectures that we want to measure. We therefore use leaf embeddings that contain the constant function, $\xi_a(z) = (1, z)$ or one-hot embeddings, and control the weight that the teacher gives to products of fewer than N blocks.

Spectra are measured, not prescribed. It is tempting to specify a teacher by prescribing its hierarchical singular values at every bond, for instance as power laws. This is not possible in general. Under the product law, all bonds carry the same total energy, $\sum_q \sigma_{e,q}^{\star 2} = \|f^\star\|^2$ for every e (Proposition 5.12), and the spectra at different bonds satisfy further compatibility conditions (Hackbusch and Uschmajew, 2017), so a prescribed family of spectra need not be realizable. The constructions that easily realize a prescribed spectrum, such as orthogonally decomposable teachers $T^\star = \sum_k s_k \bigotimes_a u_{a,k}$ with orthonormal families $(u_{a,k})_k$ at every leaf, have the same spectrum (s_k) at every bond. Every bond then sees the same atoms, there is no hierarchy, and with fixed leaf features the dynamics collapses to that of a diagonal DLN (Diagonal TTN note, 2026, Proposition 4.1). We therefore specify teachers through their cores, and always compute and report their data-weighted HSVD (Remark 5.7) rather than an intended one.

Teacher families. We use four families, each for a specific purpose.

(T1) *Gapped teachers*, with at most four modes per bond and ratios of at least two between consecutive singular values at the bonds of interest. They are drawn as orthonormal TTNs (Proposition 2.14) in whitened coordinates (Proposition 5.3), with prescribed singular values at the root bond, and draws whose measured gaps at the other bonds are too small are rejected; teachers with designed rank profiles are built explicitly (Experiment E1.2). Gapped teachers resolve individual atoms and transitions: sequential dynamics and acquisition times are only defined when time scales are separated, and singular subspaces are only stable relative to their gaps (Remark 2.17).

(T2) *Residual teachers* T_ρ, which control the difficulty. Every bond carries a constant channel in addition to r feature channels. The leaf cores map the embedding to the constant and to centred features, the constant channel is propagated, and on the feature channels an internal

Component	Base configuration	Varied in
Tree	balanced binary tree with $N = 8$ leaves, hence $\lvert V\rvert = 15$ cores	E1.4, E1.5
Data	product law; uniform one-hot leaves over $q = 4$ symbols	E1.3, E1.4, E2.3
Teacher	residual teacher T_ρ with $\rho = 0.3$, $\gamma = 1$ and two feature channels (bond dimension three); gapped teachers with $r^\star_e \le 4$	E1.2–E1.6, E2.4
TTN student	teacher's tree and leaf embeddings; $D = 2\max_e r^\star_e$	E1.5
Initialization	canonical configurations of E1.1	E1.1
Training	square loss; population gradient flow (P0) for TTN students and online SGD with $B = 4096$ (P1) for the other students; no weight decay and no label noise; learning rates from pilot runs, with $\eta\lambda_{\max} \le 1$	Phase 3
Seeds	five for screening, twenty for confirmatory claims	–

Table 4: Base configuration. Every experiment departs from it only in the factors that it varies.

vertex with children messages h_L, h_R computes

$$h = \rho\,(Uh_L + Vh_R) + \gamma\,B(h_L, h_R).$$

This is the residual teacher of Surrogate note (2026, Section 6.3), written as a TTN on augmented bonds. For $\rho = 0$, only the full product survives, and T_0 is as hard as the homogeneous teachers above. As ρ grows, components involving fewer blocks gain weight, from a staircase in which every interaction order is present to a teacher dominated by its additive part. On a comb tree, T_ρ contains the staircase teachers of Diagonal TTN note (2026, Section 5) and Surrogate note (2026, Section 6.2).

(T3) *Power-law teachers*, with many modes and singular values close to $q^{-\nu}$ at the root bond. They are used for scaling laws in time (Experiment E1.3), not for individual atoms: the gaps between consecutive singular values shrink like $q^{-\nu-1}$, and neighbouring transitions overlap.

(T4) *Parity and read-once formulas* on ± 1 leaves, as hard controls. The parity of N bits is realized on the balanced tree with bond dimension two and has the high-order saddles of Furman et al. (2026) (Remark 2.21).

Input laws. Phases 1 to 3 use the product law (5.3), under which the population quantities of TTN students are exact and the data-weighted HSVD is the Schmidt decomposition in the geometry of the loss (Proposition 5.2(iv)). The leaves are uniform one-hot variables over q symbols, which is the setting of Furman et al. (2026), ± 1 variables, or Gaussian blocks $x^{(a)} \sim \mathcal{N}(0, \Sigma_a)$ in $\mathbb{R}^b$ with $\xi_a(z) = (1, z)$. Discrete leaves have bounded embeddings and avoid the heavy tails of products of Gaussian variables; Gaussian blocks connect with the theory of SGD on Hermite expansions, and are used with $N \le 8$. Leaf covariances are isotropic, except in Experiment E1.3, which needs anisotropic ones. A full covariance Σ_x with correlations across blocks would put us in the dependent-block regime of Section 5, where exact computations are lost and the observables need Gram corrections; it is deferred to Phase 4. Singular leaf covariances are handled by the pseudoinverses of Definition 5.1, but add nothing at this stage.

7.2.2 Students and their initialization

Initializations of TTNs. For a DNN with a given parameterization, the rich, intermediate and lazy regimes are reached by changing a single scale. For a TTN, a single scale does not determine

the regime, for four reasons.

(i) *The order of the saddle at the origin grows with the tree.* The predictor g_ϑ is multilinear in the $|V|$ cores, so all derivatives of the loss of order less than $|V|$ vanish at $\vartheta = 0$. From cores of scale ς, the time needed to leave the origin grows like $\varsigma^{-(|V|-2)}$, as computed by Diagonal TTN note (2026, eq. (4.7)) in the diagonal case. A balanced binary tree with $N = 8$ leaves has $|V| = 15$ cores. The range of scales in which the dynamics is sequential and still completes within a budget is therefore narrow, and it shrinks as the tree grows.

(ii) *Balance is conserved.* By Proposition 2.12, the loss is invariant under $C_u \mapsto C_u \times_e X$, $C_v \mapsto C_v \times_e X^{-\top}$, where C_u and C_v are the two cores incident to a bond e. Write $(C)_{(e)}$ for the matricization of a core whose columns are indexed by the mode of e. Differentiating the invariance at $X = I$ gives $(C_u)_{(e)}^\top \nabla_{(C_u)_{(e)}} \mathcal{L} = \big(\nabla_{(C_v)_{(e)}} \mathcal{L}\big)^\top (C_v)_{(e)}$, and hence, along gradient flow,

$$Q_e := (C_u)_{(e)}^\top (C_u)_{(e)} - (C_v)_{(e)}^\top (C_v)_{(e)} \in \mathbb{R}^{D_e \times D_e} \tag{7.1}$$

is constant; with weight decay of strength λ, $\dot{Q}_e = -2\lambda Q_e$ (cf. Du et al., 2018), (Furman et al., 2026, Appendix F). How the initial scale is distributed over the cores is therefore a conserved property of the trajectory, not a detail, and unbalanced initializations are known to change the regime of DLNs and two-layer networks (Kunin et al., 2024).

(iii) *Laziness requires centring.* A large initialization gives lazy training only if the initial predictor is controlled, as in the scaling $\alpha(g_\vartheta - g_{\vartheta(0)})$ with $\alpha \to \infty$ of Chizat et al. (2019). The centred model is again a TTN: the difference of two TTNs on the same tree is a TTN with block-diagonal cores and twice the bond dimensions, initialized with equal blocks.

(iv) *Isotropic initializations lack the low-degree bias of DNNs.* A DNN at a standard initialization computes a function dominated by components of low degree, and its gradient has access to the low-degree components of the teacher. A TTN with isotropic Gaussian cores does neither. With $\xi_a(z) = (1, z)$ and whitened z, or with uniform one-hot leaves, each leaf message puts a fraction $1/d$ of its expected energy on the constant ($d = d_a$), and these fractions multiply along the tree. The expected fraction of the energy of the initial predictor carried by the ANOVA components that involve k blocks (Section 7.2.3) is therefore $\binom{N}{k}(1 - 1/d)^k d^{-(N-k)}$: the initial predictor is dominated by interactions of order about $N(1 - 1/d)$. Moreover, the gradient with respect to a leaf core contracts the residual with the message sent by the rest of the tree, which spans a random D-dimensional subspace of the functions of the other $N - 1$ blocks; for a generic teacher, the part of the residual that it sees has a relative size of about $(D/d^{N-1})^{1/2}$.

We therefore consider three families of initializations, each with a balance profile, i.e. a distribution of the scales over the heights of the tree, and a bond dimension D.

(Z) *Near zero*: all cores of scale ς. This is the analogue of the small initializations of DLNs, with a saddle of order $|V|$ at the origin.

(I) *Isotropic*: cores with independent entries of variance 1/fan-in, so that the bond messages have unit second moments, and an output multiplier α. With centring, $\alpha \to \infty$ is the lazy limit.

(C) *Near the constant*: cores that propagate a constant channel, so that $T(\vartheta_0) = c \bigotimes_a \hat{e}_a$, where $\langle \hat{e}_a, \xi_a \rangle = 1$, plus a perturbation of scale ς. The initial predictor is the constant c, at a rank-one point where the gradient depends on the teacher only through $\mathbb{E} f^\star$ and the components

Connectivity	Free multilinear core	Bilinear core	Quadratic node	SiLU or ReLU node
Teacher's tree	**TTN** (reference)	bilinear tree network	**quadratic tree network**	**SiLU tree network**
Other tree	**wrong-tree TTN**; tensor train			
Dense (copies of the input)	**untied copy-TTN**	bilinear MLP	**quadratic MLP**	**MLP**; residual MLP

Table 5: The grid of students: connectivity in rows, node map in columns. Neighbouring students differ in a single respect. The core students, in bold, are run in every comparison; the others are run when a comparison between neighbours requires them.

$\mathbb{E}[f^\star \mid X_a]$ involving a single block; components involving $k \geq 2$ blocks enter only through derivatives of higher order. This family is the TTN counterpart of the low-degree bias of DNNs, and of residual networks initialized near the identity.

Bond dimension plays the role of width. How α, ς and the learning rates must scale with D for a regime to persist as D grows, as the neural-tangent and maximal-update parameterizations do for DNNs (Yang and Hu, 2021), is not known for TTNs; Experiment E1.1 determines it empirically, so that results obtained at one bond dimension transfer to others.

The grid of students. A TTN on the teacher's tree differs from a DNN in two respects: its *connectivity*, which is the teacher's tree, so that locality is built in rather than discovered; and its *node map*, which is multilinear. If the two disagree, the disagreement cannot be attributed to either respect. We therefore arrange the students on the grid of Table 5, in which neighbouring students differ in a single respect, and ask which change first breaks the agreement with the TTN on the teacher's tree. In the language of Section 2.6, the first row contains TTNs without copies of the input, and the third row TTNs with copies.

A bilinear node computes $W[(Ah_L) \odot (Bh_R)]$ from the messages h_L, h_R of its children: it is a TTN core whose CP rank is at most the width (Pearce et al., 2024). A quadratic or SiLU node computes $\sigma(W[h_L; h_R] + b)$. Besides the bilinear cross terms, the quadratic node contains the self-interactions of h_L and of h_R and terms of lower degree, and it is a tied TTN vertex acting on copies of the augmented messages of its children (Theorem 2.30). Tree networks are locally connected networks without weight sharing, i.e. one-dimensional convolutional networks with kernel size and stride two and untied filters. In the dense row, every student receives the whole input. The untied copy-TTN is a binary TTN with N leaves, all of which receive $\bar{x} = (1, x)$, with free cores; the quadratic MLP is a deep polynomial network with $K = 2$, i.e. the tied TTN of Theorem 2.30; the bilinear MLP has layers $(Ah) \odot (Bh)$.

The grid is read as follows. If the TTN agrees with the tree networks but not with the dense students, the gap lies in the discovery of locality, and the copy-TTN or the tied TTN, rather than the TTN on the teacher's tree, is the appropriate surrogate of a dense network. If the TTN already disagrees with the SiLU tree network, the gap lies in the node map, and the quadratic tree network tells whether polynomial nodes suffice to reproduce the SiLU node, which is the premise of the polynomial surrogates of Section 2.7. The wrong-tree TTN measures how much of the behaviour of the TTN is due to the alignment of its tree with the teacher's.

7.2.3 Training protocols and observables

Training protocols. We distinguish four protocols:

(P0) population gradient flow, integrated by gradient descent with a small step on the exact population loss; it is available for TTN students under the product law;

(P1) online SGD, with fresh samples at every step and batch size B; with large B and small η, it is the closest approximation of (P0) available for DNN students;

(P2) training on a fixed sample of size n, by full-batch gradient descent or multi-epoch SGD;

(P3) Adam, AdamW or Muon in place of SGD in (P1) or (P2).

Protocols (P0) and (P1) separate optimization from generalization, (P2) adds the effects of a finite sample, and (P3) changes the geometry of the updates. Unless stated otherwise, the learning rate is the largest value on a predeclared grid for which $\eta\lambda_{\max} \le 1$ along the whole trajectory, where $\lambda_{\max}$ is the largest eigenvalue of the Hessian of the loss. The time $t = \sum_n \eta_n$ of Hypothesis 1.2 is then the time of an approximate gradient flow, and the edge-of-stability regime (Cohen et al., 2021) is avoided until it is studied on purpose. Learning rates and widths are fixed on pilot teachers with separate seeds, before the main runs (Section 5.1). Small initializations of TTNs require double precision, since the gradients scale like $\varsigma^{|V|-1}$.

Observables. At every checkpoint we record:

(O1) *Output observables* (Section 5.4): the loss, the coupling matrices $C_e[f_t]$, the acquisition coefficients $a_{e,q}(t)$ and the four terms of (5.11) at every bond, and the acquisition times of Definition 5.32 for $\delta \in \{0.1, 0.25, 0.5\}$.

(O2) *Decompositions*: the data-weighted HSVD of TTN students on the teacher's tree, computed exactly, and the functional HSVD of every student, estimated on crossed grids (5.13); their singular values, the alignments $\mathrm{Al}_{e,r}(t)$ at gaps, participation ratios, and the rank profile $(r_e(t))_e$, where $r_e(t)$ is the number of singular values at e above $10^{-3}\|f_t\|$.

(O3) *ANOVA observables.* Under the product law, every $f \in L^2(\nu_x)$ has a Hoeffding, or ANOVA, decomposition $f = \sum_{S\subseteq[N]} f_S$ into mutually orthogonal components, where f_S depends only on x_S and $\mathbb{E}[f_S \mid X_A] = 0$ unless $S \subseteq A$ (Hoeffding, 1948; Efron and Stein, 1981). We record the closed subtree energies $V_{A_e}(f) := \|\mathbb{E}[f \mid X_{A_e}]\|^2 = \sum_{S\subseteq A_e}\|f_S\|^2$ at every bond, and the noise-stability curve

$$\mathrm{Stab}_f(\omega) := \mathbb{E}\big[f(X)\, f(Y_\omega)\big] = \sum_{k=0}^{N} \omega^k E_k(f), \qquad E_k(f) := \sum_{|S|=k} \|f_S\|^2, \qquad \omega \in [0,1],$$

where Y_ω is obtained from X by resampling each block independently with probability $1-\omega$, and $(E_k(f))_k$ is the degree profile of f (cf. O'Donnell, 2014, Chapter 2). For TTN students, both are computed exactly by the moment recursion of Remark 5.7, with the leaf Gram matrices Σ_a replaced by $m_a m_a^\top$ for the leaves outside A_e, respectively by $m_a m_a^\top + \omega(\Sigma_a - m_a m_a^\top)$ for all leaves, where $m_a := \mathbb{E}\xi_a(X^{(a)})$. For the other students, they are estimated by Monte Carlo with one additional evaluation per sample (Sobol', 2001); the coefficients E_k are fitted from the curve only for small N. These observables are architecture-free and only need evaluations of the predictor, and they measure the structure that controls difficulty (Section 7.2.1). Unlike the HSVD, which decomposes the predictor across one cut at a time, they say how much of it is explained by each subtree alone, and how its energy is distributed over interaction orders.

(O4) *Dynamical diagnostics*: the kernel movement $\Delta_K(t) := \|K_t - K_0\|_F / \|K_0\|_F$ of the empirical tangent kernel on an audit set (Remark 1.4), the sharpness $\lambda_{\max}$, estimated with Hessian–vector products, the norms of the cores and, for TTN students, the conserved quantities (7.1), whose drift measures the departure from gradient flow.

Representation and use observables (Sections 5.7 and 5.8) are deferred to Phase 4, with the exception of interventions on the teacher, which require no probe.

7.3 Phase 0: validation

The observables involve whitenings, pseudoinverses, projections, crossed grids and sample splitting, and the simulators involve exponentially large tensors that are never formed. An error in any of them would be indistinguishable from a scientific finding. Phase 0 therefore checks them on cases with known answers, before any comparison.

E0.1 Algebra of the tied TTN

Question. Do the polynomial recurrence, the tied-tree contraction and the explicit coefficient tensor of a deep polynomial network agree, in values and in gradients?

Design. On a tiny network in double precision, with widths $(3, 4, 2)$ and quadratic activations, compare the three computations of Theorem 2.30, and their gradients with respect to all weights, biases and the readout (suggested tolerances: 10^{-10} for values and 10^{-8} for gradients, relative). An exactly polynomial activation, such as $\sigma(z) = z + z^2$, must give the same polynomial for every centre, and untying the cores, or averaging instead of summing the gradients of repeated occurrences, must break the agreement of gradients (Surrogate note, 2026, Section 8.1).

Outcome. Required before quadratic MLPs, polynomial proxies or local surrogates are used.

E0.2 Known answers

Question. Do the observables and the simulators reproduce results that are known exactly?

Design. Observables: for a TTN student equal to the teacher, all acquisition coefficients equal one and the four terms of (5.11) vanish; for the truncation of the teacher at a bond e to its k leading modes, $a_{e,q} = 1$ for $q \le k$ and $a_{e,q} = 0$ otherwise; for a student whose coupling block at a degenerate block B is $\sigma_B^\star R$, with an orthogonal $R \neq I$, the four terms take the values of Proposition 5.16(iv), in particular $a_{e,B} = \operatorname{tr} R/|B|$; for parity, the whole energy of the ANOVA decomposition is at order N. Dynamics: the exact amplitude dynamics of homogeneous diagonal TTNs and its acquisition times (Diagonal TTN note, 2026, Section 4), and the acceleration factor of the three-leaf residual TTN (Diagonal TTN note, 2026, Proposition 6.1).

Outcome. Tolerances for every observable, used to flag numerical problems in later runs.

E0.3 Exact against sampled

Question. Do the sample-based estimators, which are the only ones available for DNN students, agree with the exact values?

Design. Apply every sample-based estimator (audit-set coefficients with sample splitting, crossed-grid functional HSVDs, Monte Carlo ANOVA energies and noise-stability curves) to TTN students along training trajectories, and compare with the exact values of Section 5.9.

Outcome. Agreement within the estimated error bars is required. Since the exact and the sampled estimators are implemented by different people (Section 7.8), this experiment is also the integration test of the two code bases.

7.4 Phase 1: the dynamics of TTNs

Phase 1 uses TTN students on the teacher's tree under population gradient flow (P0), unless stated otherwise. Its runs are cheap, so it can afford dense sweeps and twenty seeds per configuration.

E1.1 Regime map

Question. Which dynamical regimes does a TTN student exhibit, as a function of its initialization family, scale, balance profile and bond dimension, and which measurable order parameters distinguish them?

Motivation. Every later experiment must state its regime, and Section 7.2.2 shows that a single initialization scale does not determine it. For DNNs, the regime depends on the scale, on the balance between layers, and on how the parameterization scales with width (Chizat et al., 2019; Woodworth et al., 2020; Yang and Hu, 2021; Kunin et al., 2024); for TTNs, the presence or absence of a low-degree bias adds a further axis.

Design. Gapped teachers and T_ρ with $\rho \in \{0.3, 1\}$, with $N \in \{4, 8\}$. Families (Z), (I) and (C); scales ς and multipliers α on logarithmic grids; three balance profiles (larger leaves, balanced, larger root); $D \in \{8, 16, 32, 64\}$; five seeds per cell. Order parameters: the kernel movement Δ_K at the end of training; the sequentiality Seq, i.e. the fraction of pairs of atoms whose acquisition windows $[\tau^{\text{out}}(0.1), \tau^{\text{out}}(0.9)]$ are disjoint; the number of plateaus of the loss; the degree profile at initialization and its evolution; the final alignments.

Predictions. Family (Z) at small ς: sequential acquisition, Seq ≈ 1, and plateaus whose lengths scale like $\varsigma^{-(k-2)}$ for saddles of order k. Family (C): acquisition by increasing interaction order, starting with the components that involve a single block. Family (I): for $N = 8$ and moderate D, slow learning of every teacher (Section 7.2.2, item (iv)); with centring and large α, $\Delta_K \to 0$, but the student only fits the projection of the teacher onto the tangent space at initialization, which is small unless D is large, so that the lazy regime is a control rather than a candidate surrogate. How unbalanced profiles and growing D move these boundaries is part of the question.

Outcome. A phase diagram; a scaling of the parameterization with D under which it is stable; and two or three *canonical configurations*, a sequential one, an intermediate one and, as a control, a lazy one, which are used in all later experiments.

E1.2 Anatomy of saddle-to-saddle dynamics

Question. In the sequential regime, which saddles does a TTN student visit, how long does it stay near each, and in what order does its rank profile grow?

Motivation. In DLNs, small initializations lead to the visit of saddles of increasing rank, as for a greedy low-rank algorithm (Saxe et al., 2014; Jacot et al., 2021), and hierarchical tensor factorizations have an incremental bias towards low hierarchical rank (Razin et al., 2022). In the setting of Furman et al. (2026, Theorem 5.2), all critical points that are not global minima are rank deficient (Remark 2.21). Two features of TTNs have no counterpart in DLNs. First, the rank is a profile over the bonds, and it cannot grow at a single bond. By Proposition 2.20(ii), applied with every vertex as the root, each of the three bond ranks at an internal vertex is at most the product of the other two. Hence the bonds of rank at least two meet every internal vertex in zero, two or

three bonds, and they form a union of paths between leaves. Starting from rank one, the smallest increment is a single path between two leaves a and b, which adds a pairwise interaction, as in $s_1 \bigotimes_c u_c + s_2 \, (w_a \otimes w_b) \otimes \bigotimes_{c \neq a,b} u_c$; the largest adds a new product term and raises the rank at every bond. Second, near a rank-one point, an increment along a path P requires the k_P cores along P to grow together. Heuristically, the plateau that precedes it then scales like $\varsigma^{-(k_P-2)}/\sigma_P$, where σ_P is the strength of the corresponding mode of the teacher. At small ς, paths between nearby leaves would then be favoured over long paths, and the order of acquisition would depend on the tree, not only on the singular values. Finally, the first escape from the origin plausibly targets a best rank-one approximation of the teacher, which is a nonconvex problem for tensors (Hillar and Lim, 2013). Unlike in DLNs, the first saddle may then depend on the initialization.

Design. The sequential configurations of families (Z) and (C), on gapped teachers with designed rank profiles: (a) rank two along the path between two siblings; (b) rank two along a path through the root; (c) two product terms, with rank two at every bond; (d) a competition between a weak short path and a strong long path. We measure the plateau lengths as functions of ς on logarithmic axes, whose slopes give the orders of the saddles. At each plateau, we record the gradient norm, the rank profile, and the loss of the best TTN with that rank profile, estimated by restarts of alternating least squares from the HSVD truncation, to test whether the plateau is a critical point of the rank-restricted problem. Finally, we compare the sequence of rank increments with two greedy algorithms, which add the best path increment or the best product term, and record the distribution of these sequences over twenty seeds.

Predictions. Rank increments along paths between leaves; plateau orders k_P; in design (d), a crossover from the short to the long path as ς grows, at $\sigma_{\text{long}}/\sigma_{\text{short}} \approx \varsigma^{-(k_{\text{long}}-k_{\text{short}})}$.

Outcome. A combinatorial description of the saddle-to-saddle path of TTNs, with plateau lengths predicted from the teacher and the tree, or a precise list of departures from it. In both cases, it is the reference against which DNN students are compared in Phase 2.

E1.3 What sets the order and the time scales

Question. Which spectrum of the teacher determines the order in which atoms are acquired and their acquisition times, and how does the loss decay when the teacher has many modes?

Motivation. At a bond e, three spectra of the teacher are candidates: the Euclidean one, of $M^\star_e$ (Definition 2.13); the data-weighted one, of $\Sigma^{1/2}_{A_e} M^\star_e \Sigma^{1/2}_{A^c_e}$, which gives the energy that each atom carries in the loss (Proposition 5.12); and the spectrum of the moment matrix $\Sigma_{A_e} M^\star_e \Sigma_{A^c_e}$, the matricization of $\mathbb{E}[y \, \Xi_{[N]}(X)]$, which drives the gradient near the origin (cf. Proposition 2.50). The three coincide for whitened leaves and differ otherwise. Example 5.4 conjectures that the data-weighted spectrum decides which atoms are acquired first, but a two-leaf computation points to the moment spectrum. With two leaves and one bond, $g_\vartheta(x) = x^{(1)\top} C_1 C_2^\top x^{(2)}$ and $\mathcal{L} = \frac{1}{2} \|\Sigma_1^{1/2} (C_1 C_2^\top - M^\star) \Sigma_2^{1/2}\|_F^2$. Suppose that Σ_1, Σ_2 and $M^\star$ are diagonal, with entries $\lambda_{1,q}$, $\lambda_{2,q}$ and m_q, and that the initialization is small, balanced and aligned. Then the modes decouple, and the product w_q of the two factors of mode q follows $\dot{w}_q = 2\lambda_{1,q}\lambda_{2,q}(m_q - w_q)w_q$, the dynamics of Saxe et al. (2014). Its acquisition time is inversely proportional, up to a logarithm, to the moment singular value $\lambda_{1,q} m_q \lambda_{2,q}$. The decrease of the loss when it is acquired, $\frac{1}{2}\lambda_{1,q}\lambda_{2,q} m_q^2$, is set by the data-weighted singular value instead. For the time scales, the theory of DLNs gives $\tau \propto 1/\sigma$ up to logarithms at fixed depth (Saxe et al., 2014), (Diagonal TTN note, 2026, eq. (4.7)), while E1.2 suggests an additional dependence on the tree. With many modes, if mode q is acquired at a time $\tau_q \propto 1/\sigma_q$ and $\sigma_q \propto q^{-\nu}$ with $\nu > 1/2$, the modes acquired by time t are those with $q \lesssim t^{1/\nu}$, and the loss decays like $\sum_{q > t^{1/\nu}} \sigma_q^2 \propto t^{-(2-1/\nu)}$. This links TTNs to dynamical models of neural scaling

laws (Michaud et al., 2023; Bordelon et al., 2024), and an exponent different from $2 - 1/\nu$ would indicate that the tree changes the time scales. Near-degenerate atoms may be acquired as a block before being resolved, as in the symmetric plateaus of soft committee machines (Saad and Solla, 1995); the block observables of Remark 5.13 are designed for this case.

Design. (a) Two and four Gaussian leaves with diagonal, then non-diagonal, covariances, and teachers for which the three spectra order the modes differently. The leading singular functions of the student at the end of each plateau are compared with the three candidates. (b) Gapped teachers with planted singular values at bonds of different heights. We measure $\tau^{\text{out}}_{e,q}$ as a function of $\sigma^\star_{e,q}$, of the height of e, and of N, D and ς. (c) Power-law teachers with $\nu \in \{0.75, 1, 1.5, 2\}$, in the sequential and intermediate configurations, and the exponent of the loss as a function of ν.

Predictions. (a) The moment spectrum sets the order, and the data-weighted spectrum the decrease of the loss at each acquisition. (b) $\tau^{\text{out}}_{e,q} \propto 1/\sigma^\star_{e,q}$ among atoms that require the same cores to grow, with corrections that depend on the tree. (c) Exponents close to $2 - 1/\nu$ in the sequential configuration.

Outcome. A law for orders and time scales in terms of a single spectrum of the teacher and of the tree. Run on DNN students in Phase 2, design (a) also tests whether the early dynamics of a DNN is driven by the same moment tensor, as one expects for any gradient method.

E1.4 Difficulty: interaction order, leaf dimension and parity

Question. How does the time needed to learn a teacher depend on the weight of its low-order interactions, on the leaf dimension and on N?

Motivation. Section 7.2.1 argued that homogeneous teachers are hard for every student: for DNNs because of their leap (Ben Arous et al., 2021; Abbe et al., 2023), for TTNs because of high-order saddles (Furman et al., 2026). If the two obstructions are one mechanism, TTNs and DNNs should depend on ρ, on the leaf dimension and on N in the same way (Experiment E2.4); if they are not, the difference measures the advantage that a TTN draws from its built-in tree. This experiment provides the TTN side.

Design. T_ρ with $\rho \in \{0, 0.1, 0.3, 1\}$, and parity, on discrete leaves with $N \in \{4, 8, 16\}$. To vary the leaf dimension: T_ρ on Gaussian blocks of dimension $b \in \{2, 4, 8, 16\}$ with $N \leq 8$, and the parity of one coordinate per block on blocks of b independent ± 1 coordinates, which is a sparse parity of N out of Nb bits. Families (I) and (C). We measure the times needed to reach relative errors 0.5 and 0.1, right-censored at the budget, the plateau lengths, and the degree profile $E_k(f_t)$ of the student over time.

Predictions. From family (C): for $\rho = 0$ and for parity, plateaus that grow quickly with N, consistently with saddles of order at least $N/2$ at the constant point (Furman et al., 2026, Proposition 5.3); for $\rho > 0$, a degree profile that fills from low to high orders, with much shorter plateaus. From family (I): slow learning for every ρ when $D \ll d^{N-1}$.

E1.5 Capacity and depth

Question. What does a TTN student learn when it cannot represent the teacher, how does extra bond dimension change its dynamics, and how do the depth of the tree and the number of factors along its bonds differ?

Motivation. (a) For an under-parameterized student, with $D < r^\star_e$ at some bonds, the HSVD truncation is quasi-optimal (Proposition 2.16). In DLNs, gradient flow from small initializations reaches the best approximation of the given rank, but for TTNs the rank-restricted landscape is

not known to be benign, so whether training reaches the truncation or a worse point is open. Real networks are under-parameterized with respect to real tasks, so this regime matters for the surrogate hypothesis. (b) Extra bond dimension adds directions and changes plateaus and implicit biases, as width does for DNNs. (c) "Depth" has two meanings for a TTN. The depth of the tree changes the function class and the bonds, and requires a new teacher. Replacing every bond by a chain of m square matrices instead leaves the function class and the bipartitions unchanged, but multiplies the number of factors, hence the orders of the saddles: it is the analogue of the depth of a DLN. Varying the two separately distinguishes the effect of hierarchical structure from that of multiplicative depth.

Design. (a) $D \in \{1, \dots, r^\star - 1\}$ on gapped and power-law teachers; the final loss is compared with that of the HSVD truncation and with the best of twenty restarts of alternating least squares. (b) $D/r^\star \in \{1, 2, 4, 8\}$. (c) Chains of $m \in \{1, 2, 4\}$ factors per bond on a fixed tree, and trees with $N \in \{4, 8, 16\}$ for teachers with matched spectra at the root bond.

Predictions. (c) Longer plateaus and sharper transitions as m grows, as for the depth of DLNs (Diagonal TTN note, 2026, eq. (4.7)).

E1.6 Acceleration and gating

Question. Does learning a feature accelerate the learning of the features built on it (Definition 1.1), and how does the effect depend on the residual strength ρ?

Motivation. Soft hierarchical learning is one of the reasons for studying TTNs. In a three-leaf channelwise residual TTN, learning the lower core divides the acquisition time of the upper one by $1 + \gamma^2\alpha^2/(2\rho^2)$ (Diagonal TTN note, 2026, Proposition 6.1), and staircases produce hard gating (Diagonal TTN note, 2026, Section 5). These results concern diagonal TTNs with fixed leaf features. Whether they survive with free cores, learned leaf directions and bond dimensions larger than one is not known, and it must be established before acceleration is compared between TTNs and DNNs (Experiment E2.5). In TTNs, the interventions of Definition 1.1 are implemented exactly, by clamping the cores that compute a feature.

Design. T_ρ on comb and balanced trees, with $\rho \in \{0, 0.1, 0.3, 1\}$ and γ fixed. For pairs (A, B) declared in advance, with B an atom at a bond above that of A, we measure τ_B with the cores that compute A clamped to the teacher's values (learned) and to their initial values (blocked). As controls, A is replaced by an atom of equal strength that B does not use. We also apply the teacher-strength intervention of Section 5.8, which replaces $f^\star$ by $f^\star - \frac{1}{2}\sigma_A^\star \chi_A^\star$ (Experiment E2.5).

Predictions. Hard gating, $\tau_B = \infty$ when A is blocked, at $\rho = 0$; finite acceleration for $\rho > 0$, close to the factor above in the setting of Diagonal TTN note (2026, Proposition 6.1); no effect in the controls.

7.5 Phase 2: TTNs and DNNs

Phase 2 compares the students of Table 5 on the same teachers and data, in the canonical configurations of E1.1. It starts as soon as these configurations are fixed, so that its first results can steer the rest of Phase 1.

E2.1 Realizability, capacity and calibration

Question. For each student and each canonical teacher, which widths fit the teacher, which learning rates are stable, and how are the time scales of different students related?

Motivation. An underfitting student says nothing about dynamics; there is no single notion of equal capacity for a TTN and a DNN; and the conventions for time must be fixed before any feature curve

is examined, since feature-specific or post hoc time warps would make Hypothesis 1.2 unfalsifiable (Section 5.1). The widths needed by tree and dense networks are not given by the bond dimensions of the teacher. By polarization, $u^\top h_L \, v^\top h_R = \frac{1}{4}\left[(u^\top h_L + v^\top h_R)^2 - (u^\top h_L - v^\top h_R)^2\right]$, so a quadratic node needs two units per bilinear term, and a quadratic network without biases cannot pass a feature unchanged through a layer.

Design. For every student and canonical teacher: a width sweep and a fit check, i.e. a predeclared relative test error under a generous budget, which gives the smallest width that fits; a capacity bracket, with parameter counts below, near and above that of the TTN (Surrogate note, 2026, Section 6.7); learning rates chosen on pilot teachers with separate seeds; for TTN students, the initialization family whose initial degree profile is closest to that of the DNN students, as a predeclared calibration, with the other families as controls; and a single time calibration per student, fixed in advance, which matches the initial function-space speed $\|\partial_t f_t\|$ at $t = 0$ (Section 5.1).

Outcome. Configurations for all students, frozen and recorded before E2.2.

E2.2 Comparison on the grid of students

Question. In the canonical configurations, do the students of the grid acquire the same atoms, in the same order and on comparable time scales, and which change of connectivity or node map first breaks the agreement with the TTN on the teacher's tree?

Motivation. This is the direct test of Hypothesis 1.2, organized so that a failure can be attributed (Section 7.2.2). The predictions of Phase 1 give it specific targets. For instance, if the dependence on the length of paths found in E1.2 holds for TTNs, it is a concrete point at which tree-structured and dense students should differ, since a dense network has no notion of distance between blocks.

Design. Two canonical teachers (a gapped teacher, and T_ρ with $\rho > 0$), the sequential and intermediate configurations (the lazy one as a control), and the core students of Table 5, with twenty paired seeds: same teacher, same data stream and, where meaningful, same initialization seed. Observables (O1)–(O4), and the comparisons of Section 5.10: distances between the trajectories of acquisition coefficients, rank correlations of acquisition times, delays, the four terms of (5.11) over time, noise-stability curves, and the alignment of tangent kernels.

Decision. Tolerances are declared before the runs, and revised only on pilots. Initially, the root-mean-square difference of the acquisition coefficients over atoms and time is at most 0.1, and the Spearman correlation of acquisition times is at least 0.8, both for the median over paired seeds. The first student of the grid that violates them locates the gap, and determines which students are used in Phases 3 and 4.

E2.3 What a TTN cannot see

Question. Does the dynamics of DNN students depend on properties of the input law that the dynamics of TTN students provably ignores?

Motivation. Under the product law, the population loss of a TTN student, and hence its population gradient flow, depends on the data only through the first and second moments of the leaf embeddings (Remark 5.5). Changing the law of the leaves at fixed moments leaves every TTN trajectory unchanged, whereas a DNN sees all the moments. Any change of the DNN trajectories under such a change is a direction in which no TTN with these leaf embeddings can be a global surrogate. Conversely, invariance would show that, in this regime, the DNN is driven by the same second-moment structure. The experiment is cheap, since its TTN side is known exactly in advance.

Design. A canonical T_ρ teacher with Gaussian blocks and $N \in \{4, 8\}$, and leaf laws with the same

mean and covariance: Gaussian, independent Rademacher coordinates, uniform on the sphere of radius $\sqrt{b}$, and a heavy-tailed law (Student-t with five degrees of freedom, rescaled to unit variance). TTN students under (P0), as a numerical check, and under SGD, where fourth moments enter through the noise; DNN students of the grid. We measure the shifts of acquisition times and trajectories relative to the Gaussian law.

Predictions. For TTN students, no change under (P0), and under SGD changes that grow with the kurtosis of the leaves. For DNN students we make no prediction: this is a measurement.

E2.4 Difficulty across students

Question. Do DNN students depend on the interaction order, the leaf dimension and N as TTN students do (Experiment E1.4)?

Motivation. The leap of a teacher and the order of the saddles of a TTN are two descriptions of difficulty. Agreement between the scaling exponents would indicate a shared mechanism, which is the strongest form of the surrogate hypothesis that does not require individual trajectories to match. For parities, the DNN side is partly known: SGD learns sparse parities of k out of n bits in a number of steps of order $n^{O(k)}$ (Barak et al., 2022).

Design. The teachers and sweeps of E1.4, on the core DNN students, with online SGD; the same observables, in particular the degree profiles $E_k(f_t)$, estimated from noise-stability curves.

Outcome. For each student, the exponents of the time needed to reach a given error as functions of b and N, and their dependence on ρ, compared with those of E1.4.

E2.5 Acceleration across students

Question. Does weakening a prerequisite atom delay the atoms above it by comparable amounts in TTN and DNN students?

Motivation. Agreement of unperturbed trajectories does not show that the same mechanism produces them, whereas agreement of the responses to interventions does (Surrogate note, 2026, Section 7.6). Interventions on the teacher need neither probes nor code directions (Section 5.8), so they can be applied to every student from the start of Phase 2; erasures and injections of internal codes are deferred to Phase 4. Controls separate dependence from staging, since atoms of different strengths are acquired at different times even when they are independent (Section 5.8).

Design. Canonical teachers. For atoms (e, q) declared in advance, replace $f^\star$ by $f^\star - \frac{1}{2}\sigma^\star_{e,q}\chi^\star_{e,q}$. In an orthonormal form of the data-weighted teacher whose orthogonality centre is the bond e (Propositions 2.14 and 5.3), this only halves one entry of the centre, and keeps the tree and the bond dimensions. Then recompute the atoms of the modified teacher, and measure the shifts of the acquisition times of the atoms at the bonds above e. As a control, weaken an atom of equal strength at a bond that is not below them. Twenty paired seeds.

Outcome. For each student and each pair of atoms, the shift, and its ratio to the shift in the TTN on the teacher's tree.

7.6 Phase 3: optimizers, noise and finite samples

Phase 3 changes the training protocol. It is run first on TTN students, where it is cheap and its effects can be compared with the exact population dynamics, and then, with the same protocol, on the students selected by E2.2.

E3.1 Gradient noise and learning rate

Question. How do gradient noise and large learning rates change plateau lengths, rank paths and regimes, and are the effects the same for TTN and DNN students?

Motivation. In the stochastic differential equation that approximates SGD (Li et al., 2017), the drift is the population gradient and the noise covariance scales like η/B; at fixed η, changing B changes only the noise. Near high-order saddles, where the drift is very small, fluctuations may dominate the escape and select among the paths of E1.2. Label noise does not change the population dynamics, since it only adds a constant to the population loss. It does change the noise of SGD, which then acquires an implicit bias towards flat minimizers (Damian et al., 2021), so it is only studied together with SGD. Beyond the gradient-flow regime, large learning rates lead to the edge of stability (Cohen et al., 2021), which may interact with plateaus.

Design. The canonical sequential configuration with discrete leaves, whose bounded embeddings keep the noise moderate. At fixed η, online SGD with $B \in \{\infty, 4096, 512, 64, 8\}$, where $B = \infty$ is population gradient descent; the pairs (η, B) and $(2\eta, 2B)$, which coincide in the SDE approximation; label noise with standard deviation 0, $0.1\|f^\star\|$ and $0.5\|f^\star\|$; and learning rates up to and beyond $\eta\lambda_{\max} = 2$, with the sharpness tracked.

Predictions. Shorter plateaus and more variable paths as B decreases, with larger effects for saddles of higher order; agreement between (η, B) and $(2\eta, 2B)$ at small η.

E3.2 Adaptive and orthogonalized optimizers

Question. Do Adam, AdamW and Muon change what is learned and in what order, and do they affect TTN and DNN students in the same way?

Motivation. The three optimizers transform differently under the symmetries of a TTN. Gradient descent is equivariant under orthogonal gauge transformations, i.e. $X \in \mathrm{O}(D_e)$ in Proposition 2.12, which are isometries of the parameter space that leave the loss invariant: gauge-equivalent initializations give the same function-space trajectory. Adam (Kingma and Ba, 2015; Loshchilov and Hutter, 2019) acts coordinate by coordinate and is equivariant only under permutations and changes of sign. Its trajectory therefore depends on the gauge of the initialization, and on the orientation of the teacher's features relative to the coordinate axes. Muon (Jordan et al., 2024) replaces the update of a matrix by the orthogonal factor of the polar decomposition of its momentum, i.e. it performs steepest descent in the spectral norm (Bernstein and Newhouse, 2024). It is equivariant under orthogonal transformations on both sides of the matrix, hence under orthogonal gauge, but an order-three core must first be matricized, and each matricization (the two children against the parent, or one child against the other child and the parent) defines a different optimizer. Since the polar factor gives the same step to all singular directions of the gradient, we expect Muon to weaken the separation of time scales responsible for sequential learning.

Design. (a) A random orthogonal gauge transformation applied at initialization: function-space trajectories under GD, Adam and Muon are compared with the untransformed ones. (b) Teachers whose leaf features are aligned with the coordinate axes or rotated at random, with rotation-invariant leaf laws, so that the dynamics of GD does not change. (c) The sequentiality Seq and the order of acquisition under the three optimizers in the canonical sequential configuration, with both matricizations for Muon. Each optimizer is tuned on pilot teachers with the same budget (Schmidt et al., 2021). Adaptive optimizers have no integrated learning rate, so trajectories are compared against the number of steps and, as secondary diagnostics, against the loss and the arc length in function space (Surrogate note, 2026, Section 8.5).

Predictions. (a) No change for GD and Muon, a change for Adam. (b) Faster acquisition of

axis-aligned features by Adam. (c) A lower sequentiality under Muon.

E3.3 Weight decay

Question. Which rank profile does training reach under weight decay, and which atoms does it keep?

Motivation. By (7.1), weight decay drives the conserved quantities to zero, $Q_e(t) = e^{-2\lambda t} Q_e(0)$, i.e. towards balanced parameters. At balanced parameters, the penalty $\sum_v \|C_v\|_F^2$ becomes a function of the represented tensor that penalizes hierarchical rank, as weight decay on the L factors of a DLN penalizes the Schatten quasi-norm of order $2/L$ of their product. Weight decay is therefore a rank-selection mechanism that competes with the fit, and it decides which atoms are kept when the data are noisy or finite.

Design. λ on a logarithmic grid, in the canonical configurations, under (P0) and under SGD with label noise. We record the final rank profile, the final acquisition coefficients and the decay of Q_e, as functions of λ and of $\sigma^\star_{e,q}$.

Predictions. A threshold strength, increasing with λ, below which atoms are not acquired or are lost after acquisition.

E3.4 Implicit bias on the computation

Question. When several computations realize the same function, which one does training select, and does the selection depend on the optimizer?

Motivation. Under the product law and on the teacher's tree, the data-weighted HSVD of a student that fits the teacher is that of the teacher (Proposition 5.2(iv)), so optimizers can only differ during training. For repeated inputs this is false: the computational HSVD is an invariant of the computation, not of the function (Remark 2.39), and Example 2.40 gives two exact realizations of $x_1^2 x_2^2$ with hierarchical ranks one and two. If different optimizers select different computational ranks at equal loss, the computational HSVD reflects the optimizer as much as the task, which bears directly on its use for interpretability (Question 2.41).

Design. Deep quadratic networks with two layers, trained to zero loss on $x_1^2 x_2^2$ and on small polynomial teachers with several exact realizations, with GD from small and standard initializations, Adam, Muon and weight decay. At convergence, we record the computational rank profile (Definition 2.36) and the data-weighted computational HSVD (Proposition 5.22), over twenty seeds.

Predictions. By analogy with the incremental rank bias of hierarchical tensor factorizations (Razin et al., 2022), GD from small initializations selects the realization of smallest computational rank. For the other optimizers, we make no prediction.

E3.5 Finite samples

Question. How many samples does each atom require, and what does training on a fixed sample change in the dynamics?

Motivation. The previous experiments use fresh samples, which separates optimization from generalization. Real training reuses a finite sample: some directions are then not determined by the data, the training and test trajectories separate, and, with one-hot leaves, learning becomes a tensor-completion problem. With weight decay, generalization can occur long after the training set is fitted (Power et al., 2022), which connects this setting with modular arithmetic (Example 2.42).

Design. Sample sizes n on a logarithmic grid; full-batch gradient descent and multi-epoch SGD, with and without weight decay; acquisition coefficients on training and test data; for each atom, the

smallest n at which it is acquired on test data. TTN students and the students selected by E2.2.

Predictions. Stronger atoms need fewer samples; with one-hot leaves and weight decay, delayed generalization at intermediate n.

7.7 Phase 4: deferred experiments

The following experiments are deferred, either because they are only well posed once Phases 1–3 have been run, or because they are more expensive and should be run only on the students and configurations that Phase 2 selects.

(i) *Internal mechanisms*: probes, sensitivities, erasures and injections (Sections 5.7 and 5.8), and the layer–bond correspondence (Hypothesis 5.27), for the students that agree with TTNs at the level of outputs. They explain the agreements and disagreements found in Phase 2.

(ii) *Dependent blocks*, with correlated Gaussian blocks or finite mixtures of product laws, which keep exact computations (Remark 5.7), and the Gram-corrected observables of Section 5.4.

(iii) *Teachers with repeated inputs*, such as deep polynomial networks, with the function and computation tracks of Section 5.6.

(iv) *Local surrogates* (Hypothesis 1.3): checkpoint approximation, blind local rollouts and chart refreshes (Surrogate note, 2026, Sections 8.2–8.4), at checkpoints of the DNN students of E2.2. They require E0.1.

(v) *Other losses*, such as logistic and cross-entropy losses, for which separable data drive the norms of the parameters to infinity, so that acquisition must be defined from normalized predictors.

(vi) *Attention and sequences*, with tokens as leaves, and discrete hierarchical data such as the random hierarchy model of Cagnetta et al. (2024), whose generative tree plays the role of the teacher's tree; attention without softmax is polynomial in its input and fits the framework of Section 2.6.

(vii) *Interpretability*: the modular-addition experiment of Question 2.41 and Example 2.42.

(viii) *Real data*, following the ladder of Surrogate note (2026, Section 8.8).

7.8 Priorities and organization

Priorities. Phase 0 blocks everything else. Within Phase 1, E1.1 comes first, since it fixes the canonical configurations; E1.2 and E1.3 follow, since they describe the dynamics in the sequential regime; and E1.4–E1.6 can run in parallel. Phase 2 starts as soon as the canonical configurations are fixed, with E2.1 and the cheap E2.3, so that the comparison steers the rest of Phase 1. Phase 3 follows the decision of E2.2, and the choice of Phase 4 experiments depends on the results of Phases 2 and 3.

Division of the work. Two people work full time on the program. Dividing the work by phenomenon (initialization, optimizers, data) would require both of them to implement every student and every observable. We divide it instead by code ownership: person A owns the TTN side, person B the other students and the comparison (Tables 6 and 7).

- Person A owns the teacher generator, for all families and leaf laws; the exact population engine for TTN students (loss, gradients and moment recursion, in double precision); and the exact observables (data-weighted HSVDs, coupling matrices, error decompositions, ANOVA energies and noise-stability curves, conserved quantities). A runs E0.2, Phase 1, and the TTN side of E3.1–E3.3.
- Person B owns the other students of Table 5; the generic training harness (optimizers, schedules, checkpoints) and the run registry; and the sample-based observables (audit-set coefficients with sample splitting, crossed grids, Monte Carlo ANOVA energies and noise stability, empirical tangent kernels, sharpness). B runs E0.1, E0.3, Phase 2, the other side of E3.1–E3.3, E3.4 and E3.5.

Four rules prevent overlap. Every module and every experiment has a single owner, even when it uses the other person's code, and the other person reviews it. Runs are stored in a shared registry, keyed by teacher, data, student, optimizer, initialization and seed, and no key is run twice; in particular, A provides the TTN trajectories used in Phase 2. Every observable is implemented once, either exactly by A or by sampling by B, and E0.3 tests each implementation against the other. Finally, the decisions at the end of each stage of Table 7 are taken jointly. The two workloads differ in kind: B carries more engineering, A more open-ended exploration.

Expected outputs. The program should produce: a phase diagram of the training regimes of TTNs, with measurable order parameters (E1.1); a description of the saddle-to-saddle path of TTNs, with laws for the order and the time scales of acquisition in terms of a spectrum of the teacher and of the tree (E1.2, E1.3); a map of where the dynamics of TTNs and DNNs agree along the grid of students (E2.2), together with the properties of the data and of the teacher that separate them (E2.3–E2.5); the effects of optimizers, noise and finite samples on both (Phase 3); and, if the global hypothesis fails, a list of measurable reasons, in the spirit of the failure taxonomy of Surrogate note (2026, Section 11).

8 Experiments (old)

The first order of buisness should be to have a much better understanding of the training dynamics of TTNs empirically and be systematic about it. Furthermore to understand the gap between TTNs and DNNs we should also compare their training dynamics in a range of setups.

We are going to work in teacher-student setups with a known ground truth coming from the teacher. We want to investigate different architectures, initiatialisations, data structure, optimizers, and eventually loss functions.

What phenomena do we want to investigate: saddle to saddle regime, intermediary regime (not too small, not too large init), lazy (large init) regime. What features of the TTN does a student TTN learns, in what order, what are the acquisition time, when we change the data distribution how do acquisition times differ. We also want to investigate other interventions: comparing the trajectories on GD and SG in a given setup, changing batch sizes, studying other optimizers (AdamW, muon), adding weight decay. We want to vary the architectures: TTNs without copy, TTNs with copying, Deep quadratic neural nets, ReLU and other realistic MLPs, adding attention (will require some thinking about how to change the data structure).

An example of a setup to start with. Consider some input data sampled from a gaussian with some covariance matrix Σ_x. Generically Σ_x is not necessarily invertible (we can start with the identity and change this). For a start, we consider a teacher matrix TTN with bond dimension e.g. somewhere between $d_e = 10 - 1e4$ (we can play with this) and hierarchical singular values

ID	Question	Owner	Requires
E0.1	Do the recurrence, the tied tree and the coefficient tensor of a deep polynomial network agree in values and gradients?	B	–
E0.2	Do the observables and the simulators reproduce known answers?	A	–
E0.3	Do the sampled estimators agree with the exact values on TTN students?	B	E0.2
E1.1	Which regimes exist, and which order parameters distinguish them?	A	E0.2
E1.2	Which saddles are visited, for how long, and how does the rank profile grow?	A	E1.1
E1.3	Which spectrum sets the order and the time scales, and how does the loss decay?	A	E1.1
E1.4	How does difficulty depend on interaction order, leaf dimension and N?	A	E1.1
E1.5	What happens below and above the teacher's ranks, and with more factors per bond?	A	E1.1
E1.6	Does learning a feature accelerate the features built on it?	A	E1.1
E2.1	Which widths fit, which learning rates are stable, which time calibration?	B	E0.3, E1.1
E2.2	Which change of connectivity or node map first breaks the agreement?	B	E2.1
E2.3	Do DNNs depend on properties of the data that TTNs provably ignore?	B	E2.1
E2.4	Do DNNs depend on difficulty as TTNs do?	B	E1.4, E2.1
E2.5	Do interventions on the teacher shift acquisition times equally?	B	E1.6, E2.1
E3.1	How do gradient noise and large learning rates change plateaus, paths and regimes?	A, B	E2.2
E3.2	How do Adam and Muon change orders and paths?	A, B	E2.2
E3.3	Which atoms and ranks survive weight decay?	A, B	E2.2
E3.4	Which computation does training select among equivalent ones?	B	E0.1
E3.5	How many samples does each atom need?	B	E2.2

Table 6: Summary of the experiments. Person A owns the TTN side and person B the other students and the comparison (Section 7.8); in E3.1–E3.3, each person runs the experiment on their own students.

Weeks	Person A (TTN side)	Person B (other students and comparison)	Decision at the end
1–2	teacher generator; exact population engine and exact observables; E0.2	students of Table 5; training harness and run registry; sampled observables; E0.1	E0.3 passed
3–5	E1.1; E1.3(a); start of E1.2	E2.1 on A's teachers; E2.3; pilot of E2.2	canonical configurations and student settings frozen
6–9	E1.2; E1.3(b, c); E1.4–E1.6	E2.2 with twenty seeds; E2.4; E2.5	first student that breaks the agreement; students for Phases 3 and 4
10–12	E3.1–E3.3 on TTN students	E3.1–E3.3 on the selected students; E3.4; E3.5	choice of Phase 4 experiments

Table 7: Indicative schedule for two people working full time.

and vectors given by power law distributions (we can try different distributions but power law is a reasonable start because they mimic the structure of more realistic data). Chose a binary tree TTN teacher with depth between 3 and 10 (start with smaller depth). Then we want to train a student on that tree. Take a binary tree student TTN with bond dimension at least twice the bond dimension of the student but more can also be tried (10x for example). We want to be in the realizable setup in which the teacher can fully learn the student. Train with square-loss with SGD (or some variant, as long as optimizer is fixed). Try with and without weight decay.

During training track at least the following observables: loss, alignment between hierarchical singular vectors of the teacher and the student, hierarchical singular values of the student, acquisition times.

Vary the data structure via the structure of the teacher (its HSVDs) to see cleanly saddle to saddle and other regimes, maybe add some label noise.

Vary the student architecture: vary the depth and bond dimension, use a TTN without copying, use a TTN with copying, a deep quadratic networks (see the previous section for the observable to track to define the features and acquisition times), a ReLU network.

For a given architecture vary the optimizer: SGD with large batch size, small batch size, change the learning rate, use AdamW, use Muon. Compare their loss trajectories, the features that they learn (singular vectors and values).

9 Experiments (slop)

This section is to be completed. For reference, the experiments suggested in the previous sections are listed below. Detailed protocols, with teachers, observables, statistical analysis and decision criteria, are given in Surrogate note (2026, Sections 6–10) and Local TN note (2026, Sections 9–11).

E0. *Algebraic check.* Verify Theorem 2.30 numerically on a tiny network in double precision: the recurrence, the tied-tree contraction and the explicit coefficient tensor must agree in value and in gradient with respect to θ. An exactly polynomial activation, such as $\sigma(z) = z + z^2$, must give the same polynomial for every centre, and untying the cores must break the agreement of gradients. This check must pass before any scientific comparison.

E1. *Global comparison* (Hypothesis 1.2). On a teacher with known features, for example the staircase teacher $f^\star(x) = \sum_m \sqrt{\lambda_m} \prod_{i \le m} u_i^\top x^{(i)}$ on independent Gaussian blocks $x^{(i)}$ with unit vectors u_i, train a DNN and TTNs (on the physical tree, and the tied TTN of the polynomial proxy) on the same data with the same optimizer. Compare losses, function-space distances, acquisition times and orders (1.1), and tangent kernels (Remark 1.4).

E2. *Local surrogates* (Hypothesis 1.3). At several checkpoints, build the order-k surrogates of Definition 2.43 for $k \in \{1, 2, 3, 4\}$; measure $\varepsilon_{0,j}$, $\varepsilon_{1,j}$ and β_j at the checkpoint; run blind rollouts from the same parameters and optimizer state. Report the largest horizon over which predictions, feature coefficients and gradients agree within tolerance, as a function of k, depth and activation spread.

E3. *Polynomial proxies* (Definition 2.51). Train proxies with $K \in \{2, 3, 4\}$ alongside the original DNN, track $\sigma_{\ell,j}(t)$ and $r_\ell(t)$ to test for saddle-to-saddle (rank-incremental) dynamics, compare them with the same diagnostics for the local surrogates of the DNN, and compare feature acquisition with the DNN.

E4. *Interpretability* (Question 2.41). Train an MLP on modular addition, with concatenated one-hot inputs and quadratic or smooth activations, for example the quadratic network of Gromov (2023). Compute the physical HSVD of its logit tensor (Example 2.42) and the HSVD of its computational TTN, or of its TTN surrogate for smooth activations, and compare both with the Fourier subspaces of the key frequencies.

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